

GLOBAL CREDIT TRADER

The Increasing Importance of Demand Technicals in Credit

- USD and EUR investment grade (IG) corporate bond markets are on pace for record amounts of primary market supply this year, and issuance in leveraged finance markets remains brisk as well. While there has been meaningful dispersion across sectors and ratings, most *index-level* spread valuations are hovering towards the tight end of their (post-financial crisis) historical ranges.
- In our view, this reflects a generally supportive demand backdrop, at least in aggregate. It also sparks an obvious question: Can this demand tailwind persist? In this *Global Credit Trader*, we take stock of several measures of credit market demand to address this question.
- At a high level, we expect the *aggregate* demand technical will be an important offset to the heavy pace of supply, which we anticipate will continue across most markets. That said, the concentrated nature of debt issuance within the AI-related ecosystem is likely to push spreads modestly wider per our existing forecasts, with the risks skewed toward more meaningful widening.
- Fund flows show very steady demand for USD IG bonds and leveraged loans so far this year, while flows for USD HY and EUR IG and HY have been more muted (but remain positive). Under the surface, there has been a marked preference for passive ETFs and for shorter duration funds, with inflows into longer duration funds lagging since May.
- TRACE data also shows strong net demand for corporate bonds in the secondary market but echoes the weakness for longer-duration paper seen in the fund flow data. Recent TIC reports show near-record demand for US domestic corporate bonds from foreign investors. New issue concessions also remain healthy in aggregate, though there have been instances of larger concessions for some recent AI-related financings. Aggregate coupon payments—which have grown to be more sizable following the 2022-2023 rate hiking cycle—also remain an important ‘passive’ demand tailwind, as they are reinvested into the corporate bond market.

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The Increasing Importance of Demand Technicals in Credit

The acceleration in new issue activity across the corporate credit market has been closely watched by market participants so far this year. Both the USD and EUR Investment Grade (IG) markets are on pace for new annual gross issuance records, with USD IG issuance approaching \$1.5 trillion year to date (up 31% year over year) and EUR IG volumes at €535 billion (up 4% YoY). Leveraged finance supply has also remained steady, albeit shy of record territory. USD HY gross supply at \$195 billion is up nearly 10% over last year's pace and EUR HY supply at €70 billion is down 13%.

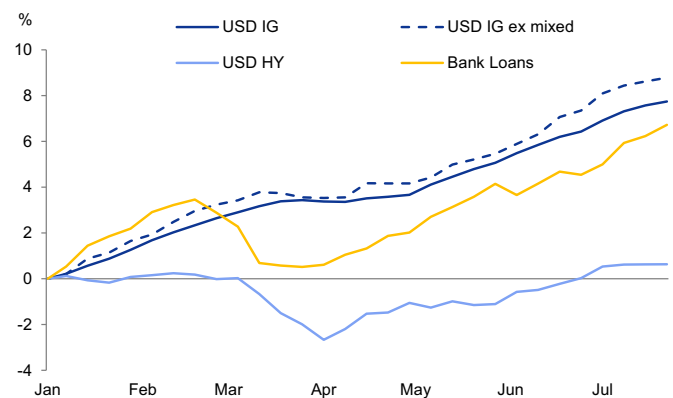
As a result of the heavy pace of primary market activity, investors have been acutely focused on the demand side of the equation. In this *Global Credit Trader* we take stock of several measures of credit market demand. At a high level, we expect the aggregate demand technical will be an important offset to the heavy pace of supply, which we anticipate will continue across most markets. That said, the concentrated nature of debt issuance within the AI-related ecosystem is likely to push spreads modestly wider per our existing forecasts, with the risks skewed toward more meaningful widening.

Fund flows: Mostly solid

The most immediate barometer for demand is fund flow data, which provides both real-time and granular insights. According to EPFR, broad USD IG bond ETFs have enjoyed net inflows for 65 consecutive weeks. Exhibit 1 illustrates that USD IG funds have added roughly 8% to their assets under management (AUM) via net inflows this year, while USD bank loan funds (including CLO-focused funds) have added over 6%. The sole laggard has been USD High Yield (HY) funds, which have generated a modest inflow of 0.6% YTD.

Exhibit 1: Funds focused on USD credit products have experienced solid inflows year to date

Year-to-date cumulative net flows into USD credit funds

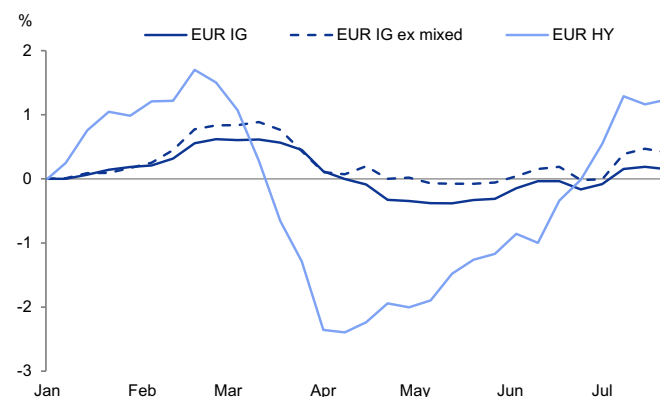


Source: EPFR, Goldman Sachs Global Investment Research

In the EUR market, flows have been softer but remain positive. Exhibit 2 highlights YTD cumulative net inflows of 0.4% for pure EUR IG corporate funds and 1.2% for EUR HY funds. While these figures may appear subdued on the surface, it is notable that every major credit market has garnered positive net inflows YTD despite tight valuations and multiple macro headwinds (most prominently, commodity market disruptions from geopolitical conflicts).

Exhibit 2: European credit funds have experienced modest net inflows year to date

Year-to-date cumulative net flows into EUR credit funds



Source: EPFR, Goldman Sachs Global Investment Research

Beneath the headline flow data, several structural themes emerge. Exhibit 3 shows cumulative YTD net flows across various vehicles: ETFs (active and passive) and mutual funds. Passive ETFs have dominated YTD flows across most asset classes, while active mutual funds have mostly experienced net outflows. Given the rising dispersion in credit markets, notably around the AI theme, it is likely that investors have migrated to using the ETF wrapper to achieve active manager selection which helps to explain the rise of the smaller, active ETF funds at the expense of mutual funds. The notable exception is in the leveraged loan market, where existing actively managed CLO ETFs have come to dominate loan-related flows over the past few years.

Exhibit 3: Most year-to-date inflows have been focused on passive ETFs

Cumulative year-to-date net fund flows (\$bn), by active/passive ETFs and active mutual funds

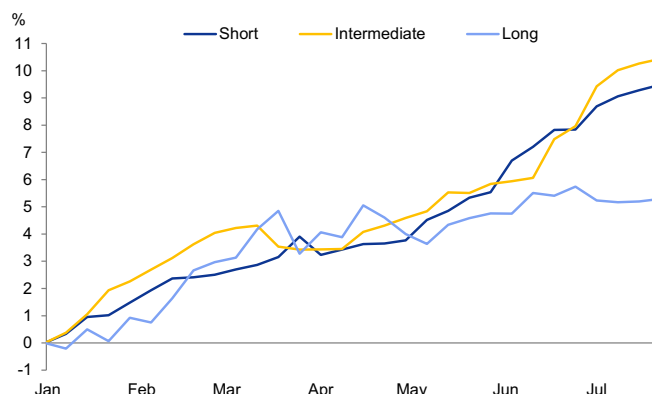
Asset Class	Passive ETF	Active ETF	Active MF	YTD Total
USD IG (corp-only)	48.0	4.8	-3.3	49.5
EUR IG (corp-only)	4.2	1.4	-2.7	2.6
USD HY Funds	3.6	3.5	-4.6	2.6
USD Loan Funds	0.4	12.1	-1.9	10.6
EUR HY Funds	1.0	0.1	0.8	2.0

Source: EPFR, Goldman Sachs Global Investment Research

Another important nuance is the recent bifurcation of IG flows across the curve. Exhibit 4 demonstrates that YTD IG corporate fund inflows have been more concentrated in short- and intermediate-duration funds over the last few months, while longer-duration funds have lagged since May (although are still posting inflows). For context, long-duration funds account for 15% of AUM in USD IG corporate funds, with short- and intermediate-duration funds accounting for 36% and 49% of AUM, respectively.

Exhibit 4: Demand for long-end corporate IG bonds has been more tepid than for front-end and intermediate durations

Cumulative year-to-date net flows into USD IG funds as a percentage of AUM, by tenor



Source: EPFR, Goldman Sachs Global Investment Research

While demand from yield-based buyers likely buoyed long duration bonds in the first four months of the year, the demand for long-duration assets has been more tepid as of late. This is likely a dual response to (1) a supply headwind at the long-end of the curve and (2) some residual uncertainty among credit investors related to the path of long-end rates, given the Fed leadership transition and potential changes in the central bank's reaction function and approach to areas such as the balance sheet. Year to date, bonds with a tenor greater than 15 years have accounted for 27.6% of USD IG gross supply, the highest share in five years.

Year-to-date performance also reflects this weakness. USD IG front-end spreads have tightened by 5bp YTD, while spreads on the 25+ year maturity index have widened by 15bp.

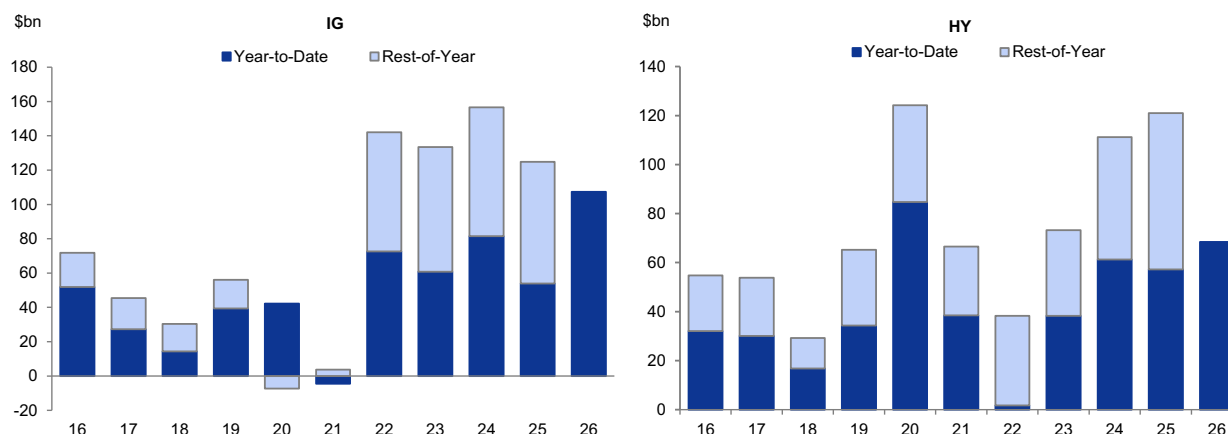
'Dealer lift' volumes reveal an investor preference for shorter maturities

Another robust metric of corporate credit demand is trade-level data, specifically the "net dealer lift" amount, which measures dealer-to-customer trades reported in TRACE, netting customer buys against customer sells. Aggregate data for both the USD IG and HY bond markets reveals a healthy appetite for corporate bonds, with the IG market on track for record net buying and the HY market pacing for its second-highest year on record after 2020 (Exhibit 5).

Zooming in on the net dealer lift data across the curve corroborates the fund flow narrative: net demand for long-end bonds has underperformed relative to the front end and belly. For instance, bonds with maturities greater than 12 years have accounted for 26% of YTD total trading volumes but account for just 22% of YTD net demand. Conversely, IG bonds with a maturity between 3 and 7 years have represented 27% of aggregate trading volumes this year but have accounted for an outsized 49% of net customer demand.

Exhibit 5: Trade-level data shows a record pace of net buying in USD IG corporate bonds, with USD HY approaching record levels

Net customer purchases of cash bonds from dealers by year, for IG (left panel) and HY (right panel)



Source: TRACE, Haver Analytics, Goldman Sachs Global Investment Research

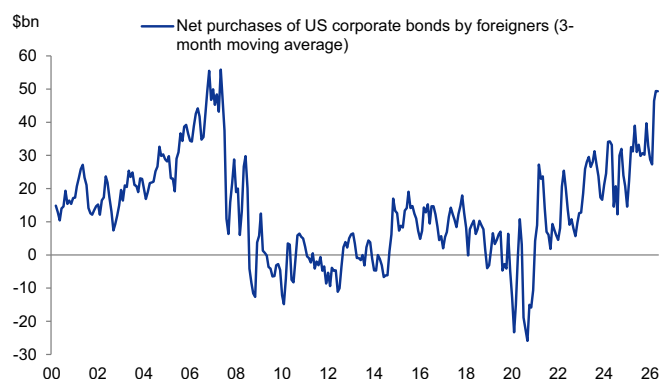
Other measures of demand also point to a solid backdrop

To complete our review of corporate bond demand, we look at three additional metrics: foreign net purchases of US bonds, new issue concessions, and passive demand from recycled coupon payments. All three point to a supportive backdrop.

Exhibit 6 shows a 3-month average of net purchases of US corporate bonds using the Treasury TIC data. (Note: this data references the residence of the issuer, not the currency of the bond.) The key takeaway is that foreign demand (with a caveat that the data is lagged by six weeks) has currently reached new post-financial crisis highs. In fact, the top five highest monthly totals for net foreign demand since the end of the financial crisis have all come in the last 18-months. As of May 2026, net long-term purchases by foreign investors totaled \$439 billion over the trailing 12-month period, pushing total foreign holdings to a record \$5.3 trillion. Looking below the surface, about half of the foreign demand so far this year has originated from Europe, with the rest spread across other regions.

Exhibit 6: TIC data shows near record foreign demand for US credit

3-month moving average of net foreign purchases of US domestic corporate bonds

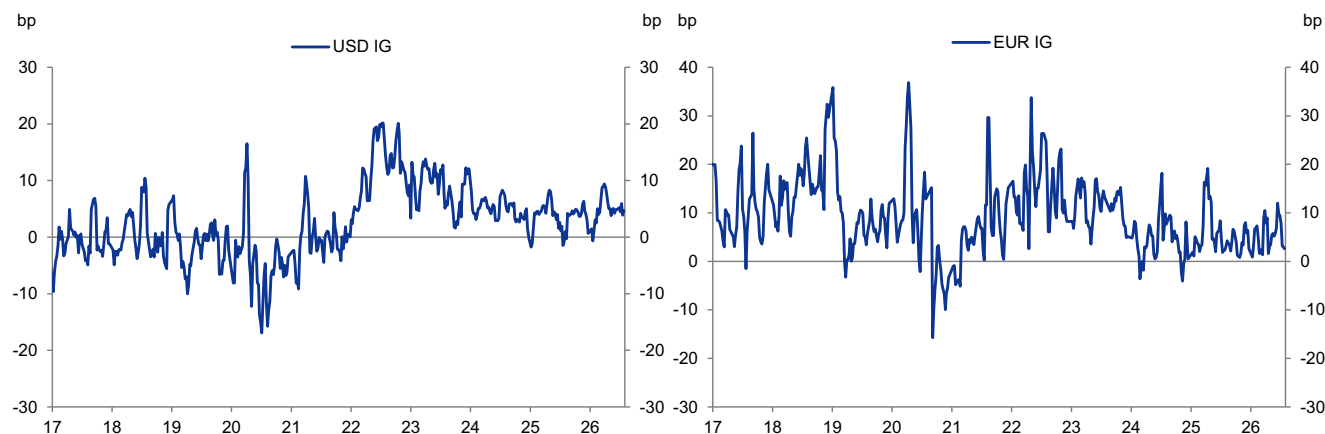


Source: US Treasury, Haver Analytics, Goldman Sachs Global Investment Research

Second, we monitor new issue concessions to gauge the market's digestion of incremental supply. Despite recent anecdotes of a few large AI-related deals pricing with higher-than-typical concessions, the *aggregate* data paints a more benign picture—at least so far. Exhibit 7 tracks the 4-week moving average of our estimate of new issue concessions for the USD and EUR IG markets. While both markets experienced upward pressure during the early stages of the Middle East conflict, they have since restabilized in the low- to mid-single digits—a range consistent with the past decade. Given our expectations for additional debt issuance from the AI-related ecosystem, we will be monitoring this metric closely.

Exhibit 7: Aggregate new issue concessions have held up well, despite the acceleration of supply

4-week moving average of USD (left panel) and EUR (right panel) IG estimated new issue concessions

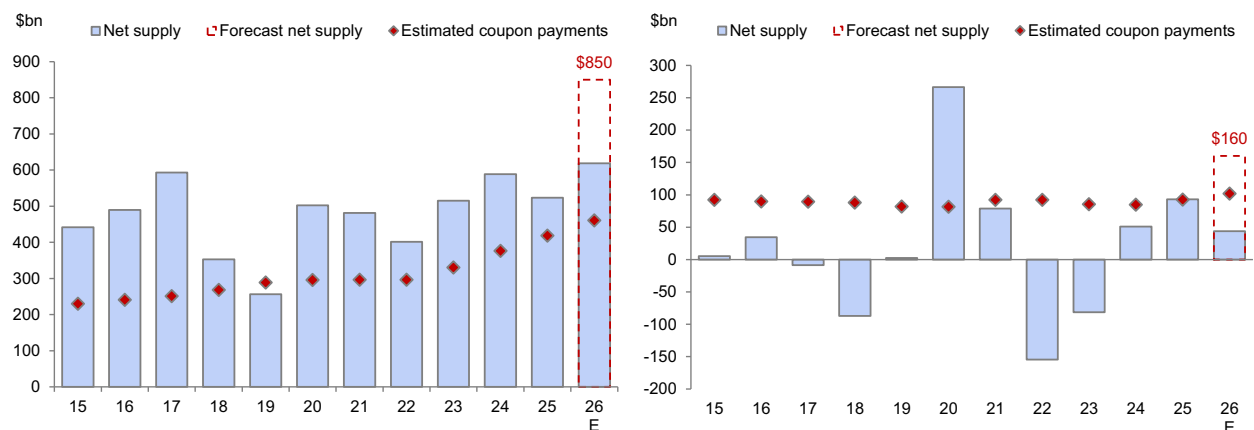


Source: iBoxx, Dealogic, Goldman Sachs Global Investment Research

Finally, we view coupon payments as a powerful source of passive demand, operating under the assumption that the majority of these payments are recycled back into the corporate bond market. Exhibits 8 and 9 plot our estimates of annual coupon payments against net supply in the USD and EUR IG and HY markets. Despite accelerating net supply this year, aggregate coupon payments have also grown, driven by higher all-in yields resulting from the 2022-2023 rate hiking cycle. We estimate that coupon payments alone will absorb more than 50% of net supply in the USD IG and HY markets (Exhibit 8) and roughly 40% of net supply in the EUR IG and HY markets (Exhibit 9). While lower than the percentages seen over the last few years, this remains very healthy compared to the past decade.

Exhibit 8: Coupon payments should still be enough to absorb more than half of this year's net supply in the USD corporate bond market

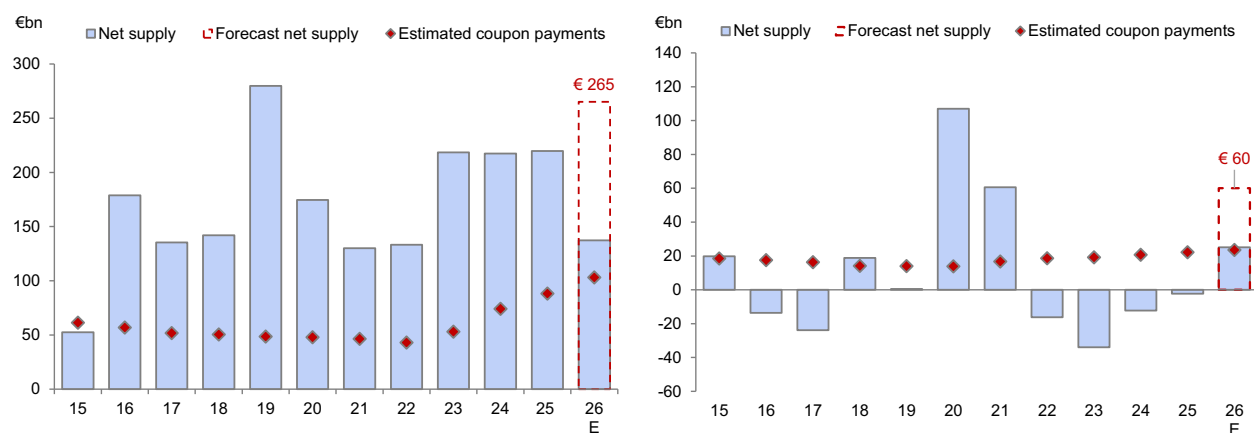
Annual net supply vs. estimated aggregate coupon payments in the USD IG (left panel) and HY markets (right panel)



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 9: Coupon payments should absorb roughly 40% of 2026 net supply in the EUR corporate bond market

Annual net supply vs. estimated aggregate coupon payments in the EUR IG (left panel) and HY markets (right panel)



Source: Bloomberg, Goldman Sachs Global Investment Research

None of these demand measures is comprehensive on its own—each has its flaws and potential blind spots. That said, this broad array of metrics points to a supportive *aggregate* demand technical. We expect this will be an important mitigant to the heavy pace of supply that we expect.

That said, the concentrated nature of debt issuance within the AI-related ecosystem is likely to push spreads modestly wider per our existing forecasts, with the risks skewed toward more meaningful widening.

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 10: USD and EUR spread forecasts

Market	Updated as of Jul. 29, 2026				
	Current	2026Q3	2026Q4	2027Q1	2027Q2
USD spreads (bp)					
USD IG	80	85	82	80	78
USD HY	289	305	295	290	280
EUR spreads (bp)					
EUR IG	79	91	88	86	84
EUR HY	291	320	315	310	305

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 12: Issuance forecasts

Market	Updated as of Jul. 29, 2026	
	2026 YTD (bn)	2026 Full-year forecast (bn)
USD market		
Gross issuance		
IG	\$1,460	\$2,100
HY	\$195	\$400
Leveraged loans	\$253	--
Net issuance		
IG	\$617	\$850
HY	\$37	\$160
EUR market		
Gross issuance		
IG	€ 535	€ 850
HY	€ 70	€ 125
Leveraged loans	€ 56	--
Net issuance		
IG	€ 136	€ 265
HY	€ 24	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 11: USD and EUR excess returns forecast

Market	Updated as of Jul. 29, 2026		
	Current duration	2026 YTD (%)	2026 Full-year forecast (%)
USD excess returns			
USD IG	6.7	0.2%	0.4%
USD HY	3.3	1.1%	1.6%
EUR excess returns			
EUR IG	4.6	0.6%	0.5%
EUR HY	3.1	1.2%	1.4%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 13: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of Jul. 29, 2026	
	2026 YTD	2026 Full-year forecast
USD market		
Rising stars (bn)	\$62	\$75
Fallen angels (bn)	\$34	\$60
HY defaults (12-month trailing)	4.0%	4.0%
Leveraged loan defaults (12-month trailing)	5.0%	4.75%
EUR market		
Rising stars (bn)	€ 10	€ 20
Fallen angels (bn)	€ 4	€ 30
HY defaults (12-month trailing)	4.6%	5.0%
Leveraged loan defaults (12-month trailing)	3.6%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 14: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	The growth, inflation and monetary policy mix is somewhat more favorable in the US, relative to Europe. As a result, and relative to the USD market, we forecast a wider peak (and a slower recovery in 2H2026) in EUR credit spreads.
	Overweight EUR IG CLO tranches vs. USD IG CLO tranches	EUR IG tranches offer carry pickup relative to USD IG CLO tranches. While the policy and macroeconomic picture is less supportive in Europe, subordination should protect tranches up in the capital structure from a continued idiosyncratic default picture.
USD market	Overweight IG vs. HY	We have a slight preference for IG vs. HY in the USD market, owing to valuations and our economists' forecasts for below-potential growth. We view the BB/BBB spread ratio as fairly priced amid a somewhat more challenging growth and inflation mix, and have a neutral view between the two rating cohorts.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
	Overweight BBs, Neutral CCCs, Underweight Bs	Current spread levels indicate that single-Bs offer the lowest credit risk premium relative to expected losses among all HY rating buckets.
	Overweight HY bonds vs. loans	HY bonds still provide a superior trade-off between carry and duration compared to leveraged loans. In addition, loans' substantial software sector exposure is likely to present a sentiment overhang, at least in the near-term.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight Hybrids vs. BBBs	We view the additional spread compensation of hybrids vs. BBBs as moderately attractive in the context of the tight spread environment. We also expect the hybrid spread premium to compress vs. BBBs, as the hybrid market grows further.
EUR market	Overweight IG vs. HY	We prefer IG over HY given the more challenging growth, inflation, and monetary policy mix in the Euro Area.
	Overweight As vs. BBBs	In the EUR market, we prefer As over BBBs, as the weaker growth backdrop warrants more of an up-in-quality stance. That said, we would not move too far up the quality spectrum, as we expect supply technicals will be a headwind in the EUR AA market.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB may also increase borrowing costs for floating rate issuers, given the central bank's focus on inflationary pressures.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.

Source: Goldman Sachs Global Investment Research

Earnings calendar

Exhibit 15: USD IG earnings report dates

Upcoming earnings report dates for the Bloomberg USD IG Index (LUACTRUU); minimum \$1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
7/31/2026			8/3/2026			8/4/2026			8/5/2026		
ABBV	AbbVie Inc	57.4	MJFG	Mitsubishi UFJ Financial Group	41.8	HSBC	HSBC Holdings PLC	89.1	LLY	Eli Lilly & Co	39.8
SUMIBK	Sumitomo Mitsui Financial Gro.	41.8	OKE	ONEOK Inc	29.3	DUK	Duke Energy Corp	68.2	DIS	Walt Disney Co/The	31.1
ENBCN	Enbridge Inc	22.9	WMB	Williams Cos Inc/The	26.5	PFE	Pfizer Inc	53.4	HNDA	Honda Motor Co Ltd	22.7
XOM	ExxonMobil Holdings Corp	18.5	MAR	Marriott International Inc/MD	15.0	ET	Energy Transfer LP	40.5	MET	MetLife Inc	15.6
NWVG	NatWest Group PLC	15.9	ARE	Alexandria Real Estate Equitie	10.5	TOYOTA	Toyota Motor Corp	31.6	HONA	Honeywell Aerospace Inc	15.5
ETN	Eaton Corp PLC	14.2	FANG	Diamondback Energy Inc	9.4	BPLN	BP PLC	30.6	PSX	Phillips 66	15.4
LYB	LyondellBasell Industries NV	10.8	TSN	Tyson Foods Inc	6.2	SPCX	Space Exploration Technologies	25.0	NI	NiSource Inc	15.1
ARESSI	Ares Management Corp	6.1	CNHI	CNH Industrial NV	4.3	MPLX	Marathon Petroleum Corp	24.7	O	Realty Income Corp	14.3
AN	AutoNation Inc	3.5	CLX	Clorox Co/The	4.0	GILD	Gilead Sciences Inc	23.0	GPN	Global Payments Inc	13.9
FTSCN	Fortis Inc/Canada	2.7	BWP	Loews Corp	3.2	CAT	Caterpillar Inc	22.9	KHC	Kraft Heinz Co/The	11.8
FRT	Federal Realty Investment Trus	2.2	CNA	Loews Corp	3.0	PEG	Public Service Enterprise Grou	20.1	OXY	Occidental Petroleum Corp	11.2
PXD	ExxonMobil Holdings Corp	2.1	JXN	Jackson Financial Inc	2.1	PRU	Prudential Financial Inc	16.6	ATO	Atmos Energy Corp	9.4
LIN	Linde PLC	1.7	GBDC	Golub Capital BDC Inc	2.0	SYF	Sysco Corp	10.9	RPRX	Royalty Pharma PLC	8.8
NVT	nVent Electric PLC	1.3	L	Loews Corp	1.8	FIS	Fidelity National Information	10.2	COR	Cencora Inc	8.6
ARES	Ares Management Corp	1.3	VNOM	Diamondback Energy Inc	1.6	DVN	Devon Energy Corp	9.6	MSI	Motorola Solutions Inc	7.6
BEN	Franklin Resources Inc	1.2	SBAC	SBA Communications Corp	1.5	RABOBK	Coöperatieve Rabobank UA	9.4	EXPE	Expedia Group Inc	5.5
SNE	Sony Group Corp	1.0	SBRA	Sabra Health Care REIT Inc	1.2	PNW	Pinnacle West Capital Corp	8.8	SOLV	Solventum Corp	4.6
						CRBG	Corebridge Financial Inc	7.2	NNN	NNN REIT Inc	4.2
						KMB	Kimberly-Clark Corp	6.0	HST	Host Hotels & Resorts Inc	4.1
						MPC	Marathon Petroleum Corp	5.4	OC	Owens Corning	4.0
8/6/2026			8/7/2026			8/10/2026			8/11/2026		
TMUS	Deutsche Telekom AG	61.3	PPL	PPL Corp	11.2	JBS	JBS NV	16.1	CAH	Cardinal Health Inc	6.3
SRE	Sempra	25.7	PAA	Plains All American Pipeline L	8.2	SPG	Simon Property Group Inc	15.4	HRB	H&R Block Inc	1.5
ED	Consolidated Edison Inc	24.1	VST	Vistra Corp	4.0	ABXCN	Barrick Mining Corp	3.2			
COP	ConocoPhillips	18.0	TE	Emera Inc	3.8	PPC	JBS NV	3.1			
TRGP	Targa Resources Corp	17.9	EMACN	Emera Inc	2.5	EMBRBZ	Embraer SA	1.7			
ONCRX	Sempra	15.8	MBGL	Mobility Global Inc	2.0	FERG	Ferguson Enterprises Inc	1.5			
FISV	Fiserv Inc	15.0	TTWO	Take-Two Interactive Software	1.9						
KDP	Keurig Dr Pepper Inc	13.9									
RSG	Republic Services Inc	10.8									
CQP	Cheniere Energy Inc	9.5									
DGELN	Diageo PLC	9.2									
EVRG	Evergy Inc	8.5									
KVUE	Kenvue Inc	7.0									
FOXA	Fox Corp	6.6									
AIG	American International Group I	6.2									
PH	Parker-Hannifin Corp	5.9									
CNQCN	Canadian Natural Resources Lt	5.8									
LNG	Cheniere Energy Inc	4.7									
S	Deutsche Telekom AG	4.5									
TAP	Molson Coors Beverage Co	4.4									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 16: USD HY earnings report dates

Upcoming earnings report dates for the Bloomberg USD HY index (LF98TRUU); minimum \$500 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
7/31/2026			8/3/2026			8/4/2026			8/5/2026		
NWL	Newell Brands Inc	4.6	NSANY	Nissan Motor Co Ltd	12.4	TDG	TransDigm Group Inc	19.8	IRM	Iron Mountain Inc	10.8
ABR	Arbor Realty Trust Inc	1.2	ECHO	EchoStar Corp	7.7	PARA	Paramount Skydance Corp	12.1	SM	SM Energy Co	6.5
PRM	Perimeter Solutions Inc	1.2	WHR	Whirlpool Corp	6.4	NRG	NRG Energy Inc	12.0	XYZ	Block Inc	5.2
EOFP	Forvia SE	1.0	DISH	EchoStar Corp	5.0	SUN	Sunoco LP	10.7	RIG	Transocean Ltd	4.3
TACN	TransAlta Corp	0.7	CRGYFN	Crescent Energy Co	4.2	CE	Celanese Corp	8.2	GT	Goodyear Tire & Rubber Co/Th	4.1
CRI	Carter's Inc	0.6	ALSN	Allison Transmission Holdings	2.4	LVT	Lumen Technologies Inc	7.3	HNDLIN	Hindalco Industries Ltd	3.9
MOGA	Moog Inc	0.5	SNAP	Snap Inc	2.1	DVA	DaVita Inc	6.3	WULF	Terawulf Inc	3.2
			JAZZ	Jazz Pharmaceuticals PLC	1.5	WYNNMAC	Wynn Resorts Ltd	4.1	TLN	Talen Energy Corp	2.7
			DHC	Diversified Healthcare Trust	1.4	BALL	Ball Corp	3.9	SBGI	Sinclair Inc	2.6
			NSCO	Custom Truck One Source Inc	0.9	ET	Energy Transfer LP	3.2	MEDIND	Medline Inc	2.5
			VNO	Vornado Realty Trust	0.9	MTCHII	Match Group Inc	3.0	MTDR	Matador Resources Co	2.4
			BWXT	BWX Technologies Inc	0.8	W	Wayfair Inc	2.6	DKL	Delek US Holdings Inc	2.2
			JELD	JELD-WEN Holding Inc	0.8	WYNNFIN	Wynn Resorts Ltd	2.6	EHC	Encompass Health Corp	2.1
			ON	ON Semiconductor Corp	0.7	CC	Chemours Co/The	2.5	KNTK	Kinetik Holdings Inc	2.1
			LIND	Lindblad Expeditions Holdings	0.7	NGL	NGL Energy Partners LP	2.2	LILAPR	Liberty Latin America Ltd	2.0
			GPOR	Gulfport Energy Corp	0.7	HOUS	Compass Inc	2.1	OUT	Outfront Media Inc	2.0
			TDW	Tidewater Inc	0.7	LNW	Light & Wonder Inc	2.1	ADNT	Adient PLC	1.8
			SBH	Sally Beauty Holdings Inc	0.6	BLKPRL	Cipher Digital Inc	2.0	PRGO	Perrigo Co PLC	1.8
			DAC	Danaos Corp	0.5	GPK	Graphic Packaging Holding Co	2.0	AXON	Axon Enterprise Inc	1.8
						ENR	Energizer Holdings Inc	1.8	ECPG	Encore Capital Group Inc	1.8
8/6/2026			8/7/2026			8/10/2026			8/11/2026		
CSCHLD	Optimum Communications Inc	15.5	GTN	Gray Media Inc	5.0	MPW	Medical Properties Trust Inc	5.1	VENLNG	Venture Global Inc	11.0
WBD	Warner Bros Discovery Inc	13.7	AXL	Dauch Corp	2.7	RAKUTN	Rakuten Group Inc	3.6	VEGLPL	Venture Global Inc	9.5
RKT	Rocket Cos Inc	10.0	MLTPLN	Claritev Corp	2.2	IHRT	iHeartMedia Inc	2.5	VENTGL	Venture Global Inc	5.5
POST	Post Holdings Inc	7.1	EMACN	Emera Inc	2.0	CODI	Compass Diversified Holdings	1.3	BGS	B&G Foods Inc	1.3
NXST	Nexstar Media Group Inc	6.1	GLP	Global Partners LP/MA	1.3	CRC	California Resources Corp	1.3	ARMK	Aramark	1.2
FSK	FS KKR Capital Corp	4.3	CTEV	Claritev Corp	1.0	ACH	Accendra Health Inc	1.2			
STWD	Starwood Property Trust Inc	4.2	FLR	Fluor Corp	0.5	SURCEN	Surgery Partners Inc	1.2			
NAVI	Navient Corp	3.8	HE	Hawaiian Electric Industries I	0.5	ACM	AECOM	1.2			
BCULC	Restaurant Brands International	3.7				BZH	Beazer Homes USA Inc	1.0			
FUN	Six Flags Entertainment Corp	3.4				SDRLNO	Seadrill Ltd	0.7			
RHP	Ryman Hospitality Properties I	3.3				INFNAR	Infinity Natural Resources Inc	0.6			
OTEXCN	Open Text Corp	3.3				CNDT	Conduent Inc	0.5			
GEL	Genesis Energy LP	3.2									
GMABDC	Genmab A/S	2.5									
LAMR	Lamar Advertising Co	2.5									
GEN	Gen Digital Inc	2.5									
BURLN	Burford Capital Ltd	2.4									
GSYCN	goeasy Ltd	2.4									
KGS	Kodiak Gas Services Inc	2.4									
HTZ	Hertz Global Holdings Inc	2.3									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 17: EUR IG earnings report dates

Upcoming earnings report dates for the Bloomberg EUR IG Index (LECP TREU); minimum €1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
7/31/2026			8/3/2026			8/4/2026			8/5/2026		
ENGIFP	Engie SA	28.3	MUFG	Mitsubishi UFJ Financial Group	4.9	HSBC	HSBC Holdings PLC	22.5	ANNGR	Vonovia SE	15.5
NWVG	NatWest Group PLC	19.7	CNH	CNH Industrial NV	1.8	BKNG	Booking Holdings Inc	15.0	NOVOB	Novo Nordisk A/S	13.0
LIN	Linde PLC	14.5	MITCO	Mitsubishi Corp	1.0	BPLN	BP PLC	12.7	HEIANA	Heineken Holding NV	11.6
AXASA	AXA SA	13.4				TOYOTA	Toyota Motor Corp	12.0	DHLGR	Deutsche Post AG	10.0
OMVAV	OMV AG	7.8				BAYNGR	Bayer AG	9.6	FREGR	Fresenius SE & Co KGaA	6.6
RBAIV	Raiffeisen Bank International	6.5				CCEP	Coca-Cola Europacific Partners	8.9	HNDA	Honda Motor Co Ltd	5.4
SUMIBK	Sumitomo Mitsui Financial Gro.	6.5				RABOBK	Coöperatieve Rabobank UA	7.9	IFXGR	Infineon Technologies AG	4.8
BKIR	Bank of Ireland Group PLC	6.0				LEGR	LEG Immobilien SE	4.4	MET	MetLife Inc	4.7
EJRAIL	East Japan Railway Co	6.0				LHAGR	Deutsche Lufthansa AG	3.9	CCHLN	Coca-Cola HBC AG	4.3
HOLNSW	Holcim AG	5.4				CONGR	Continental AG	3.3	LLY	Eli Lilly & Co	4.2
ETN	Eaton Corp PLC	3.3				PFE	Pfizer Inc	3.3	GLENLN	Glencore PLC	3.9
ABBV	AbbVie Inc	3.1				FIS	Fidelity National Information	2.6	WKLNA	Wolters Kluwer NV	3.8
XOM	ExxonMobil Holdings Corp	3.0				EVKGR	Evonik Industries AG	2.3	FMEGR	Fresenius SE & Co KGaA	3.6
UCAJLN	Unicaja Banco SA	2.9				CAJAMA	Banco de Crédito Social Coope	2.2	BAMIM	Banco BPM SpA	3.5
ALPHA	Alpha Bank SA	2.8				CAT	Caterpillar Inc	2.0	KHC	Kraft Heinz Co/The	3.4
AMSSM	Amadeus IT Group SA	2.0				DUK	Duke Energy Corp	1.9	ADNA	Koninklijke Ahold Delhaize NV	3.3
IAGLN	International Consolidated Air	1.5				EMR	Emerson Electric Co	1.5	O	Realty Income Corp	2.4
COVFIN	Covestro AG	1.0				APTV	Aptiv PLC	1.3	SDZSW	Sandoz Group AG	2.4
UKPONE	Engie SA	1.0				SNLN	Smith & Nephew PLC	1.0	OTPHB	OTP Bank Nyrt	1.6
									FLTR	Flutter Entertainment PLC	1.4
8/6/2026			8/7/2026			8/10/2026			8/11/2026		
SIEGR	Siemens AG	22.8	ALVGR	Allianz SE	13.9	SPG	Simon Property Group Inc	1.3	IHGLN	InterContinental Hotels Group	2.2
CMZB	Commerzbank AG	15.4	MUNRE	Muenchener Rueckversicherung	6.3				ISSDC	ISS A/S	1.9
DT	Deutsche Telekom AG	13.5	BACRED	Banca Monte dei Paschi di Sier	3.3				IWGLN	International Workplace Group	1.1
NTT	NTT Inc	11.3	MONTE	Banca Monte dei Paschi di Sier	2.8						
KBCBB	KBC Group NV	11.2	UNIIM	Unipol Assicurazioni SpA	2.8						
GIM	Generali	10.9	DTRGR	Daimler Truck Holding AG	2.4						
TMUS	Deutsche Telekom AG	7.3	MITHCC	Mitsubishi HC Capital Inc	1.0						
DGELN	Diageo PLC	7.1									
FISV	Fiserv Inc	5.0									
MRKGR	Merck KGaA	4.9									
HBRLN	Harbour Energy PLC	3.6									
NNGRNV	NN Group NV	3.5									
WPPLN	WPP PLC	3.0									
BPEIM	BPER Banca SPA	3.0									
KDP	Keurig Dr Pepper Inc	3.0									
JDEPNA	Keurig Dr Pepper Inc	2.9									
SRENVX	Swiss Re AG	2.8									
ZURNVX	Zurich Insurance Group AG	2.3									
HENKEL	Henkel AG & Co KGaA	2.1									
ONCRTX	Sempre	2.1									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 18: EUR HY earnings report dates

Upcoming earnings report dates for the Bloomberg Pan-Euro HY index (LP01TREU); minimum €100 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
7/31/2026			8/3/2026			8/4/2026			8/5/2026		
EOFP	Forvia SE	3.7	NSANY	Nissan Motor Co Ltd	2.1	BAYNGR	Bayer AG	4.6	PPCGA	Public Power Corp SA	1.9
ALPHA	Alpha Bank SA	1.0	WHR	Whirlpool Corp	0.5	CE	Celanese Corp	1.8	BAMIIM	Banco BPM SpA	1.5
						AMSSW	ams-OSRAM AG	1.7	IRM	Iron Mountain Inc	1.2
						IFIM	Banca IFIS SpA	1.5	MKSI	MKS Inc	1.0
						BALL	Ball Corp	0.9	DOBIM	doValue SpA	0.7
						CAJAMA	Banco de Credito Social C	0.8	UGI	UGI Corp	0.7
						ENR	Energizer Holdings Inc	0.7	HNDLIN	Hindalco Industries Ltd	0.5
						ASMDEE	Asmodee Group AB	0.6	PRMWCN	Primo Brands Corp	0.4
						LHAGR	Deutsche Lufthansa AG	0.5	GT	Goodyear Tire & Rubber Co/The	0.4
						YULCLN	Synthomer PLC	0.4	PRGO	Perrigo Co PLC	0.4
						BOCHLN	Bank of Cyprus Holdings F	0.3	BFFIM	BFF Bank SpA	0.3
						SHILN	SIG PLC	0.3	ECPG	Encore Capital Group Inc	0.3
						GPK	Graphic Packaging Holding	0.3			
8/6/2026			8/7/2026			8/10/2026			8/11/2026		
SOFTBK	SoftBank Group Corp	7.9	LXSGR	LANXESS AG	2.2	MPW	Medical Properties Trust Ir	2.1	ARMK	Aramark	0.4
HBRLN	Harbour Energy PLC	1.8	ETLFP	Eutelsat Communicatio	2.1	MOGOFI	Eleving Group SA	0.3			
MYTIL	Metten Energy & Metals P	1.4	BACRED	Banca Monte dei Pasct	1.9						
BPEIM	BPER Banca SPA	1.3	MONTE	Banca Monte dei Pasct	0.5						
FOUR	Shift4 Payments Inc	1.1									
DAN	Dana Inc	0.6									
WBD	Warner Bros Discovery Inv	0.6									
NOVALJ	Nova Ljubljanska Banka d	0.5									
ADENVX	Adecco Group AG	0.5									
PRAA	PRA Group Inc	0.3									
TATTGA	CrediaBank SA	0.2									

Source: Bloomberg, Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 19: Performance of USD and EUR markets

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	80	5.5	-1.60	-1.60	-0.76	2	0.64
AA	60	5.3	-2.15	-2.15	-1.66	11	0.67
A	68	5.3	-1.40	-1.40	-0.53	3	0.56
BBB	100	5.6	-1.46	-1.46	-0.23	-1	0.55
HY	289	7.5	-0.42	-0.42	1.53	23	0.95
BB	176	6.5	-0.46	-0.46	1.37	7	0.54
B	303	7.8	-0.21	-0.21	2.23	2	0.44
CCC	1013	14.6	-1.00	-1.00	-0.86	128	1.63
USD Leveraged Loans							
Leveraged loan index	437	8.2	0.00	0.00	-0.46	42	0.4
BB	257	6.3	0.00	0.00	0.70	2	-0.1
B	448	8.3	0.00	0.00	-0.93	57	0.5
CCC	1877	23.9	0.00	0.00	-5.09	314	3.3
USD Securitized products							
Agency MBS	28	5.2	-1.46	-1.46	-0.03	6	0.6
ABS (Autos)	50	4.7	0.30	0.30	1.77	-16	0.4
ABS (Credit cards)	21	4.3	0.18	0.18	1.39	-12	0.4
ABS (Student loans)	93	4.9	0.48	0.48	2.76	120	-0.1
USD Macro indices							
CDX IG	55	--	-0.13	-0.13	0.34	5	--
CDX HY	323	--	-0.51	-0.51	1.08	7	--
EUR Corporate bonds							
IG	91	3.9	-0.94	-0.94	0.43	2	0.4
AA	69	3.8	-1.25	-1.25	0.03	7	0.5
A	84	3.8	-0.94	-0.94	0.41	0	0.4
BBB	105	4.0	0.00	0.00	0.00	3	0.4
HY	264	5.9	-0.49	-0.49	1.41	-6	0.3
AT1	263	0.0	0.00	0.00	0.00	-23	0.0
EUR Leveraged loans							
Leveraged loan index	497	7.2	0.00	0.00	-0.87	47	0.6
BB	282	5.0	0.00	0.00	0.75	-70	-0.6
B	471	6.9	0.00	0.00	-1.30	56	0.7
EUR Macro indices							
iTraxx Main	55	--	-0.14	-0.14	0.4	5	--
iTraxx Xover	269	--	-0.94	-0.94	2.3	25	--
iTraxx SenFin	57	--	-0.10	-0.10	0.6	3	--
iTraxx SubFin	92	--	-0.06	-0.06	1.4	-1	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96. EUR IG we use the iBoxx QW5A index with history back to Jan-99. EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 20: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	178.0	80		5.5	7.3	BAA1/BAA2	0	1	1	-9	0.1	-1.3	-1.3	0.0
Chemicals	1%	80.1	84		5.6	7.5	BAA1/BAA2	0	-1	-1	-14	0.1	-1.3	-1.3	0.5
Metals & Mining	1%	72.0	72		5.5	7.5	A3/BAA1	0	3	3	-4	0.0	-1.5	-1.5	-0.4
Paper	0%	25.9	90		5.5	5.9	BAA2/BAA3	1	2	2	-6	0.2	-0.9	-0.9	0.0
Capital Goods	5%	408.8	61		5.2	6.6	A3/BAA1	0	7	7	-4	0.1	-1.4	-1.4	-0.4
Aerospace & Defense	2%	149.6	68		5.4	8.0	BAA1/BAA2	0	6	6	-5	0.0	-1.9	-1.9	-0.7
Building Materials	1%	39.9	76		5.4	6.5	BAA1/BAA2	0	4	4	0	0.2	-1.3	-1.3	-0.4
Diversified Manufacturing	2%	116.6	60		5.2	6.3	A3/BAA1	1	7	7	-2	0.1	-1.4	-1.4	-0.4
Construction Machinery	1%	57.4	37		4.8	4.6	A1/A2	0	9	9	-9	0.2	-0.7	-0.7	0.1
Packaging	0%	14.2	68		5.1	4.4	BAA2/BAA3	0	1	1	-14	0.3	-0.4	-0.4	0.5
Environmental	0%	31.1	46		5.0	5.8	A3/A3	1	8	8	-6	0.2	-0.9	-0.9	-0.3
Communications	8%	570.5	115		5.9	8.3	A3/BAA1	2	16	16	22	-0.4	-3.0	-3.0	-2.9
Cable & Satellite	2%	123.8	174		6.6	8.7	BAA1/BAA2	2	19	19	33	-0.5	-3.9	-3.9	-3.9
Media & Entertainment	2%	132.4	102		5.8	8.2	A1/A2	0	20	20	20	-0.6	-3.3	-3.3	-4.5
Wireless	2%	129.7	89		5.5	7.0	BAA1/BAA2	2	12	12	10	-0.2	-2.0	-2.0	-1.2
Wirelines	2%	184.7	104		5.9	8.9	BAA1/BAA2	2	13	13	17	-0.3	-2.9	-2.9	-2.4
Consumer Cyclical	7%	556.0	73		5.3	6.2	A2/A3	1	10	10	3	0.1	-1.5	-1.5	-0.7
Automotive	2%	168.5	82		5.3	4.2	A3/BAA1	-1	3	3	-8	0.3	-0.5	-0.5	0.7
Leisure	0%	4.7	101		5.6	6.2	BAA2/BAA2	-3	11	11	-1	0.4	-1.5	-1.5	-0.8
Gaming	0%	22.6	108		5.6	4.6	BAA2/BAA3	1	4	4	-2	0.2	-0.7	-0.7	0.4
Home Construction	0%	9.6	82		5.3	5.0	BAA1/BAA2	0	4	4	-2	0.3	-0.8	-0.8	0.1
Lodging	0%	19.5	71		5.2	4.7	BAA2/BAA3	0	8	8	-4	0.3	-0.7	-0.7	0.1
Retailers	4%	270.8	65		5.3	7.5	A1/A2	2	22	22	23	-0.2	-2.3	-2.3	-1.8
Restaurants	0%	31.9	62		5.3	7.5	BAA1/BAA2	0	8	8	-3	0.1	-1.7	-1.7	-0.6
Consumer Cyc Services	0%	28.4	76		5.3	6.0	A3/BAA1	0	5	5	7	0.2	-1.1	-1.1	-0.6
Consumer Non-Cyclical	15%	1107.6	66		5.4	7.6	A2/A3	0	6	6	-2	0.1	-1.7	-1.7	-0.8
Consumer Products	1%	59.1	46		5.0	5.8	A2/A3	-1	4	4	-9	0.2	-0.9	-0.9	-0.1
Food & Beverage	3%	228.4	70		5.4	7.5	A3/BAA1	0	5	5	2	0.1	-1.6	-1.6	-0.9
Healthcare	4%	323.3	72		5.4	7.7	A3/BAA1	-1	4	4	-5	0.1	-1.7	-1.7	-0.6
Pharmaceuticals	5%	398.7	59		5.4	8.2	A1/A2	0	9	9	0	0.0	-2.0	-2.0	-1.1
Supermarkets	0%	13.1	88		5.7	9.3	BAA1/BAA2	0	5	5	3	0.0	-2.3	-2.3	-1.6
Tobacco	1%	84.9	71		5.4	6.4	A3/BAA1	0	4	4	-8	0.1	-1.3	-1.3	-0.3
Energy	7%	522.5	81		5.5	7.5	A3/BAA1	2	3	3	-10	0.0	-1.5	-1.5	0.1
Independent Energy	1%	90.8	83		5.5	7.3	BAA1/BAA2	1	3	3	-13	0.0	-1.5	-1.5	0.8
Integrated Energy	1%	101.4	56		5.3	8.0	AA3/A1	1	2	2	-5	0.0	-1.7	-1.7	-0.7
Oil Field Services	0%	24.1	83		5.5	7.5	A3/BAA1	1	2	2	-14	0.0	-1.5	-1.5	0.1
Refining	0%	29.9	84		5.5	7.0	BAA1/BAA2	2	-1	-1	-20	0.0	-1.1	-1.1	1.0
Midstream	4%	276.3	89		5.6	7.4	BAA1/BAA2	2	4	4	-8	0.0	-1.5	-1.5	0.1
Technology	10%	771.6	94		5.6	6.9	A2/A3	3	19	19	23	-0.2	-2.4	-2.4	-1.9
Transportation	2%	140.4	68		5.5	8.9	A3/BAA1	1	5	5	-5	0.0	-2.1	-2.1	-0.8
Airlines	0%	17.3	80		5.2	3.8	A2/A3	7	7	7	-11	0.1	-0.3	-0.3	0.9
Railroads	1%	86.9	66		5.6	10.7	A3/BAA1	-1	4	4	-2	-0.1	-2.6	-2.6	-1.4
Transportation Services	0%	36.2	69		5.4	7.2	A3/BAA1	1	7	7	-8	0.0	-1.8	-1.8	0.1
Other Industrial	1%	39.1	62		5.5	10.0	AA3/A1	4	5	5	-2	0.0	-2.1	-2.1	-0.8
Utility	10%	759.5	82		5.6	8.0	A3/BAA1	2	4	4	-3	0.0	-1.8	-1.8	-0.6
Electric	9%	685.0	82		5.6	8.0	A3/BAA1	2	5	5	-3	0.0	-1.8	-1.8	-0.6
Natural Gas	1%	57.9	89		5.6	7.7	A3/BAA1	1	3	3	-8	0.0	-1.6	-1.6	-0.3
Other Utility	0%	16.6	69		5.4	7.8	BAA1/BAA2	0	4	4	0	0.1	-1.7	-1.7	-1.0
Financial Institutions	32%	2413.3	80		5.3	5.3	A2/A3	1	6	6	3	0.2	-0.9	-0.9	-0.2
Banking	22%	1655.0	75		5.2	4.8	A2/A3	1	8	8	5	0.2	-0.8	-0.8	-0.1
Brokerage Asset Managers Exchanges	2%	119.6	80		5.4	6.0	A3/BAA1	0	3	3	9	0.1	-1.1	-1.1	-0.6
Finance Companies	1%	109.2	145		5.8	3.5	BAA2/BAA3	0	0	0	2	0.3	-0.1	-0.1	0.9
Insurance	4%	327.8	86		5.6	8.1	A3/BAA1	0	5	5	-3	0.0	-1.9	-1.9	-0.9
REITS	3%	193.8	71		5.2	5.3	A3/BAA1	0	5	5	-7	0.2	-0.8	-0.8	0.1
Other Financial	0%	7.9	103		5.5	4.9	A3/BAA1	1	5	5	2	0.2	-0.6	-0.6	0.9
US Corporate IG		7467.3	80		5.4	6.7	A2/A3	1	8	8	4	0.0	-1.6	-1.6	-0.8

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 21: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	93.0	263		7.3	3.3	BA3/B1	-1	1	1	-9	0.0	0.0	0.0	2.6
Chemicals	3%	47.7	237		7.0	3.4	BA3/B1	-4	-2	-2	-26	-0.1	-0.2	-0.2	3.8
Metals & Mining	3%	38.3	215		6.8	3.2	BA3/B1	4	6	6	3	0.0	0.0	0.0	1.7
Paper	0%	7.0	708		11.6	2.7	B1/B2	-1	1	1	41	0.4	1.4	1.4	-0.3
Capital Goods	10%	154.0	240		7.1	3.3	BA3/B1	3	15	15	13	0.1	-0.7	-0.7	0.1
Aerospace & Defense	2%	30.8	113		5.9	3.2	BA3/B1	3	10	10	13	0.1	-0.3	-0.3	1.5
Building Materials	3%	39.9	369		8.4	3.5	B1/B2	5	14	14	29	0.1	-1.2	-1.2	-1.2
Diversified Manufacturing	2%	23.7	188		6.5	3.0	BA2/BA3	0	22	22	79	0.2	-0.2	-0.2	1.1
Construction Machinery	2%	23.1	141		6.1	3.1	BA3/B1	5	6	6	7	0.1	-0.4	-0.4	1.7
Packaging	2%	27.4	345		8.0	3.3	B1/B2	1	15	15	-17	0.3	-0.8	-0.8	-1.7
Environmental	1%	9.0	171		6.3	3.4	BA3/B1	3	24	24	15	0.2	-1.0	-1.0	0.7
Communications	14%	203.0	471		9.3	3.8	B1/B2	5	10	10	36	-0.5	-1.2	-1.2	0.2
Cable & Satellite	5%	70.5	675		11.2	3.0	B1/B2	9	17	17	69	-0.9	-1.7	-1.7	-1.8
Media & Entertainment	5%	78.5	344		8.1	4.2	BA3/B1	2	4	4	3	-0.2	-0.5	-0.5	1.4
Wireless	1%	17.0	338		8.1	4.3	B1/B2	8	16	16	35	-1.0	-1.5	-1.5	-0.9
Wirelines	3%	37.1	412		8.8	4.2	B1/B2	4	5	5	32	-0.2	-1.5	-1.5	2.5
Consumer Cyclical	18%	263.7	246		7.0	3.1	BA3/B1	1	4	4	-2	0.2	-0.1	-0.1	2.1
Automotive	3%	47.2	271		7.2	3.0	BA3/B1	1	11	11	-2	0.2	-0.3	-0.3	1.7
Leisure	1%	21.0	241		6.9	3.0	B2/B3	-4	12	12	35	0.4	-0.5	-0.5	1.5
Gaming	3%	44.5	274		7.4	3.0	BA3/B1	2	8	8	13	0.1	-0.2	-0.2	1.8
Home Construction	1%	18.5	183		6.4	2.9	BA3/B1	1	7	7	-10	0.3	0.0	0.0	2.7
Lodging	1%	16.4	127		5.9	4.0	BA2/BA3	3	13	13	-4	0.2	-0.5	-0.5	1.1
Retailers	4%	60.7	230		6.7	3.3	BA3/B1	2	-4	-4	-30	0.1	0.1	0.1	3.0
Restaurants	1%	11.3	163		6.1	3.9	BA3/B1	3	7	7	0	0.1	-0.3	-0.3	1.7
Consumer Cyc Services	3%	44.3	306		7.6	2.7	B1/B2	1	-1	-1	11	0.2	0.2	0.2	2.2
Consumer Non-Cyclical	11%	165.1	277		7.4	3.2	B1/B2	0	10	10	13	0.2	-0.4	-0.4	1.8
Consumer Products	2%	26.9	332		7.9	3.9	BA3/B1	-1	4	4	2	0.6	-1.0	-1.0	-0.7
Food & Beverage	2%	29.1	189		6.5	3.6	B1/B2	-4	9	9	11	0.2	-0.4	-0.4	1.4
Healthcare	5%	73.5	263		7.2	2.9	BA3/B1	-1	16	16	26	0.1	-0.4	-0.4	2.3
Pharmaceuticals	2%	25.7	393		8.6	3.0	B2/B3	1	1	1	-10	0.2	0.3	0.3	4.4
Supermarkets	1%	9.6	193		6.4	3.6	BA2/BA3	7	23	23	180	-0.1	-1.2	-1.2	-0.6
Tobacco	0%	0.3	220		7.0	3.2	BA3/BA3	-1	13	13	28	0.3	-0.8	-0.8	-0.2
Energy	11%	166.7	170		6.4	3.4	BA2/BA3	9	-3	-3	-37	-0.2	0.1	0.1	4.3
Independent Energy	3%	46.3	188		6.6	3.2	BA3/B1	9	-5	-5	-25	-0.2	0.5	0.5	4.4
Oil Field Services	2%	27.5	201		6.9	3.3	BA3/B1	11	3	3	-33	-0.3	0.0	0.0	4.1
Refining	0%	5.4	57		5.0	2.3	BA3/B1	18	-76	-76	-84	0.0	1.4	1.4	7.3
Midstream	6%	87.5	157		6.2	3.6	BA2/BA3	8	4	4	-42	-0.1	-0.1	-0.1	4.2
Technology	10%	146.5	390		8.5	3.2	BA3/B1	6	16	16	24	-0.3	-1.1	-1.1	0.1
Transportation	2%	33.6	438		9.0	3.0	BA3/B1	2	8	8	21	0.2	-1.0	-1.0	-1.2
Airlines	1%	13.9	391		8.4	2.8	BA3/B1	-3	20	20	41	0.9	-1.3	-1.3	0.1
Railroads	0%	1.8	974		14.7	2.5	B1/B2	4	9	9	1	-0.4	0.1	0.1	-6.8
Transportation Services	1%	17.9	419		8.9	3.2	BA3/B1	6	0	0	19	-0.3	-0.9	-0.9	-1.4
Other Industrial	1%	15.5	243		5.9	3.2	B1/B2	-1	3	3	-6	0.3	0.5	0.5	1.6
Utility	3%	44.0	176		6.4	4.1	BA2/BA3	4	10	10	20	0.0	-0.6	-0.6	2.0
Financial Institutions	13%	195.5	245		7.1	3.1	BA3/B1	1	0	0	10	0.2	0.2	0.2	1.3
Banking	0%	3.0	180		6.3	4.2	BA1/BA2	4	0	0	-16	0.0	-0.4	-0.4	0.5
Brokerage Asset Managers Exchanges	1%	18.2	177		6.5	3.1	BA3/B1	1	-1	-1	-6	0.3	0.2	0.2	2.2
Finance Companies	4%	66.6	258		7.2	3.1	BA2/BA3	1	4	4	24	0.2	0.0	0.0	0.7
Insurance	4%	60.6	232		6.9	3.3	BA3/B1	1	-5	-5	21	0.2	0.3	0.3	0.5
REITS	2%	29.4	252		7.0	2.7	B1/B2	0	4	4	-7	0.2	0.0	0.0	2.6
Other Financial	1%	17.7	307		7.7	3.0	BA3/B1	0	-4	-4	-3	0.4	0.5	0.5	3.1
US Corp High Yield		1480.7	289		7.5	3.3	BA3/B1	3	7	7	8	0.0	-0.4	-0.4	1.5

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 22: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.6	€ 396.3	87.6		3.9	4.6	A	2	0	0	3	0.12	-0.85	-0.85	0.34
Automobiles & Parts	5.2	€ 164.7	110.2		4.0	3.7	A	2	1	1	9	0.11	-0.66	-0.66	0.30
Beverages	2.9	€ 92.0	80.8		3.9	5.8	A	2	-1	-1	0	0.12	-1.09	-1.09	0.45
Food Producers	1.9	€ 61.0	68.9		3.7	5.1	A	2	-1	-1	-1	0.14	-0.99	-0.99	0.30
Household Goods	0.2	€ 6.2	50.2		3.7	5.9	AA	2	0	0	-1	0.15	-1.15	-1.15	0.24
Personal Goods	1.7	€ 53.0	65.8		3.6	4.3	A	1	0	0	3	0.15	-0.83	-0.83	0.20
Tobacco	0.6	€ 19.4	106.7		4.0	4.8	BBB	2	0	0	-5	0.11	-0.94	-0.94	0.71
Consumer Services	4.7	€ 146.5	90.5		4.0	5.2	A	4	4	4	8	0.04	-1.20	-1.20	0.12
Food & Drug Retailers	0.6	€ 19.7	85.9		3.8	4.4	BBB	2	2	2	0	0.12	-0.86	-0.86	0.49
General Retailers	0.6	€ 19.7	102.1		4.2	6.4	AA	8	10	10	23	-0.21	-2.05	-2.05	-0.66
Media	1.1	€ 35.3	90.7		3.8	4.3	BBB	3	3	3	7	0.10	-0.96	-0.96	0.33
Travel & Leisure	2.3	€ 71.8	87.7		4.0	5.6	A	4	2	2	6	0.05	-1.19	-1.19	0.11
Energy	3.1	€ 98.2	96.4		4.0	4.9	A	3	0	0	-1	0.09	-0.94	-0.94	0.68
Oil & Gas Producers	2.8	€ 87.6	96.5		4.0	4.8	A	3	0	0	-2	0.08	-0.93	-0.93	0.68
Oil Equipment / Services & Distribution	0.3	€ 8.2	96.2		4.0	5.5	A	2	4	4	20	0.16	-1.10	-1.10	0.69
Renewable Energy	0.1	€ 2.4	94.1		3.9	5.0	BBB	1	-5	-5	-7	0.18	-0.76	-0.76	0.81
Industrials	9.5	€ 299.7	81.1		3.8	4.5	A	2	0	0	0	0.13	-0.85	-0.85	0.39
Aerospace & Defense	0.5	€ 15.7	69.3		3.7	4.5	A	2	-3	-3	5	0.12	-0.80	-0.80	0.09
Construction & Materials	1.5	€ 48.4	84.8		3.8	4.3	BBB	2	2	2	1	0.12	-0.87	-0.87	0.40
Electronic & Electrical Equipment	0.9	€ 28.0	72.6		3.7	4.9	A	2	1	1	3	0.11	-1.02	-1.02	0.19
General Industrials	1.2	€ 38.2	69.3		3.7	5.2	A	2	-1	-1	-5	0.11	-1.05	-1.05	0.35
Industrial Engineering	1.1	€ 34.8	76.0		3.6	3.3	A	1	-1	-1	-1	0.15	-0.53	-0.53	0.62
Industrial Transportation	3.3	€ 104.3	84.4		3.8	4.7	BBB	2	0	0	0	0.13	-0.87	-0.87	0.33
Support Services	1.0	€ 30.4	101.1		3.9	4.1	BBB	1	6	6	6	0.16	-0.70	-0.70	0.67
Health Care	5.5	€ 173.6	78.1		3.9	5.8	A	2	1	1	-1	0.13	-1.15	-1.15	0.39
Health Care Equipment & Services	2.4	€ 75.0	84.4		4.0	5.7	A	3	2	2	1	0.11	-1.15	-1.15	0.32
Pharmaceuticals & Biotechnology	3.1	€ 98.6	73.5		3.9	5.8	A	2	0	0	-3	0.14	-1.14	-1.14	0.45
Basic Materials	2.7	€ 84.0	81.8		3.8	4.8	A	2	0	0	-4	0.12	-0.91	-0.91	0.59
Chemicals	2.0	€ 63.7	81.2		3.8	4.9	A	2	0	0	-3	0.11	-0.95	-0.95	0.53
Forestry & Paper	0.2	€ 5.6	74.0		3.6	4.0	BBB	2	-4	-4	-13	0.11	-0.63	-0.63	0.93
Industrial Metals	0.1	€ 3.6	86.7		3.8	5.0	BBB	1	1	1	-6	0.16	-1.01	-1.01	0.67
Mining	0.4	€ 11.0	87.9		3.8	4.1	BBB	2	1	1	-9	0.13	-0.83	-0.83	0.68
Technology	2.8	€ 88.6	101.5		4.1	5.6	A	7	6	6	15	-0.12	-1.61	-1.61	-0.56
Software & Computer Services	2.4	€ 76.0	105.1		4.2	5.9	A	7	7	7	16	-0.15	-1.75	-1.75	-0.69
Technology Hardware & Equipment	0.4	€ 12.6	66.0		3.5	3.7	A	3	4	4	6	0.07	-0.75	-0.75	0.21
Telecommunications	5.4	€ 170.8	98.0		4.0	5.4	BBB	5	6	6	6	-0.03	-1.38	-1.38	0.09
Fixed Line Telecommunications	4.4	€ 137.5	97.5		4.0	5.4	BBB	6	7	7	7	-0.06	-1.41	-1.41	0.05
Mobile Telecommunications	1.1	€ 33.3	99.9		4.0	5.3	BBB	3	4	4	4	0.08	-1.25	-1.25	0.27
Financials	42.2	€ 1,328.7	94.9		3.9	4.0	A	2	2	2	0	0.14	-0.79	-0.79	0.56
Banks Sen	26.6	€ 836.7	83.8		3.7	3.8	A	1	1	1	0	0.16	-0.74	-0.74	0.53
Banks Sub	4.1	€ 128.6	125.5		4.1	3.6	BBB	3	1	1	-1	0.07	-0.64	-0.64	0.84
Financial Services Sen	2.8	€ 87.5	81.1		3.8	4.5	A	1	0	0	-2	0.15	-0.87	-0.87	0.59
Financial Services Sub	0.0	€ 1.0	112.4		3.9	1.9	A	5	0	0	-26	0.04	-0.18	-0.18	1.55
Insurance Sen	1.4	€ 42.7	78.8		3.7	4.3	A	1	2	2	3	0.14	-0.87	-0.87	0.26
Insurance Sub	2.7	€ 86.0	136.6		4.3	5.0	A	5	5	5	3	0.01	-1.10	-1.10	0.58
Real Estate Sen	4.6	€ 145.7	111.2		4.1	4.7	BBB	2	2	2	2	0.16	-0.97	-0.97	0.48
Real Estate Sub	0.0	€ 0.5	176.5		4.7	5.6	A	7	-8	-8	0	-0.08	-0.66	-0.66	0.00
Utilities	11.5	€ 361.3	95.1		4.0	5.2	BBB	2	0	0	1	0.12	-1.04	-1.04	0.48
Electricity	6.9	€ 216.6	97.4		4.0	5.2	BBB	3	0	0	1	0.10	-1.05	-1.05	0.52
Gas / Water & Multiutilities	4.6	€ 144.7	91.6		3.9	5.2	BBB	2	0	0	1	0.14	-1.03	-1.03	0.43
Index	100.0	€ 3,147.7	91.4		3.9	4.6	A	2	1	1	2	0.11	-0.93	-0.93	0.43

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 23: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	26.8	€ 120.7	291		5.8	2.7	BB	16	1	1	16	-0.25	-0.29	-0.29	1.68
Consumer Goods	15.1	€ 67.9	255		5.4	2.8	BB	8	-15	-15	19	-0.10	-0.08	-0.08	1.78
Consumer Services	11.7	€ 52.7	343		6.3	2.5	BB	28	27	27	25	-0.45	-0.57	-0.57	1.56
Energy	10.1	€ 45.6	253		5.4	3.0	BB	11	-3	-3	-5	-0.10	-0.20	-0.20	1.10
Energy	2.3	€ 10.5	238		5.2	2.6	BB	16	-6	-6	-63	-0.16	-0.04	-0.04	3.37
Utilities	7.8	€ 35.1	257		5.4	3.1	BB	10	-2	-2	10	-0.07	-0.25	-0.25	0.46
Industrials	31.3	€ 141.2	317		6.0	2.7	BB	15	-11	-11	-28	-0.21	-0.11	-0.11	2.52
Basic Materials	6.1	€ 27.7	416		7.0	2.7	B	32	-1	-1	-147	-0.65	-0.30	-0.30	6.24
Construction & Materials	1.5	€ 6.6	505		7.9	3.0	BB	1	-24	-24	-103	-0.04	0.76	0.76	1.83
Health Care	8.9	€ 40.4	287		5.7	2.6	B	10	-25	-25	20	-0.07	-0.17	-0.17	1.93
Industrials	14.7	€ 66.5	273		5.6	2.8	BB	12	-3	-3	-13	-0.12	-0.09	-0.09	1.62
TMT	19.8	€ 89.5	360		6.5	3.4	BB	26	30	30	74	-0.70	-1.15	-1.15	0.06
Technology	4.5	€ 20.2	487		7.8	3.6	B	36	56	56	109	-1.09	-1.23	-1.23	-1.52
Telecommunications	15.4	€ 69.4	320		6.1	3.3	BB	25	20	20	48	-0.59	-1.13	-1.13	0.40
Financials	12.0	€ 54.1	360		6.4	3.1	BB	10	-4	-4	33	-0.09	-0.09	-0.09	0.65
Core Financials	4.6	€ 20.8	193		4.8	3.1	BB	6	-1	-1	15	0.00	-0.34	-0.34	0.87
Financial Services	1.5	€ 6.8	940		12.2	2.0	B	30	-28	-28	338	-0.08	1.71	1.71	-2.65
Real Estate	5.9	€ 26.4	391		6.8	3.3	BB	13	8	8	13	-0.17	-0.35	-0.35	1.47
Index	100.0	€ 451.1	319		6.1	2.9	BB	17	4	4	17	-0.29	-0.37	-0.37	1.44

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

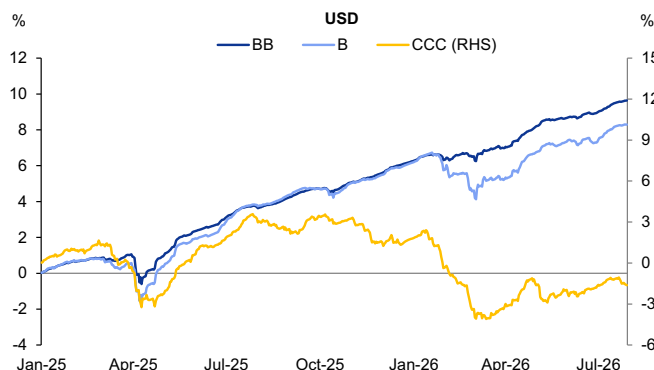
Exhibit 24: USD ETF performance

Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$32.85B	-1427	13.3%	309	12.24	106.47	0.05%	-1.2	-2.0	-2.0	iBoxx USD Liquid Investment Grade Index
	VCSH	\$44.71B	71	13.0%	569	0.83	78.71	0.00%	0.9	0.0	0.0	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$67.58B	123	16.9%	832	2.36	81.45	-0.01%	-0.4	-1.1	-1.1	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$10.32B	87	14.5%	202	--	51.07	0.02%	2.5	0.4	0.4	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$23.B	50	7.8%	441	0.24	52.20	0.03%	1.0	0.0	0.0	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.57B	118	10.6%	355	1.39	52.40	0.00%	-0.4	-1.1	-1.1	Bloomberg Barclays U.S. Interim
	SPSB	\$10.66B	230	27.7%	356	2.94	29.96	0.02%	1.3	0.2	0.2	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$11.49B	-53	6.9%	347	4.06	33.19	-0.02%	0.3	-0.4	-0.4	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCIT	\$8.92B	0	23.5%	124	13.19	72.24	-0.01%	-2.1	-3.6	-3.6	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$16.54B	705	-15.4%	209	38.93	79.45	0.15%	1.5	-0.2	-0.2	iBoxx USD Liquid High Yield Index
	JNK	\$6.89B	-153	-7.7%	72	9.41	95.63	0.07%	1.7	-0.2	-0.2	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.74B	-126	-7.5%	192	0.97	24.85	-0.01%	1.6	-0.2	-0.2	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.59B	0	-3.0%	181	0.41	42.13	0.11%	1.7	-0.1	-0.1	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.74B	0	17.2%	19	0.17	92.61	-0.04%	1.3	-0.4	-0.4	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.62B	14	-5.4%	40	0.08	40.49	-0.06%	0.6	0.2	0.2	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.05B	-33	-1.5%	105	0.56	28.91	-0.18%	1.7	-0.6	-0.6	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$6.75B	-221	0.8%	330	16.54	20.39	-0.27%	0.7	0.6	0.6	S&P/LSTA U.S. Leveraged Loan 100 Index
	SRLN	\$5.34B	0	-16.7%	132	5.75	40.39	-0.18%	1.4	0.8	0.8	Markit iBoxx USD Liquid Leverage Loans Total Return Index
Loans	FTSL	\$2.32B	0	1.0%	52	0.46	44.75	0.00%	1.1	0.5	0.5	S&P/LSTA US Leveraged Loan 100 Market Value Index
	EMB	\$14.63B	-360	-10.0%	154	9.95	94.76	-0.28%	1.0	-1.3	-1.3	J.P. Morgan EMBI Global Core Index
	EMLC	\$4.76B	0	15.9%	188	2.23	25.52	0.13%	2.0	0.4	0.4	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
CLOs	JAAA	\$29.27B	132	20.2%	578	5.33	50.66	0.00%	2.5	0.3	0.3	
	PAAA	\$11.24B	225	80.9%	218	--	51.47	-0.02%	2.7	0.4	0.4	
	JBBB	\$1.41B	7	18.9%	30	--	47.47	0.25%	2.5	0.3	0.3	
	CLOI	\$1.51B	21	13.6%	29	27.61	52.95	0.02%	2.8	0.4	0.4	
	CLOA	\$2.19B	10	58.4%	42	--	51.98	0.06%	2.8	0.5	0.5	

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

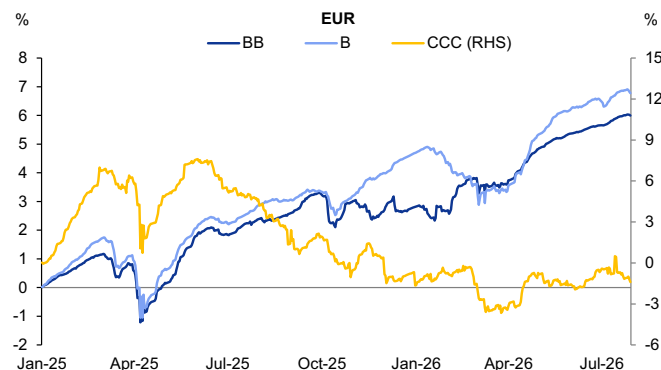
Performance dashboard: Leveraged Loans

Exhibit 25: Cumulative total return by rating for USD Leveraged Loans



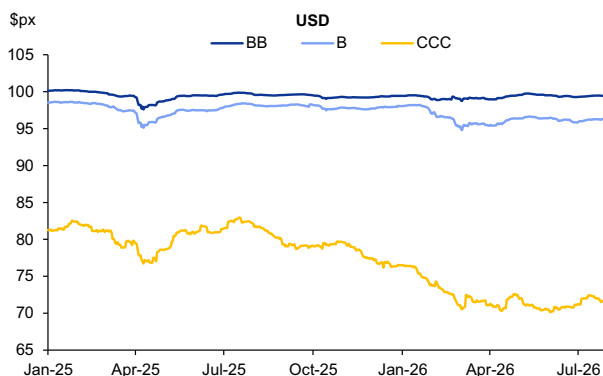
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 26: Cumulative total return by rating for EUR Leveraged Loans



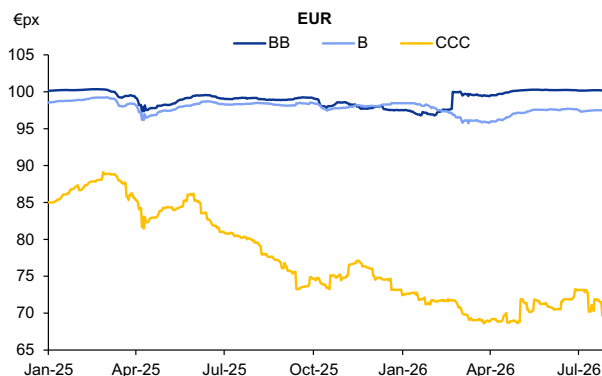
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 27: Bid price for USD Leveraged Loans by rating



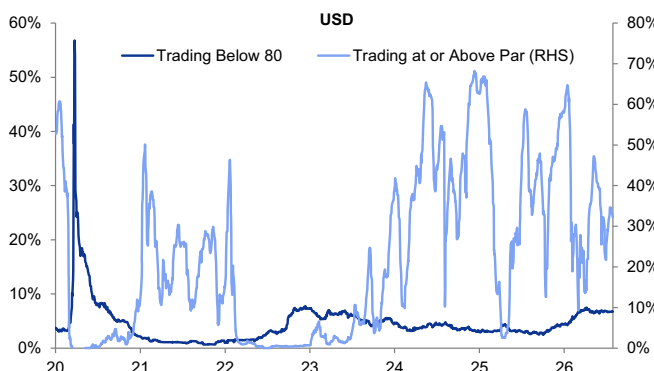
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 28: Bid price for EUR Leveraged Loans by rating



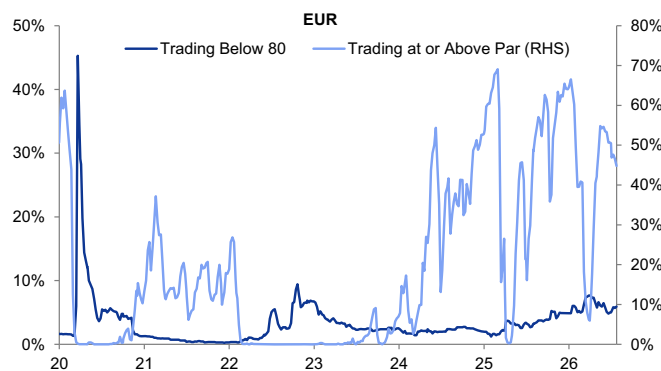
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 29: Price breakdown for USD Leveraged Loans



Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 30: Price breakdown for EUR Leveraged Loans

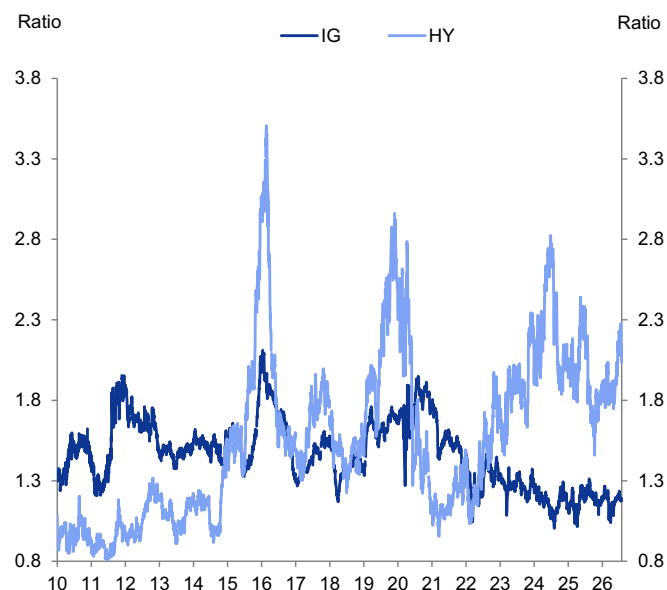


Source: PitchBook, Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

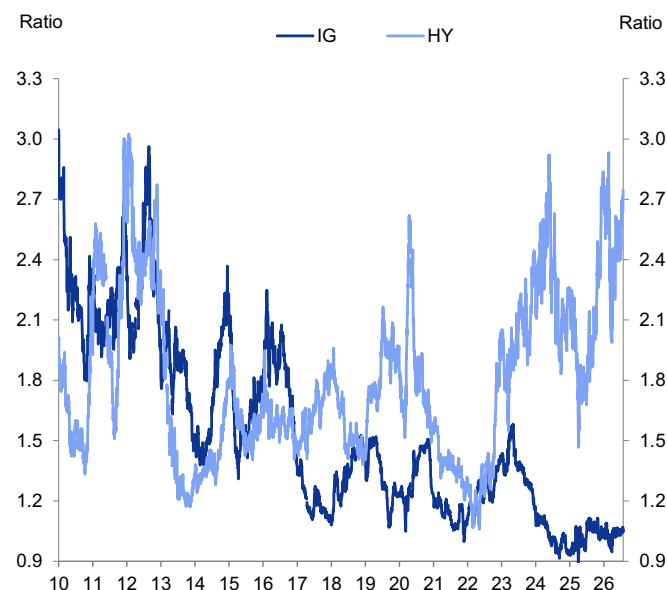
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 31: USD spread dispersion



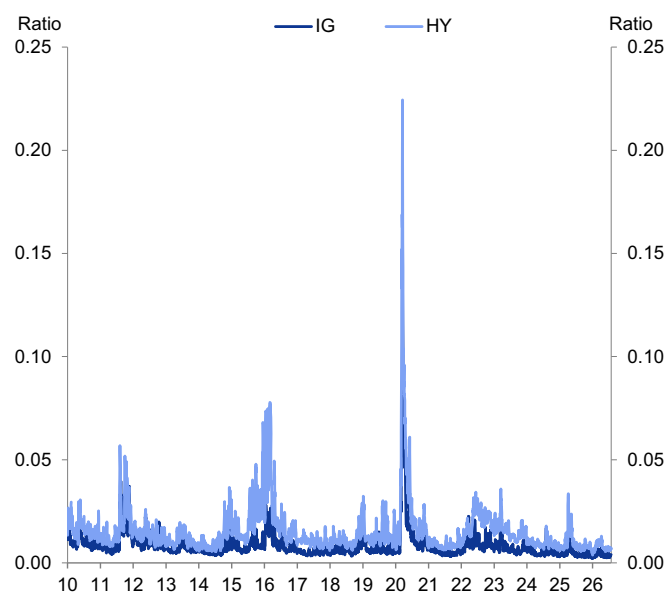
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 32: EUR spread dispersion



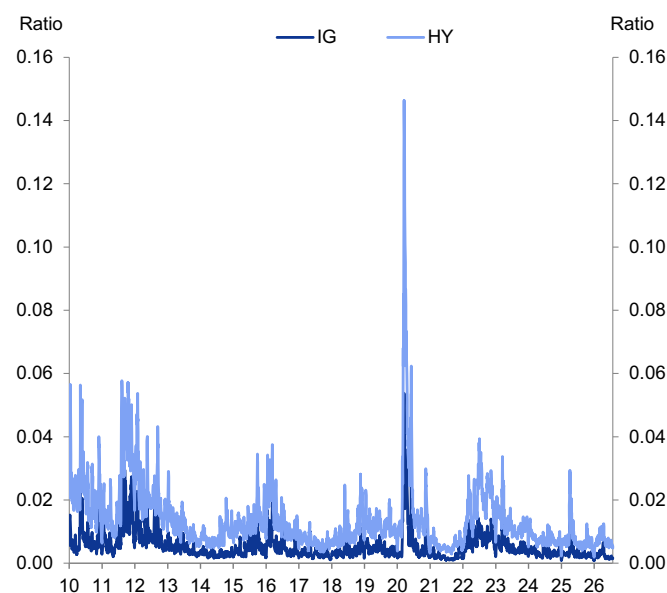
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 33: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 34: EUR return dispersion

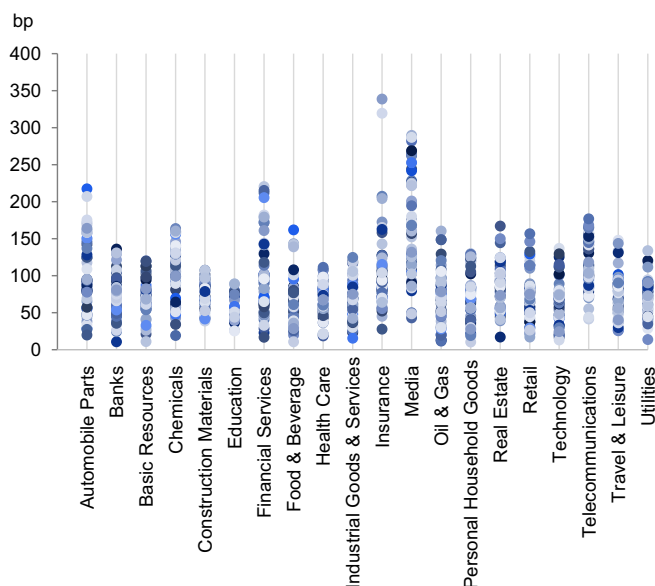


Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Sector dispersion

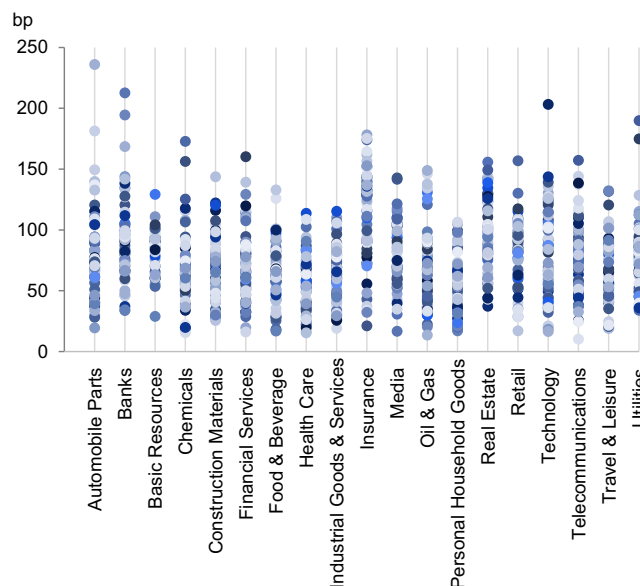
Issuer-level spreads are calculated using a market-value weighted average across the capital structure.

Exhibit 35: Distribution of issuer-level spreads by sector in the USD IG market



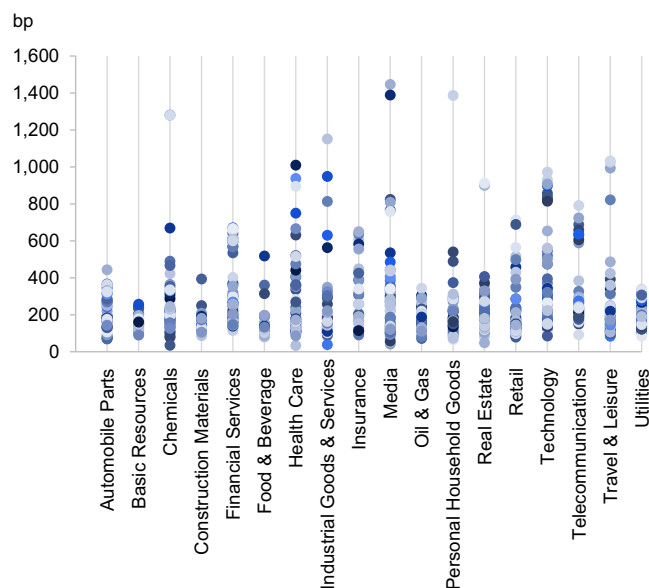
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 36: Distribution of issuer-level bond spreads by sector in the EUR IG market



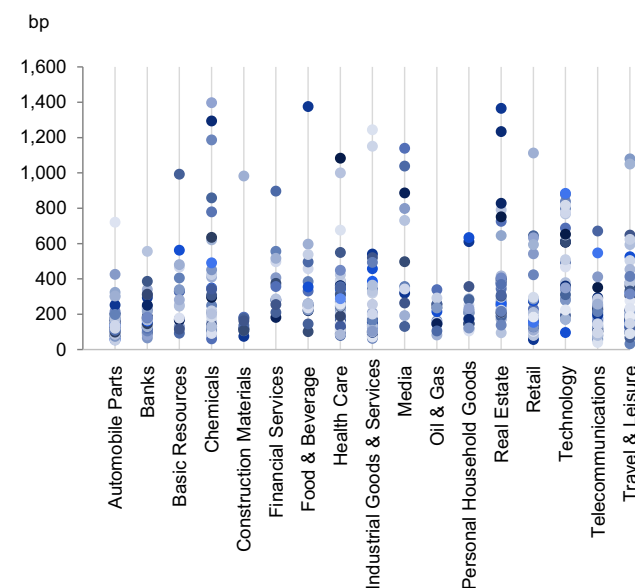
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 37: Distribution of issuer-level bond spreads by sector in the USD HY market



Source: Goldman Sachs Global Investment Research

Exhibit 38: Distribution of issuer-level bond spreads by sector in the EUR HY market



Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 39: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Oracle Corp	Software	ORCL 4.1 03/25/61	1,500	Baa2/BBB-/BBB	3.7%	2.1	59.13
Intel Corp	Semiconductors	INTC 5.05 08/05/62	900	Baa2/BBB/BBB	2.7%	2.1	78.58
Jefferies Fin Group Inc	Diversified Finan Serv	JEF 6 1/2 01/20/43	400	Baa2/BBB/BBB+	2.2%	2.2	98.32
Hess Corp	Oil&Gas	HES 5.6 02/15/41	1,250	WR/AA-/--	2.0%	2.0	98.70
Concentrix Corp	Software	CNXC 6 1/2 03/01/29	600	Baa3/BBB-/BBB	1.9%	1.9	96.50
Verizon Communications	Telecommunications	VZ 3 11/20/60	2,000	Baa1/BBB+/A-	1.8%	1.0	53.96
Amazon.Com Inc	Internet	AMZN 4.1 04/13/62	1,250	A1/AA/AA-	1.8%	1.2	68.72
Lear Corp	Auto Parts&Equipment	LEA 5 1/4 05/15/49	625	Baa2/BBB/BBB	1.8%	1.5	86.21
Oracle Corp	Software	ORCL 6.55 02/04/46	2,250	Baa2/BBB-/BBB	1.7%	1.5	87.00
Brighthouse Financial In	Insurance	BHF 4.7 06/22/47	1,014	Baa3/BBB/BBB-	1.7%	1.1	65.66
Toyota Motor Corp	Auto Manufacturers	TOYOTA 5.053 06/30/35	500	A1/A+/--	1.6%	1.6	99.10
Pfizer Inc	Pharmaceuticals	PFE 4.3 06/15/43	750	A2/A/--	1.6%	1.3	83.22
Intel Corp	Semiconductors	INTC 5.15 02/21/34	900	Baa2/BBB/BBB	1.6%	1.5	97.91
Amazon.Com Inc	Internet	AMZN 3 7/8 08/22/37	2,750	A1/AA/AA-	1.5%	1.3	86.77
Comcast Corp	Media	CMCSA 4.95 10/15/58	1,035	A3/A-/A-	1.5%	1.1	75.20
Commonspirit Health	Healthcare-Services	CATMED 5.318 12/01/34	725	A3/A-/A-	1.5%	1.4	98.60
Meta Platforms Inc	Internet	META 5.55 08/15/64	2,750	Aa3/AA-/--	1.4%	1.1	80.28
General Motors Finl Co	Auto Manufacturers	GM 5.45 09/06/34	750	Baa2/BBB/BBB	1.3%	1.3	98.40
Meta Platforms Inc	Internet	META 4 7/8 11/15/35	6,500	Aa3/AA-/--	1.3%	1.2	93.34
Jbs Nv/Usa Foods/Food Co	Food	JBS 6 1/2 12/01/52	1,548	Baa3/BBB-/BBB-	1.3%	1.2	96.47
Bottom 20 performers							
Broadcom Inc	Semiconductors	AVGO 5.7 01/15/56	1,250	A3/A-/A-	-3.1%	-2.8	91.12
Intel Corp	Semiconductors	INTC 3 1/4 11/15/49	2,000	Baa2/BBB/BBB	-2.6%	-1.6	62.13
At&T Inc	Telecommunications	T 6.2 10/30/56	2,000	Baa2/BBB/BBB+	-2.5%	-2.3	93.93
Nvidia Corp	Semiconductors	NVDA 3 1/2 04/01/50	2,000	Aa1/AA/--	-2.4%	-1.7	69.14
Florida Power & Light Co	Electric	NEE 5.8 03/15/65	700	Aa2/A+/AA-	-2.4%	-2.3	97.82
At&T Inc	Telecommunications	T 3.65 09/15/59	6,496	Baa2/BBB/BBB+	-2.4%	-1.4	60.30
Intel Corp	Semiconductors	INTC 3.1 02/15/60	1,000	Baa2/BBB/BBB	-2.3%	-1.2	53.98
At&T Inc	Telecommunications	T 5.85 04/30/46	1,350	Baa2/BBB/BBB+	-2.3%	-2.1	92.63
Charter Comm Opt Llc/Cap	Media	CHTR 4.8 03/01/50	2,800	Ba1/BBB-/BBB-	-2.2%	-1.5	71.20
Charter Comm Opt Llc/Cap	Media	CHTR 3.95 06/30/62	1,400	Ba1/BBB-/BBB-	-2.0%	-1.1	55.62
At&T Inc	Telecommunications	T 3.55 09/15/55	7,495	Baa2/BBB/BBB+	-2.0%	-1.2	60.90
Dell Int Llc / Emc Corp	Computers	DELL 8.35 07/15/46	646	Baa2/BBB+/BBB+	-2.0%	-2.4	123.60
Vertiv Holdings Co	Machinery-Constr&Mining	VRT 5.8 03/15/56	500	Baa3/BBB-/BBB-	-2.0%	-1.8	93.50
Kla Corp	Semiconductors	KLAC 5 1/4 07/15/62	800	A2/A-/A	-1.9%	-1.7	88.42
Nvidia Corp	Semiconductors	NVDA 5 5/8 06/15/56	3,500	Aa1/AA/--	-1.9%	-1.8	91.80
Nvidia Corp	Semiconductors	NVDA 5.55 06/15/46	3,000	Aa1/AA/--	-1.9%	-1.8	93.61
Intel Corp	Semiconductors	INTC 5.6 02/21/54	1,150	Baa2/BBB/BBB	-1.9%	-1.7	88.71
Verizon Communications	Telecommunications	VZ 5 7/8 11/30/55	3,250	Baa1/BBB+/A-	-1.9%	-1.8	92.53
Oracle Corp	Software	ORCL 6 1/8 08/03/65	1,000	Baa2/BBB-/BBB	-1.9%	-1.5	79.20
Texas Instruments Inc	Semiconductors	TXN 5.05 05/18/63	1,550	Aa3/A+/--	-1.9%	-1.6	85.37

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 40: Top and bottom 20 performers in the USD HY market

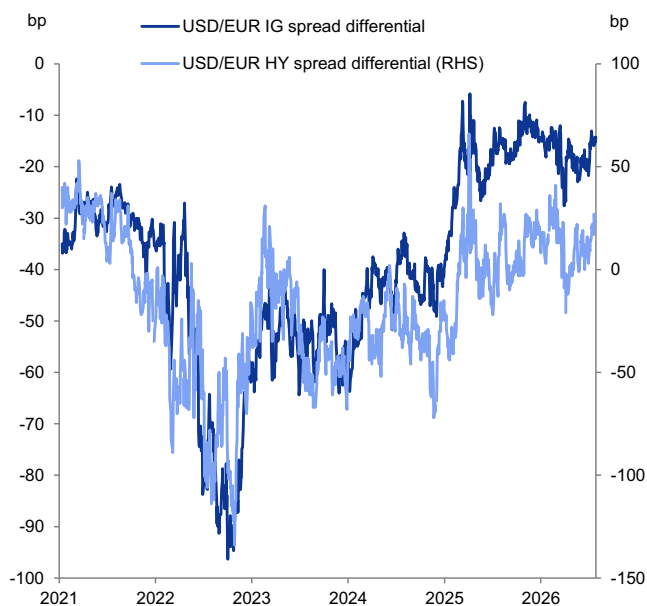
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Jeldwen Holding	Building Materials	JELD 7 09/01/32	350	Caa2/CCC/--	12.9%	6.6	51.38
Whirlpool Corp	Home Furnishings	WHR 4.6 05/15/50	500	B2/B+/BB-	3.7%	2.0	53.43
Ziprecruiter Inc	Internet	ZIP 5 01/15/30	255	B3/B/B+	3.1%	2.4	76.32
Virtusa Corp	Computers	VRTU 7 1/8 12/15/28	350	Caa1/B/--	2.7%	2.0	75.50
Whirlpool Corp	Home Furnishings	WHR 6 1/2 06/15/33	600	B2/B+/BB-	2.2%	1.9	82.80
Sabre Gbl Inc	Leisure Time	SABHLD 10 3/4 11/15/29	446	B3/B/--	2.1%	2.0	92.75
Sabre Gbl Inc	Leisure Time	SABHLD 11 1/8 07/15/30	1,325	B3/B/--	1.8%	1.6	92.39
Gray Media Inc	Media	GTN 5 3/8 11/15/31	1,198	Caa1/CCC/--	1.7%	1.3	72.75
Cco Hldgs Llc/Cap Corp	Media	CHTR 7 02/01/33	1,750	B1/BB-/BB+	1.6%	1.5	94.50
Cco Hldgs Llc/Cap Corp	Media	CHTR 4 1/4 02/01/31	3,000	B1/BB-/BB+	1.6%	1.4	87.50
Clearwater Paper Corp	Packaging&Containers	CLW 4 3/4 08/15/28	275	B3/CCC/--	1.5%	1.3	84.50
Jetblue Airways/Loyalty	Airlines	JBLU 9 7/8 09/20/31	2,000	Caa1/B/B	1.5%	1.3	85.50
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 5/8 04/30/33	450	B2/B+/BB-	1.4%	1.2	86.88
Lcpr Sr Secured Fin Dac	Media	LILAPR 5 1/8 07/15/29	820	Caa3/CCC+/CCC+	1.3%	0.8	56.50
Cco Hldgs Llc/Cap Corp	Media	CHTR 4 1/2 06/01/33	1,750	B1/BB-/BB+	1.3%	1.1	84.13
Vistajet Malta/Vm Holds	Airlines	VSTJET 8 3/4 01/15/32	525	B3/B/--	1.3%	1.3	96.45
Everforth Inc	Computers	EFOR 4 5/8 05/15/28	550	Ba3/BB/--	1.3%	1.2	94.00
Chs/Community Health Sys	Healthcare-Services	CYH 9 3/4 01/15/34	1,790	B3/B-/B	1.2%	1.3	100.25
Kohl'S Corporation	Retail	KSS 5.55 07/17/45	427	B3/BB-/BB-	1.2%	0.8	68.61
Vistajet Malta/Vm Holds	Airlines	VSTJET 6 3/8 02/01/30	1,000	B3/B/--	1.2%	1.1	92.87
Bottom 20 performers							
Virgin Medi O2 Vi Dac	Media	VMED 8 1/2 03/15/33	500	B3/B-/B-	-11.8%	-8.0	67.50
Hertz Corp/The	Commercial Services	HTZ 5 12/01/29	1,000	Caa1/CCC/CCC-	-8.4%	-2.5	29.46
Virgin Media Finance Plc	Media	VMED 5 07/15/30	925	B3/B-/B-	-7.9%	-5.0	63.50
Coreweave Inc	Software	CRWV 9 5/8 07/15/32	1,250	B1/B/BB-	-7.3%	-6.6	91.13
Camelot Return Merger Su	Building Materials	CNR 8 3/4 08/01/28	710	Caa1/CCC+/--	-7.2%	-4.0	55.52
Vz Secured Financing Bv	Media	ZIGGO 7 1/2 01/15/33	1,150	B1/B+/BB	-7.0%	-6.6	95.13
Hertz Corp/The	Commercial Services	HTZ 12 5/8 07/15/29	1,250	Ba3/B-/B+	-6.4%	-4.6	72.50
Coreweave Inc	Software	CRWV 9 3/4 10/01/31	2,750	B1/B/BB-	-6.0%	-5.5	91.50
Scripps Escrow li Inc	Media	SSP 5 3/8 01/15/31	392	Caa3/CCC-/CC	-5.7%	-3.5	61.50
Vz Secured Financing Bv	Media	ZIGGO 5 01/15/32	1,525	B1/B+/BB	-5.5%	-4.8	86.75
Coreweave Inc	Software	CRWV 9 1/4 06/01/30	2,000	B1/B/BB-	-5.3%	-5.0	94.50
Vmed O2 Uk Financing I	Telecommunications	VMED 7 3/4 04/15/32	750	Ba3/B+/BB	-4.6%	-4.0	87.05
Coreweave Inc	Software	CRWV 9 02/01/31	1,750	B1/B/BB-	-4.6%	-4.2	91.50
Ziggo Bv	Media	ZIGGO 4 7/8 01/15/30	991	B1/B+/BB	-4.5%	-4.2	93.00
Tronox Inc	Chemicals	TROX 4 5/8 03/15/29	1,075	Caa2/CCC+/--	-4.3%	-2.8	64.50
Cleveland-Cliffs Inc	Iron/Steel	CLF 6 1/4 10/01/40	235	B3/B-/B-	-4.2%	-3.6	84.35
Vmed O2 Uk Finan	Telecommunications	VMED 4 1/4 01/31/31	1,350	Ba3/B+/BB	-4.1%	-3.3	80.75
Virgin Media Secured Fin	Media	VMED 4 1/2 08/15/30	915	Ba3/B+/BB	-4.1%	-3.4	84.70
Vmed O2 Uk Finan	Telecommunications	VMED 4 3/4 07/15/31	1,400	Ba3/B+/BB	-3.7%	-3.0	81.50
Ew Scripps Co	Media	SSP 9 7/8 08/15/30	750	Caa2/CCC+/CCC-	-3.2%	-2.9	89.63

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

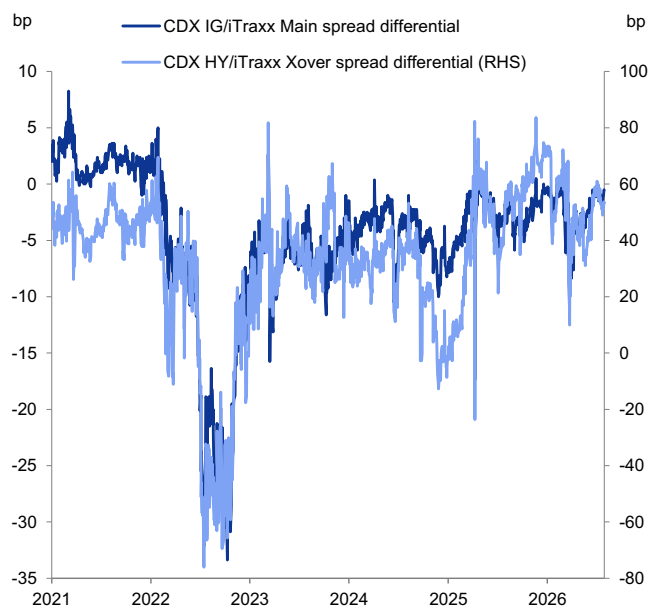
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 41: USD/EUR IG and HY spread differential



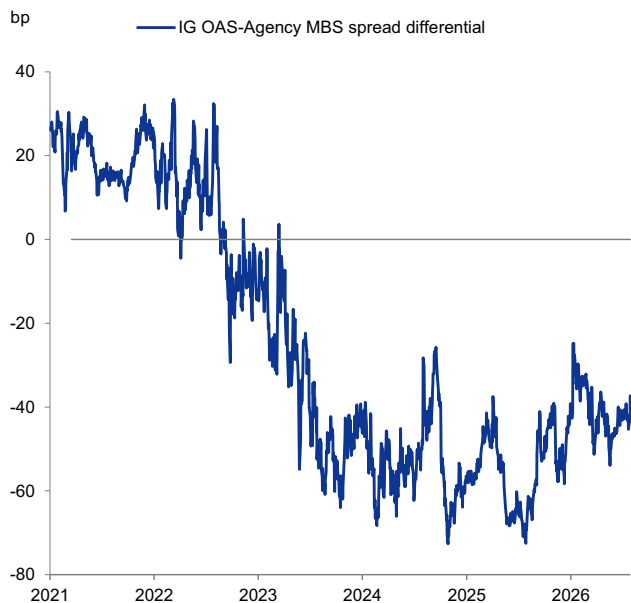
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 42: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



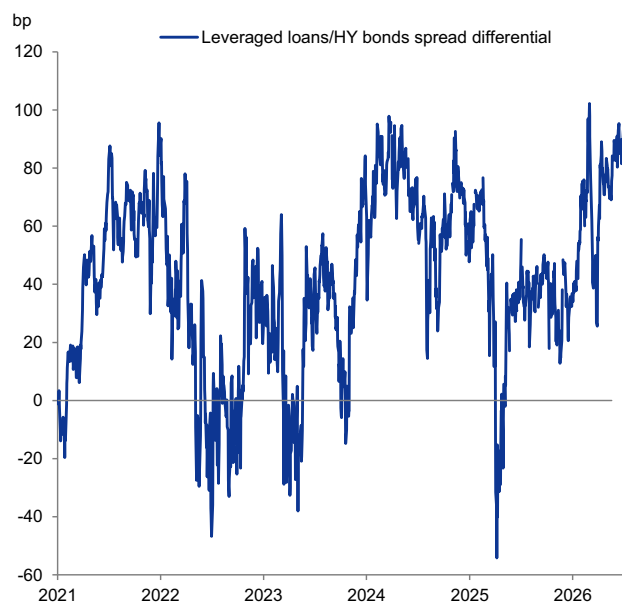
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 43: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

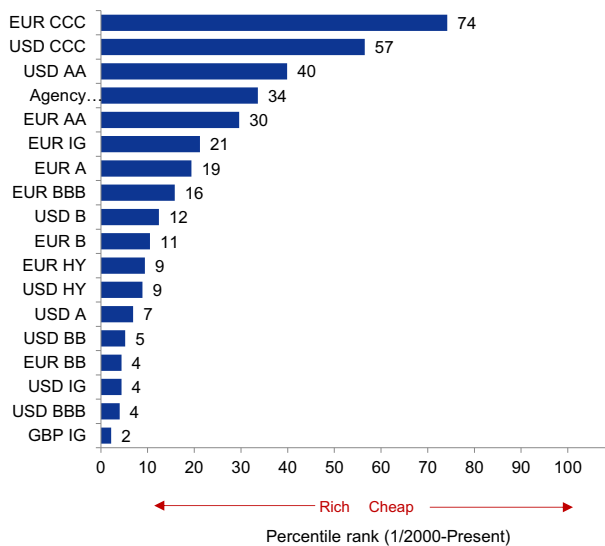
Exhibit 44: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

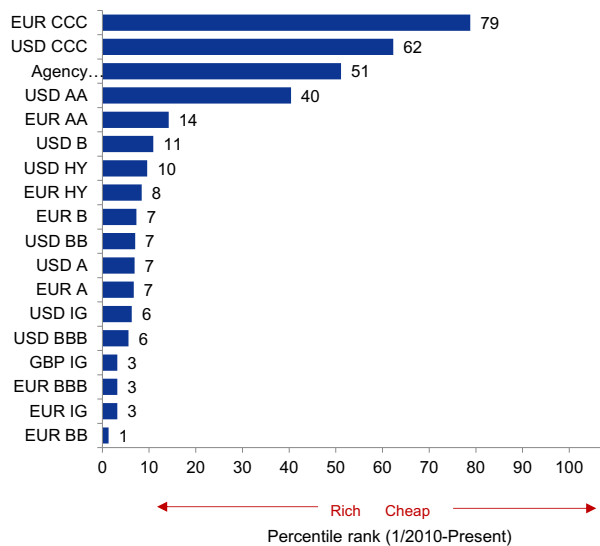
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 45: Percentile ranks for spreads across markets (2000-current)



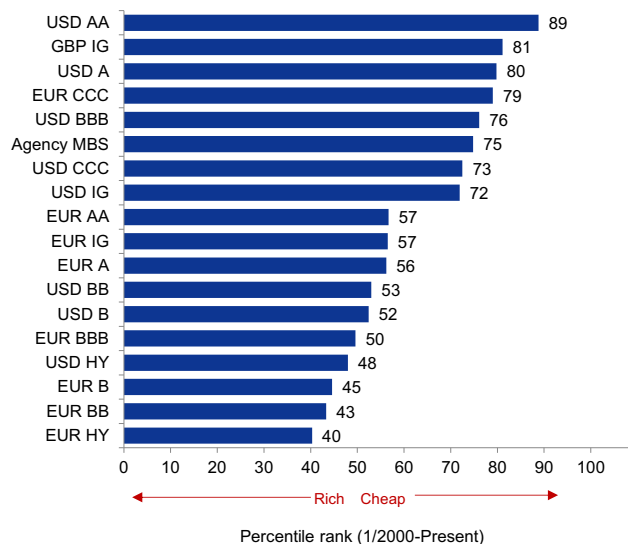
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 46: Percentile ranks for spreads across markets (2010-current)



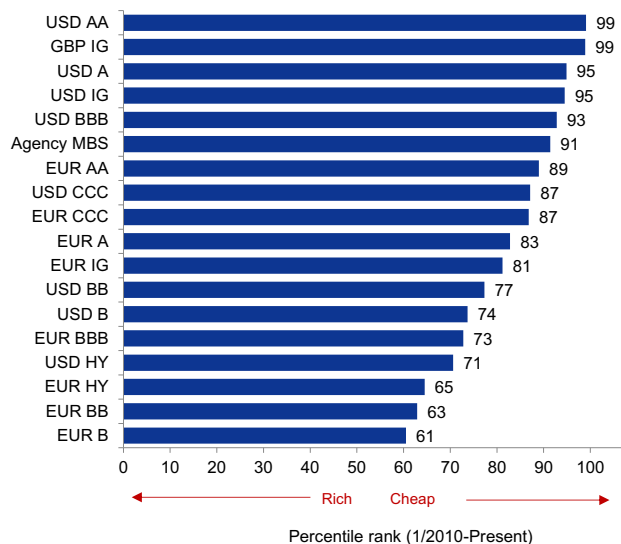
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 47: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

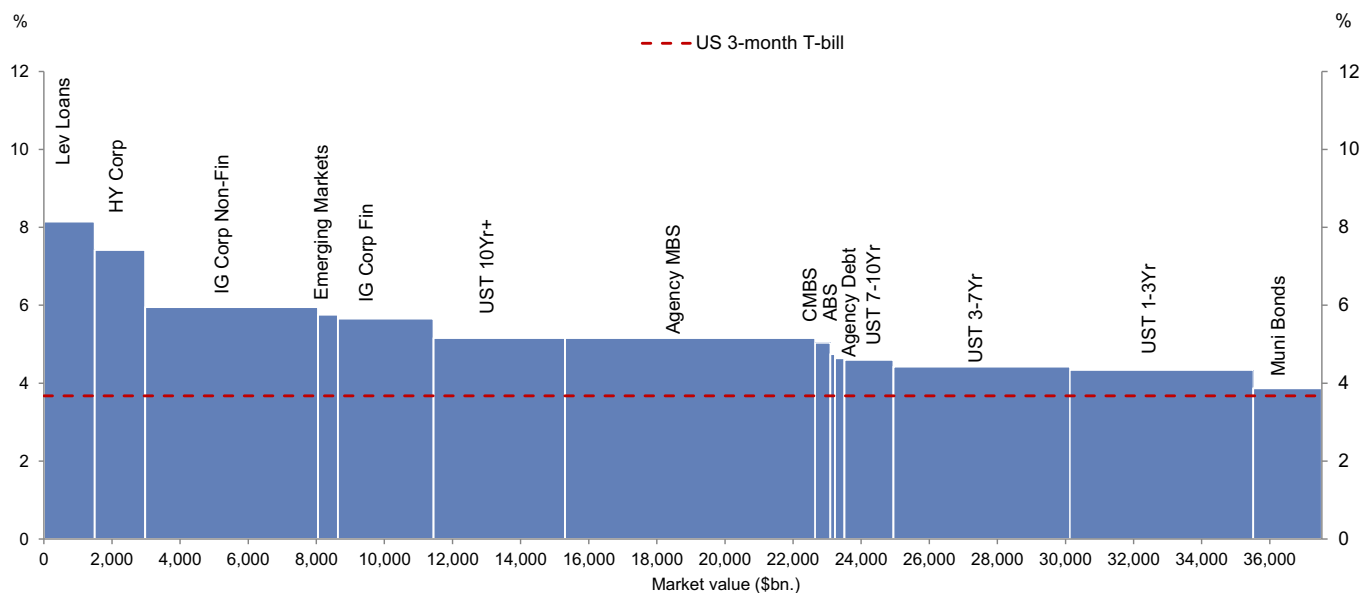
Exhibit 48: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

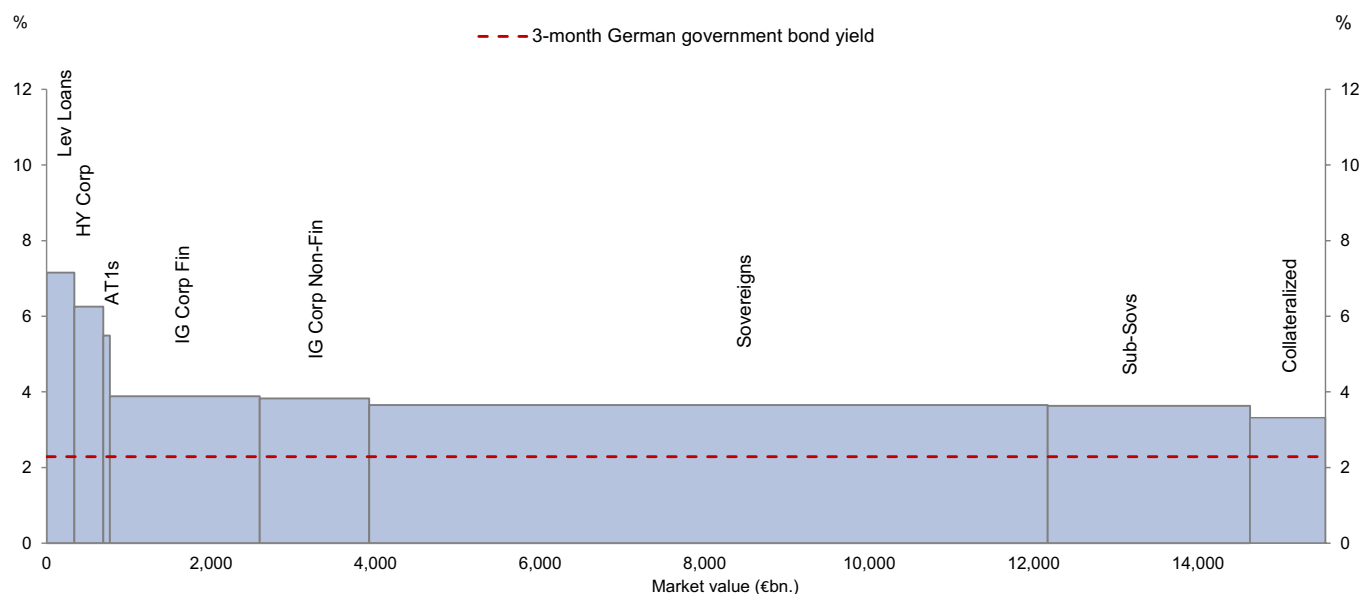
Exhibit 49: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 50: Average yield vs. total market value outstanding for the EUR fixed income markets

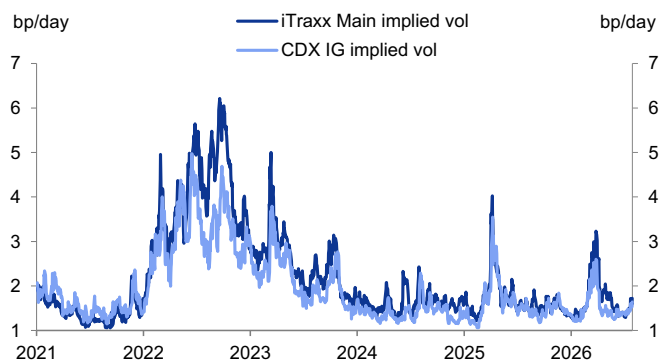


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

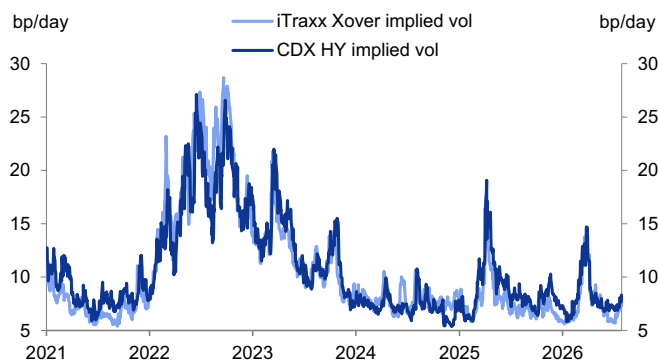
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 51: CDX IG and iTraxx Main vol 3m ATM implied volatility



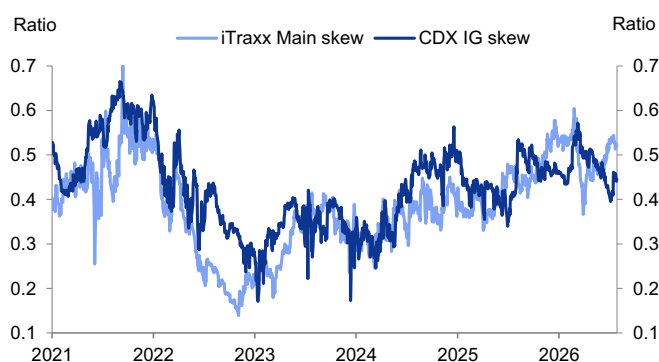
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 52: CDX HY and iTraxx Xover vol 3m ATM implied volatility



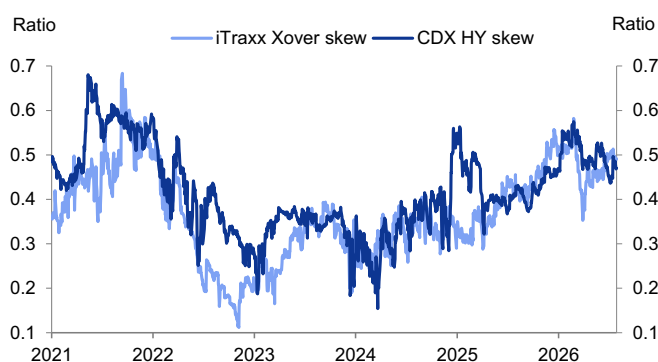
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 53: CDX IG and iTraxx Main skew



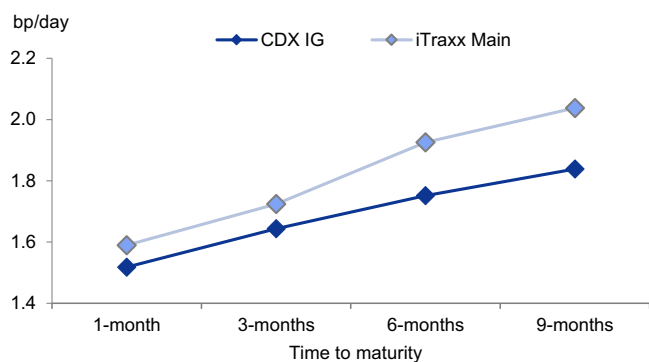
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 54: CDX HY and iTraxx Xover skew



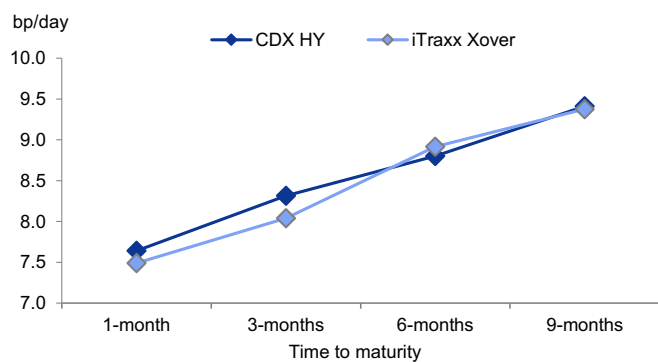
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 55: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 56: CDX HY and iTraxx Xover ATM implied volatility term structure



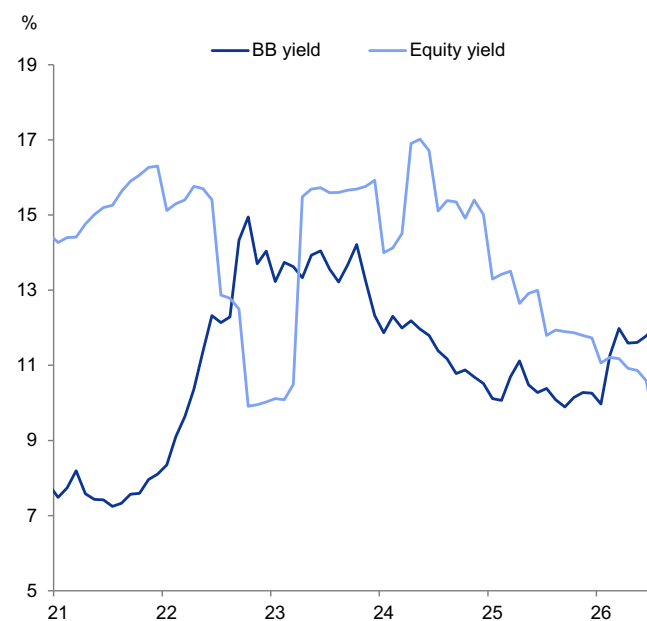
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

Exhibit 57: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	113	35%	23%	16%
AA	145	11%	7%	3%
A	169	8%	5%	2%
BBB	300	46%	28%	21%
BB	779	76%	64%	72%

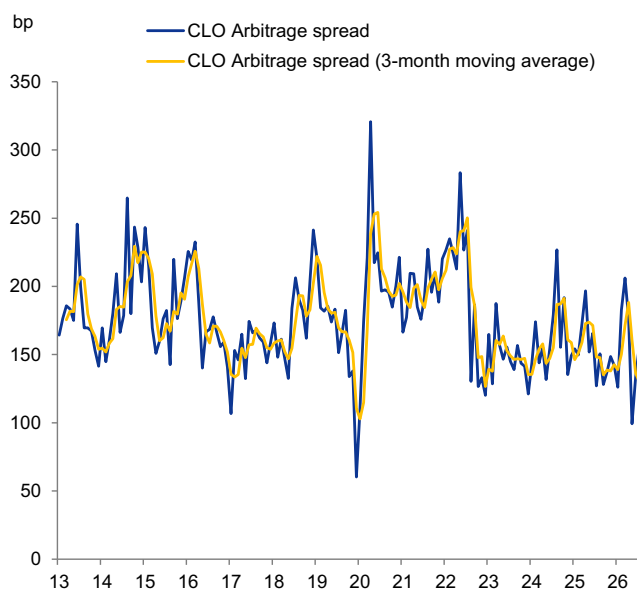
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 58: CLO equity vs. CLO BB yield

Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 59: CLO new issue arbitrage spread

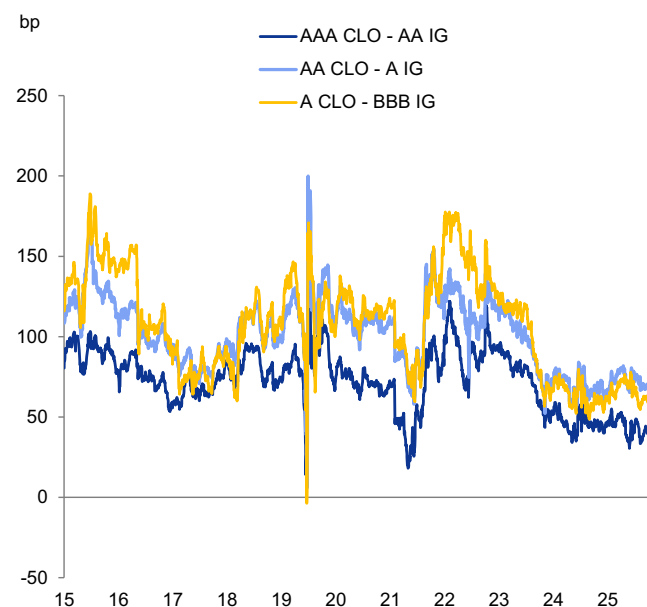
New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 60: CLO vs. USD IG spread differential

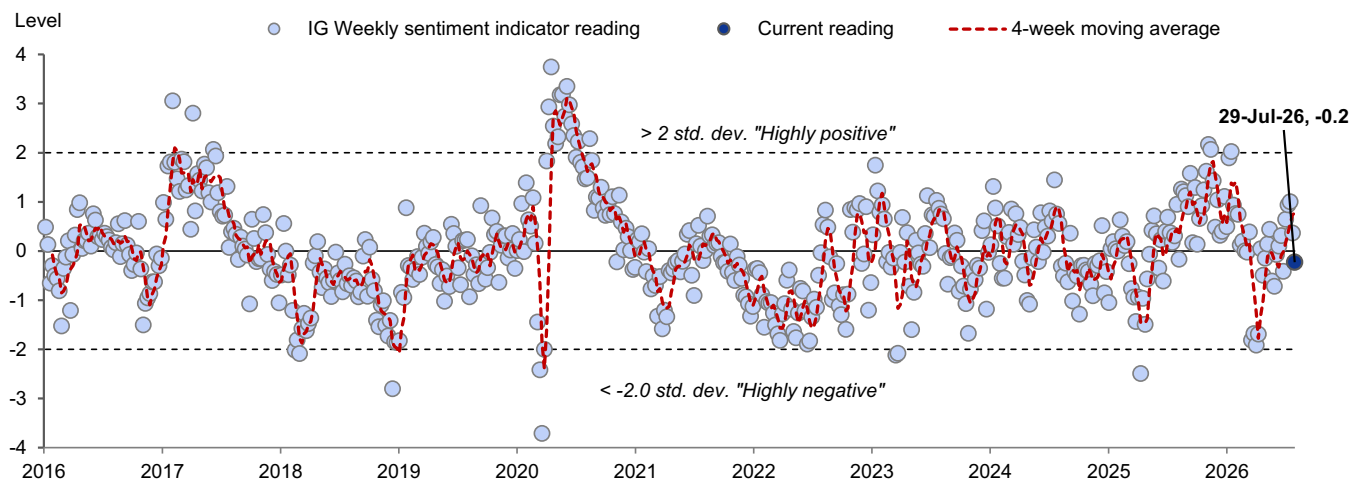
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

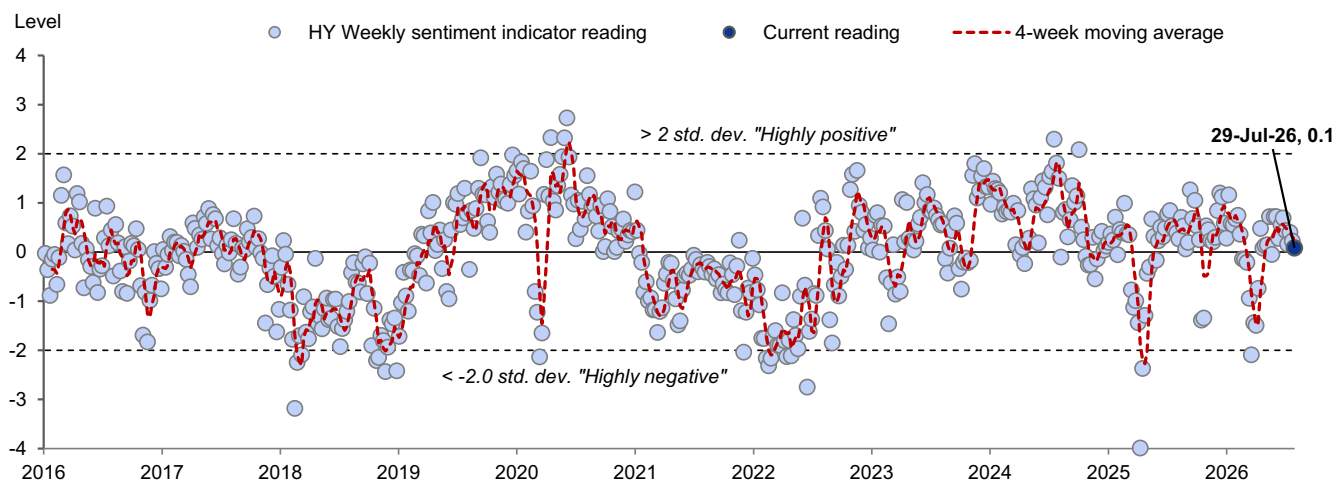
USD IG and HY Sentiment indicators

Exhibit 61: IG Weekly sentiment indicator



Source: See table below.

Exhibit 62: HY Weekly sentiment indicator



Source: See table below.

Exhibit 63: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 64: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
NVDA	NVIDIA Corp	32.5	2426.4	7.5	AA	1.4	7.1
GOOGL	Alphabet Inc	51.0	1875.6	3.7	AA	1.5	9.5
META	Meta Platforms Inc	84.0	2679.9	3.2	AA-	1.3	9.3
NOW	ServiceNow Inc	5.5	133.8	2.4	A	1.7	6.0
AVGO	Broadcom Inc	54.1	1299.0	2.4	A-	2.2	6.2
CEG	Constellation Energy Generation LLC	12.3	252.1	2.1	BBB+	3.1	6.8
ORCL	Oracle Corp	117.2	2374.0	2.0	BBB	4.3	8.4
DOW	The Dow Chemical Company	12.7	227.8	1.8	BBB-	7.7	9.2
BHI	Baker Hughes Holdings LLC / Baker Hughes Co-Obligor Inc	11.4	191.7	1.7	A-	4.4	7.8
PYPL	PayPal Holdings Inc	9.5	149.6	1.6	A-	3.8	7.1
RSG	Republic Services Inc	9.3	131.8	1.4	A-	3.9	4.7
CRM	Salesforce Inc	33.5	469.4	1.4	A	1.6	7.0
MSFT	Microsoft Corp	30.8	426.6	1.4	AAA	7.4	14.0
KLAC	KLA Corp	5.3	72.1	1.4	A	4.7	10.1
UBER	Uber Technologies Inc	7.8	104.9	1.4	BBB+	2.1	6.3
FANG	Diamondback Energy Inc	9.3	123.3	1.3	BBB	3.0	8.0
SNPS	Synopsys Inc	9.0	116.0	1.3	BBB	1.3	6.8
VOD	Vodafone Group PLC	13.5	172.1	1.3	BBB	5.4	11.4
DVN	Devon Energy Corp	7.7	93.7	1.2	BBB+	7.9	9.2
INTU	Intuit Inc	6.0	73.1	1.2	A-	2.5	6.4

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

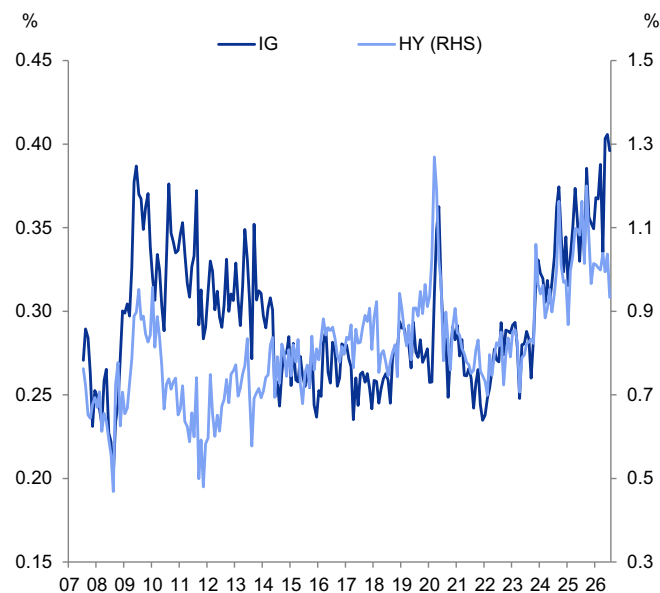
Exhibit 65: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
CRWV	CoreWeave Inc	7.8	411.9	5.3	B+	0.6	3.8
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	102.3	5.1	B	1.8	3.9
APLD	APLD ComputeCo LLC	2.4	87.0	3.7	BB-	0.6	2.8
PFORGE	APLD ComputeCo 2 LLC	2.2	74.0	3.4	BB-	0.3	3.2
CORZ	Core Scientific Finance I LLC	3.3	110.2	3.3	BB-	0.2	3.0
VMED	Vmed O2 UK Financing I PLC	8.1	251.4	3.1	BB-	4.7	3.9
BLKPRL	Black Pearl Compute LLC	2.0	55.1	2.8	BB-	0.4	3.9
ACI	Albertsons Cos Inc / Safeway Inc / New Albertsons LP / Alberts	7.3	182.2	2.5	BB	2.7	4.0
WHR	Whirlpool Corp	3.9	74.9	1.9	BB-	1.7	4.4
VOLTAG	VoltaGrid LLC	2.0	38.3	1.9	B+	0.6	3.7
BURLN	Burford Capital Global Finance LLC	2.1	38.2	1.8	BB-	2.3	4.2
WULF	WULF Compute LLC	3.2	58.1	1.8	BB	0.7	2.9
ZIGGO	VZ Secured Financing BV	4.2	69.3	1.7	B+	4.2	4.3
CYH	CHS/Community Health Systems Inc	8.7	138.7	1.6	B-	3.8	3.0
TRACTC	SV RNO Property Owner 1 LLC	3.8	59.9	1.6	BB	0.3	3.7
MERIDI	Meridian Arc Holdco LLC	5.7	85.9	1.5	BB	0.2	4.1
GT	The Goodyear Tire & Rubber Company	4.0	59.9	1.5	B+	3.0	4.0
MIK	Michaels Cos Inc/The	2.8	39.3	1.4	B-	0.3	5.0
CLF	Cleveland-Cliffs Inc	6.0	82.9	1.4	B	1.8	2.8
SABHLD	Sabre GLBL Inc	3.2	44.2	1.4	B-	0.9	2.7

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

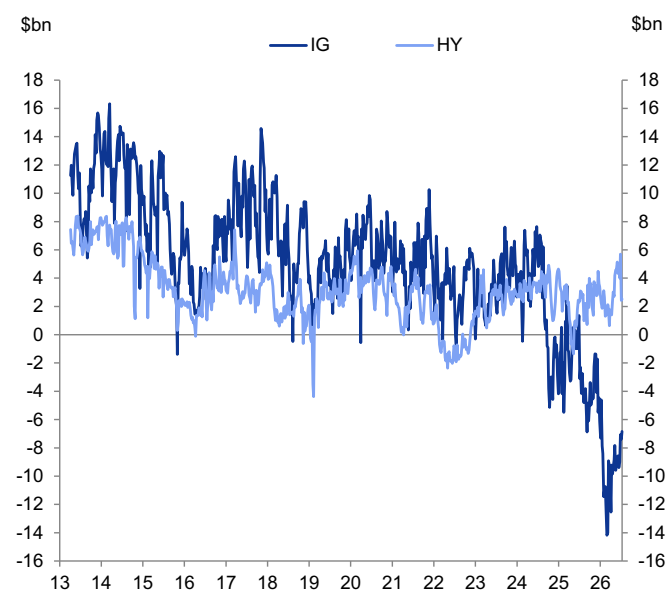
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 66: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



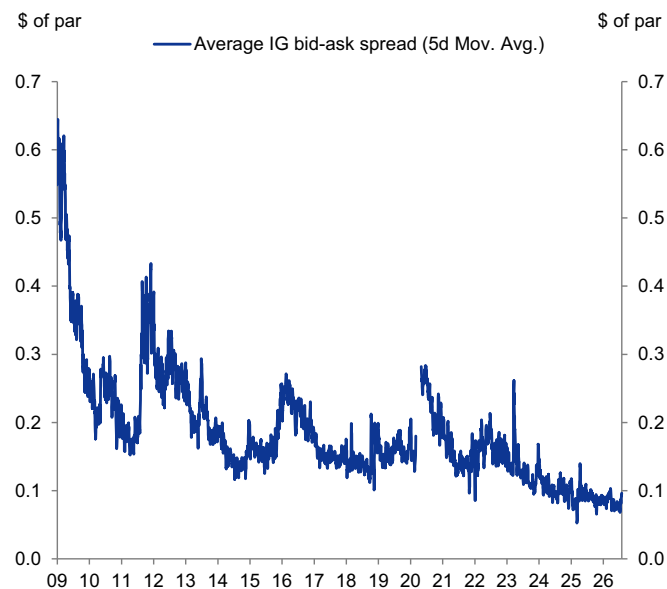
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 67: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

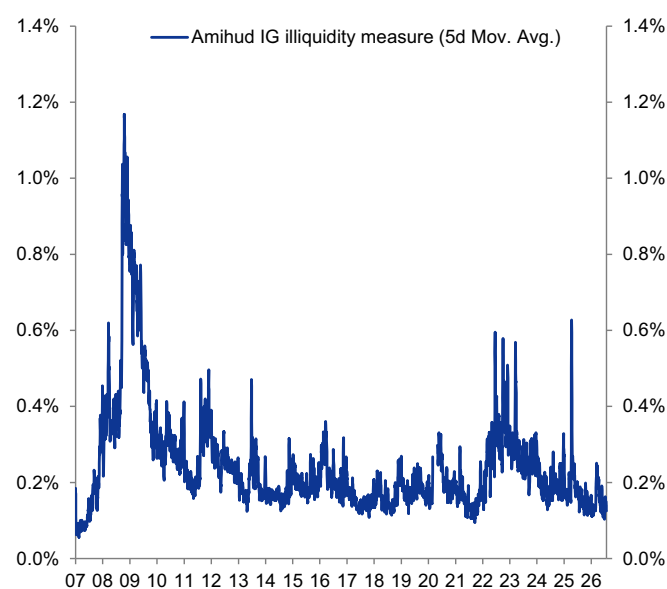
Exhibit 68: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 69: Amihud IG illiquidity measure 5-day moving average

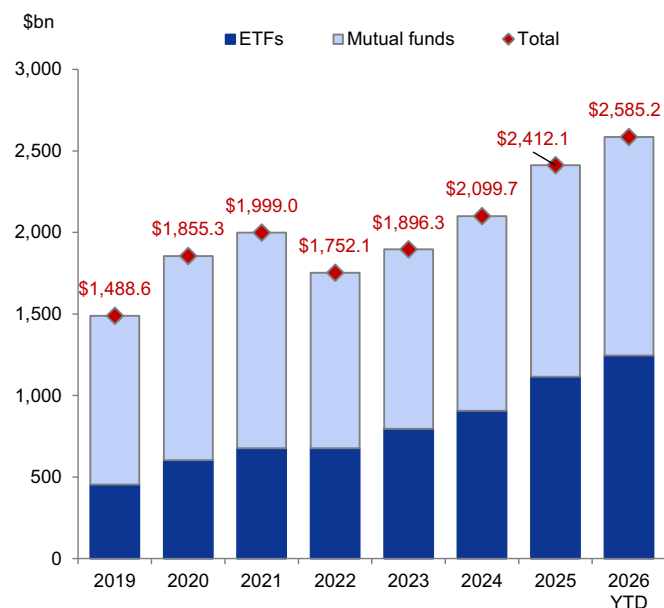


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

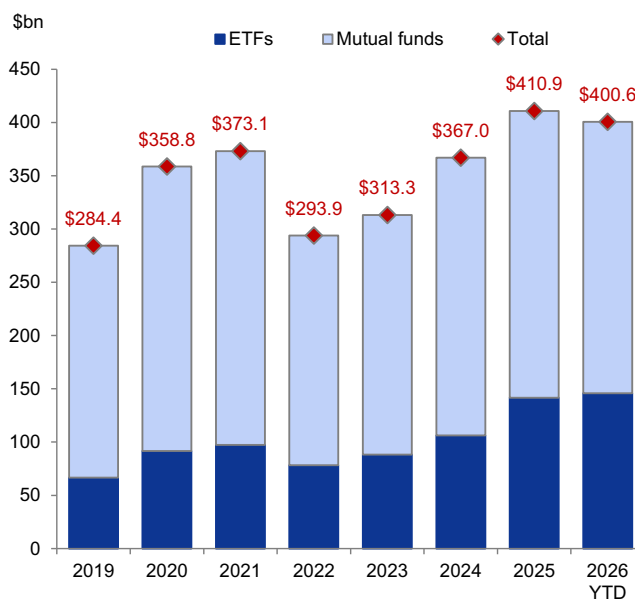
Flow dashboard: Mutual fund and ETF AUM

Exhibit 70: AUM in USD IG-focused Mutual Funds and ETFs



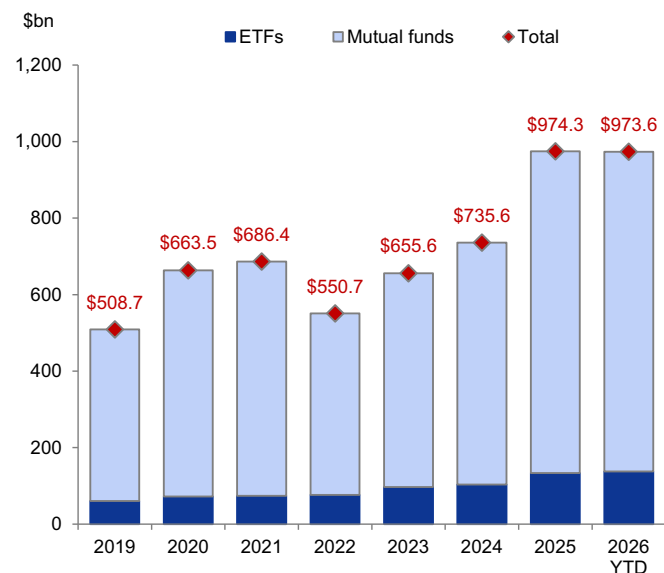
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 71: AUM in USD HY-focused Mutual Funds and ETFs



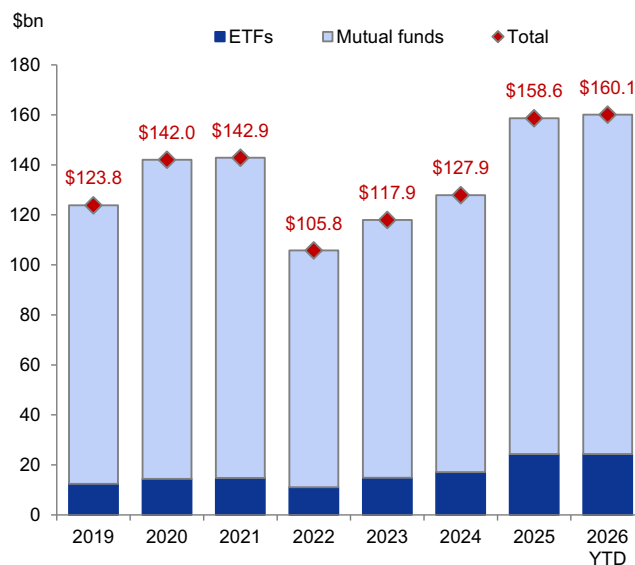
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 72: AUM in EUR IG-focused Mutual Funds and ETFs



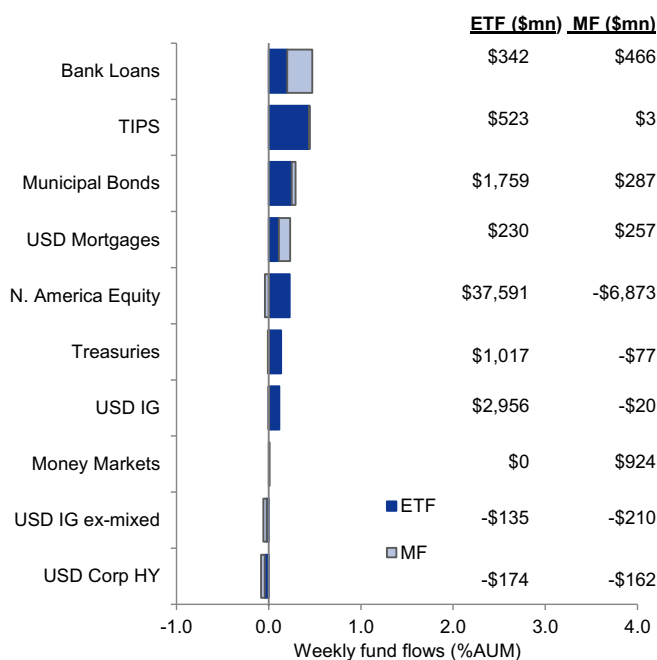
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 73: AUM in EUR HY-focused Mutual Funds and ETFs

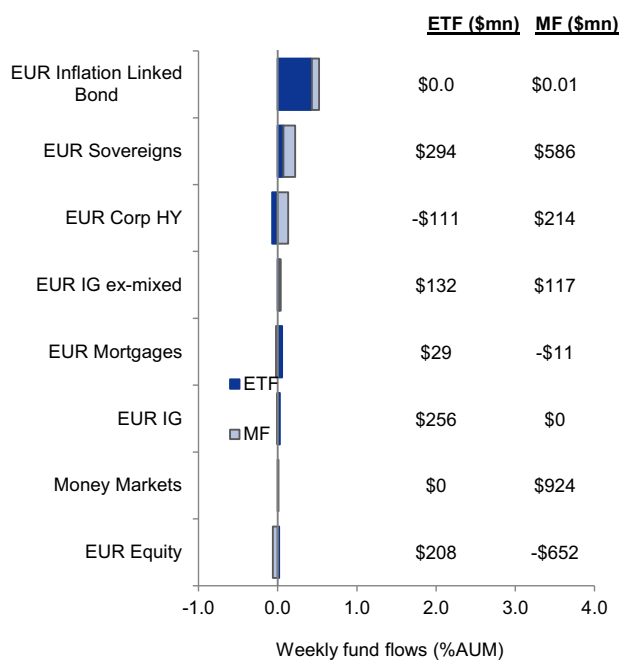


Source: EPFR, Goldman Sachs Global Investment Research

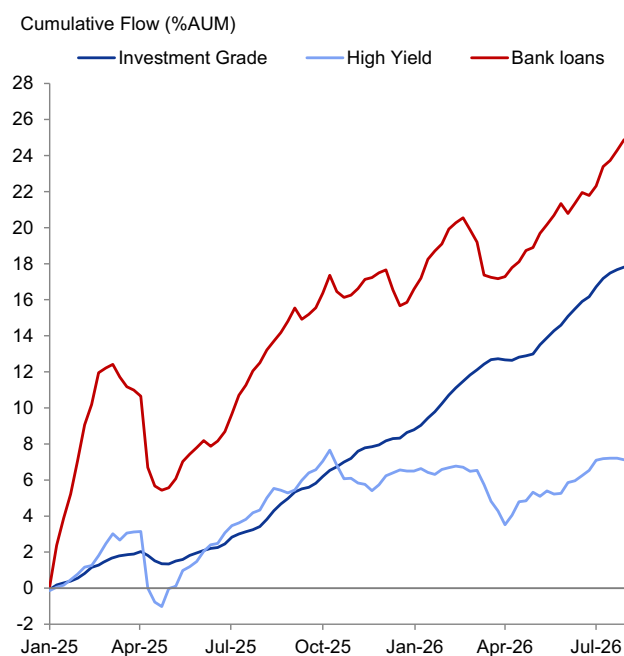
Flow dashboard: Mutual fund and ETF flows

Exhibit 74: Weekly mutual fund flows into US fixed income and equity markets

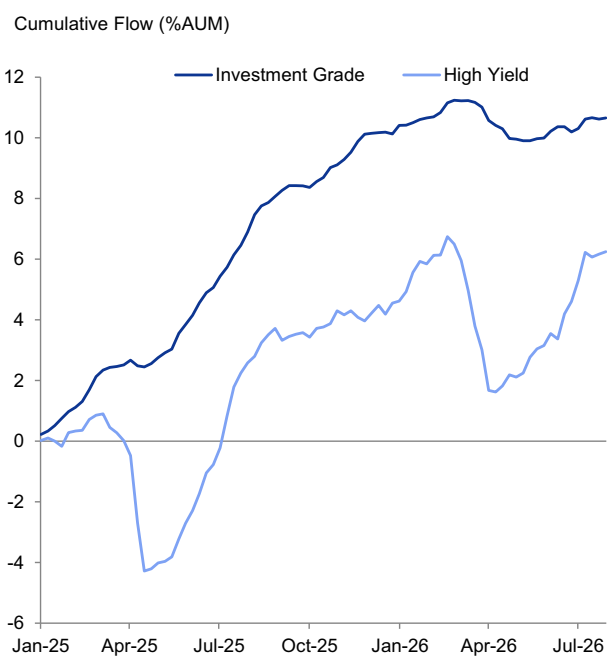
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 75: Weekly mutual fund flows into European fixed income and equity markets

Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 76: Cumulative flows as a percentage of beginning of the year AUM for US credit funds

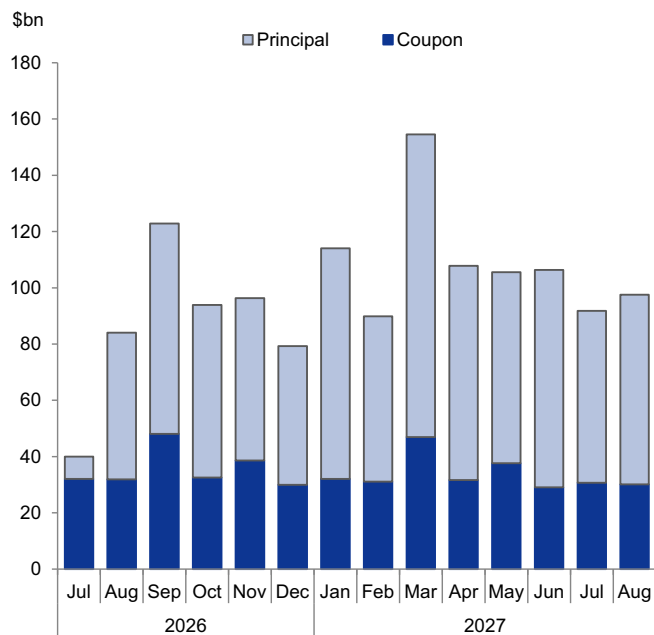
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 77: Cumulative flows as a percentage of beginning of the year AUM for European credit funds

Source: EPFR, Goldman Sachs Global Investment Research

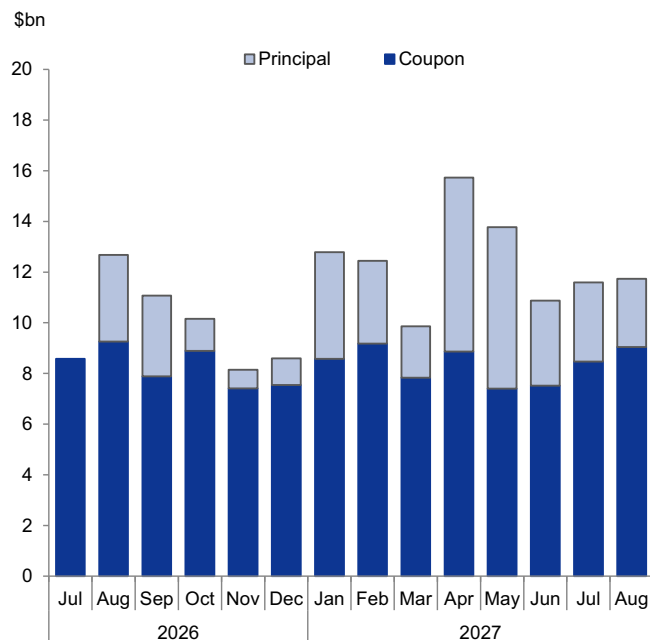
Flow dashboard: Upcoming coupon and principal payments

Exhibit 78: Our estimates of upcoming coupon and principal payments in the USD IG market



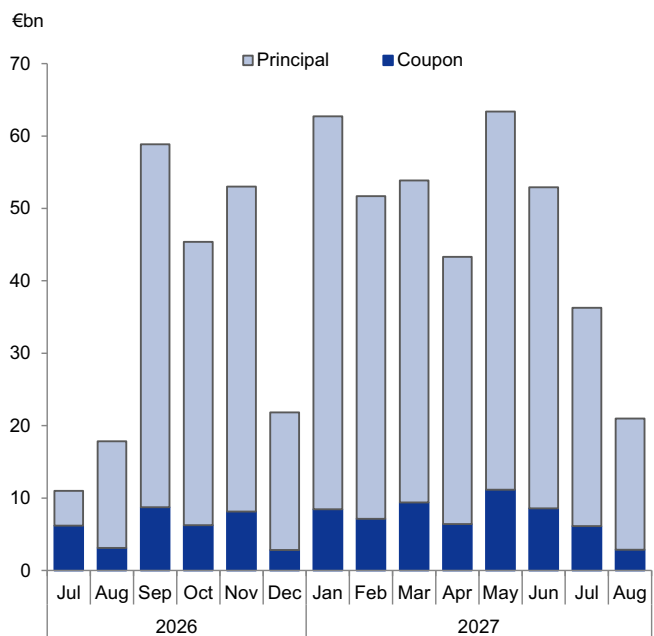
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 79: Our estimates of upcoming coupon and principal payments in the USD HY market



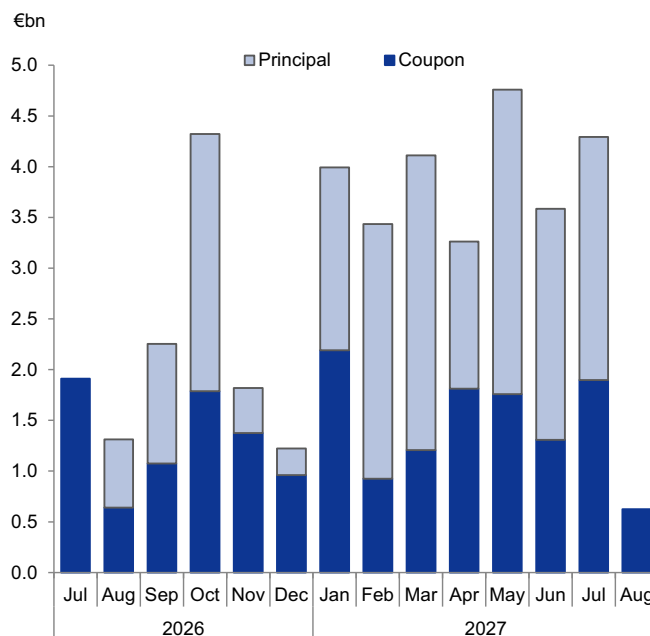
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 80: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

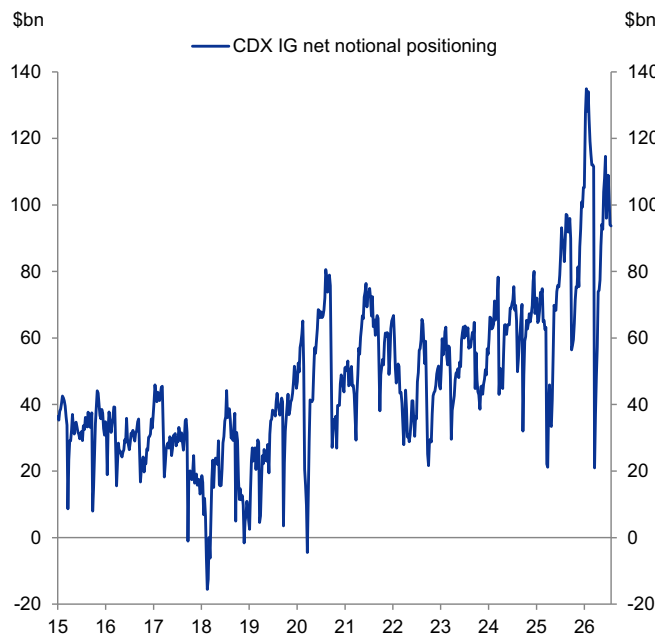
Exhibit 81: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 82: Net notional positioning in the CDX IG 5-year on-the-run contract



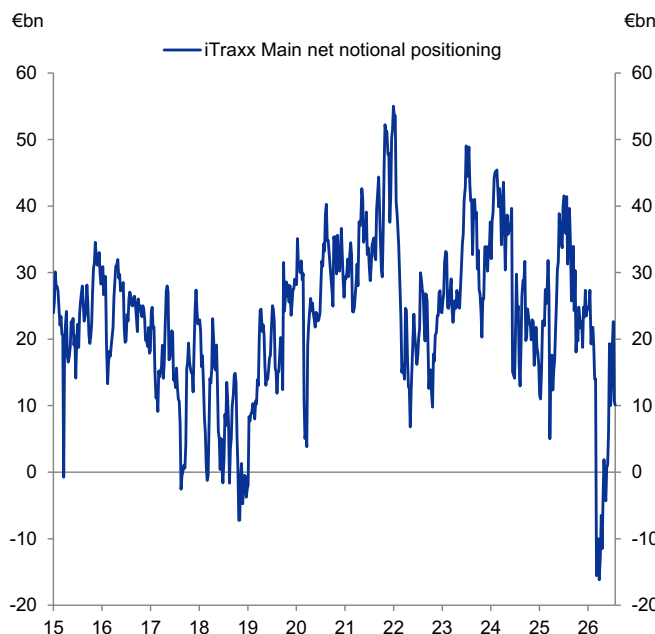
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 83: Net notional positioning in the CDX HY 5-year on-the-run contract



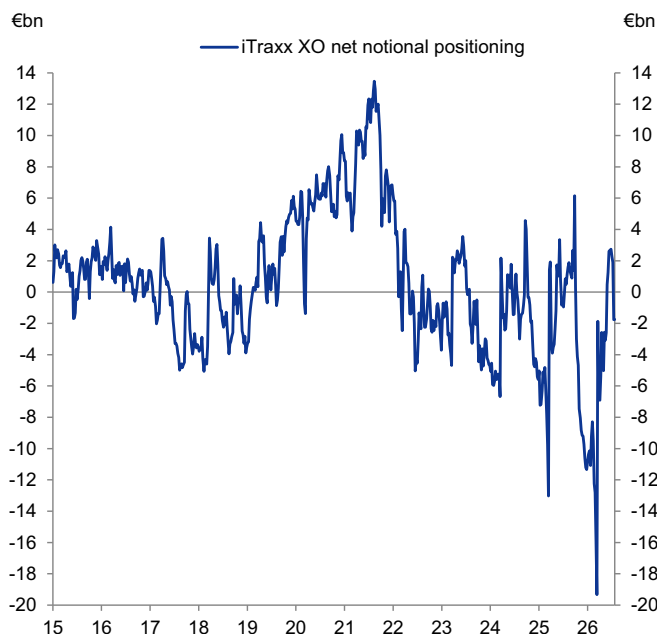
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 84: Net notional positioning in the iTraxx Main 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

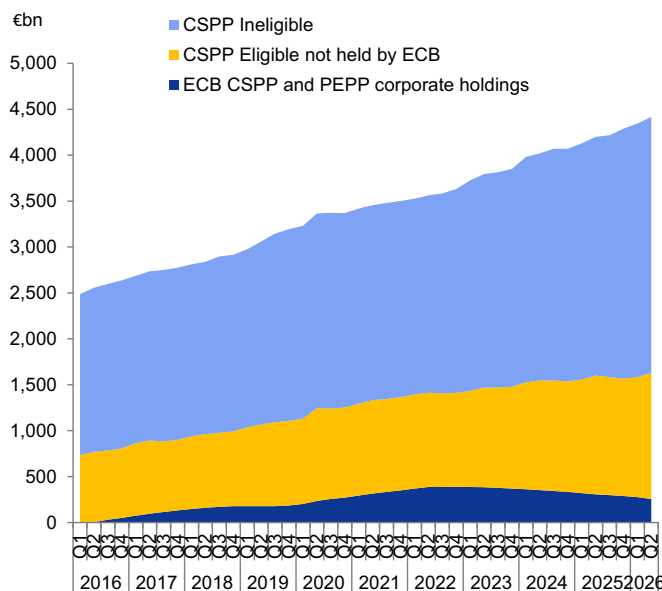
Exhibit 85: Net notional positioning in iTraxx over 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

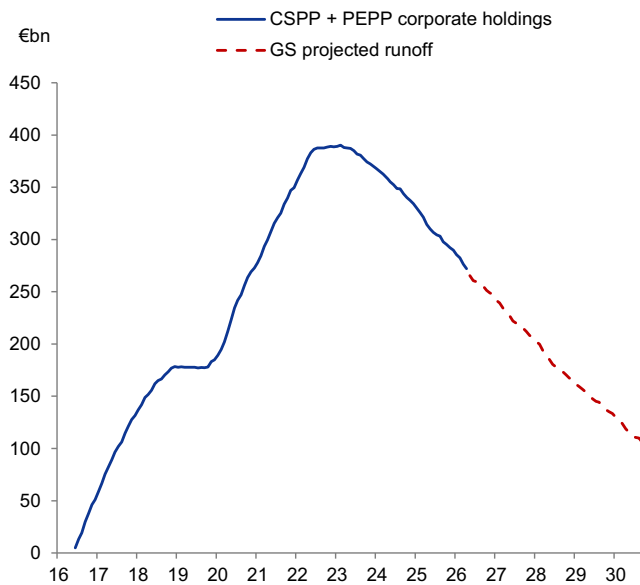
Flow dashboard: ECB QT

Exhibit 86: Ownership share of the ECB in the EUR IG market



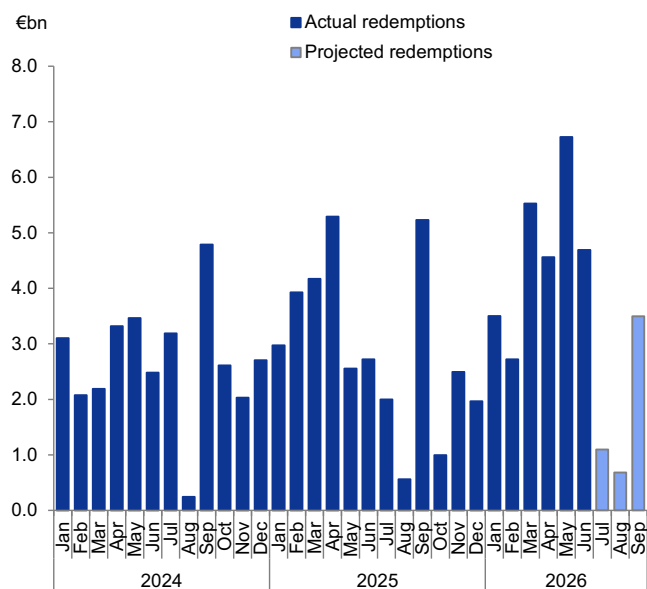
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 87: Run-off projections for the ECB corporate bond holdings



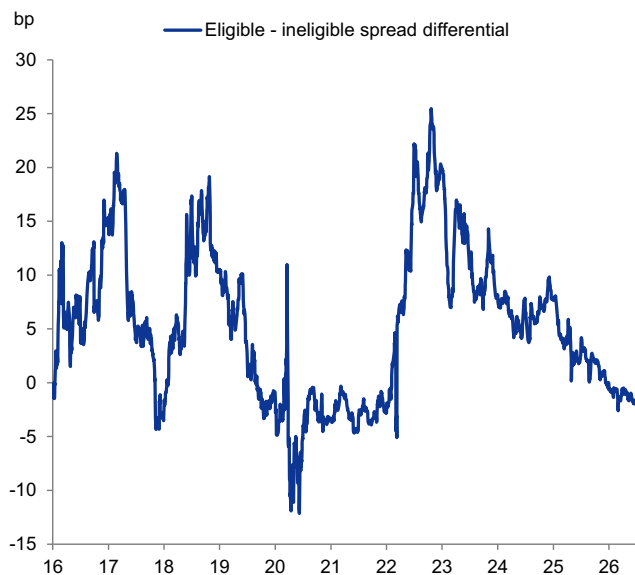
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 88: CSPP monthly run-off pace



Source: ECB, Goldman Sachs Global Investment Research

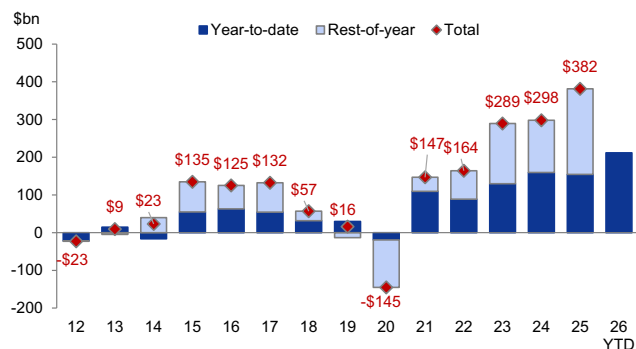
Exhibit 89: Performance of CSPP-eligible bonds vs. non-eligible bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

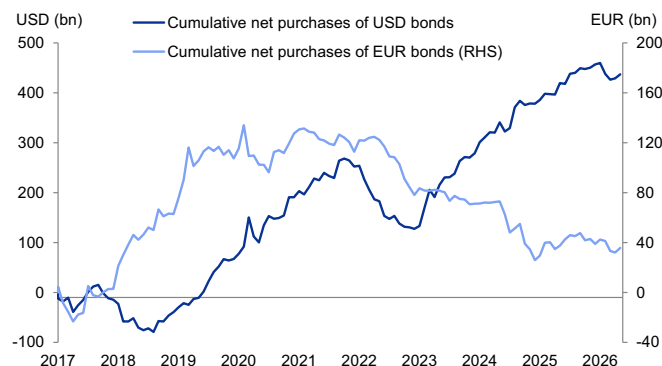
Exhibit 90: Net purchases of USD corporate bonds by foreign investors



Note: As of May 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

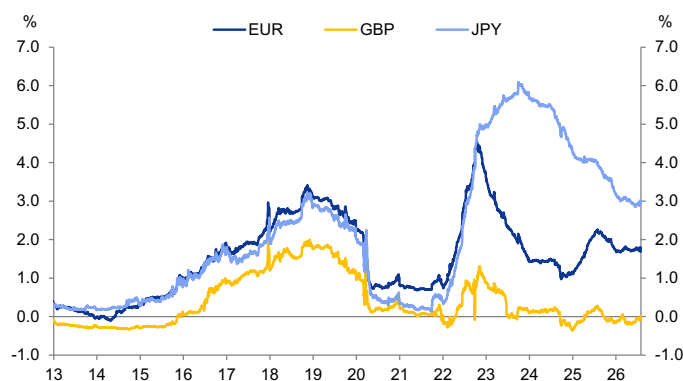
Exhibit 91: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of May 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 92: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

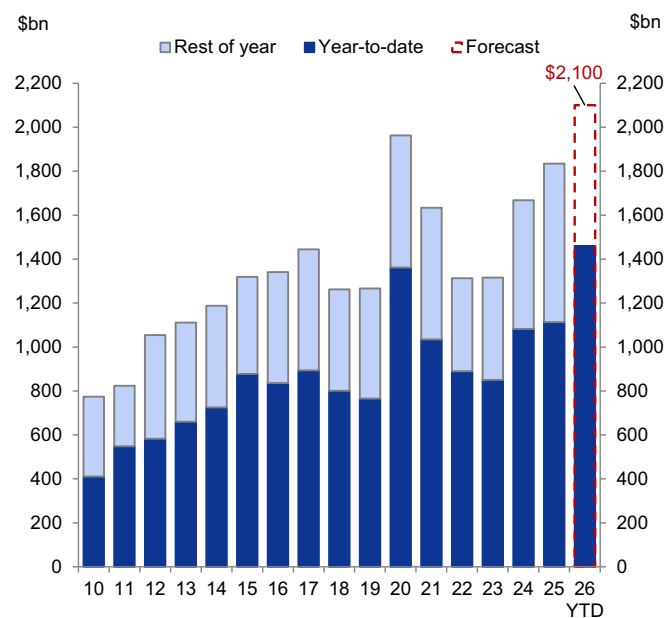


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 93: USD IG gross issuance

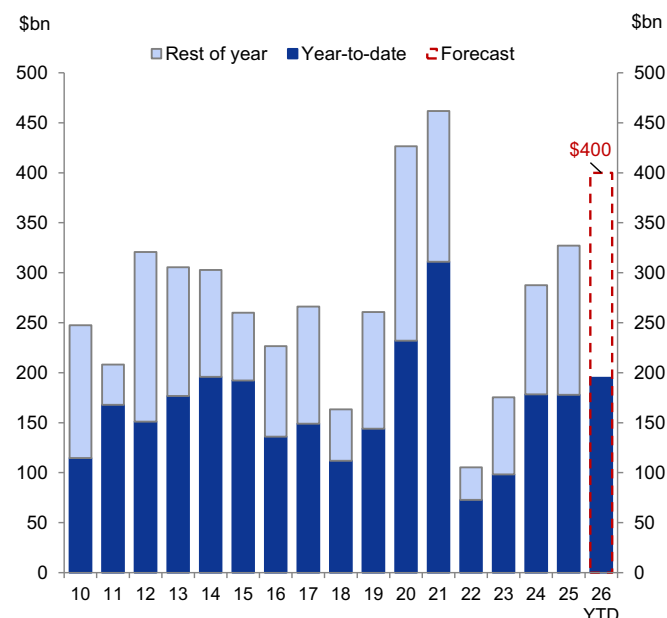
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 94: USD HY gross issuance

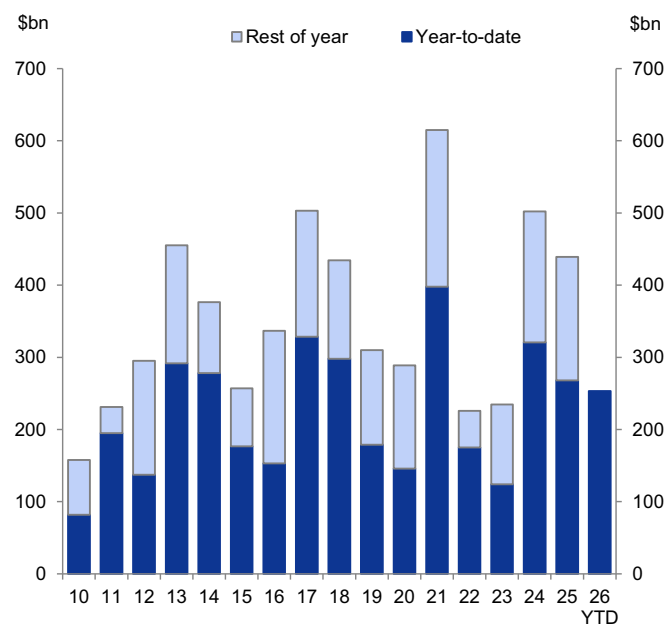
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 95: USD institutional leveraged loan gross issuance

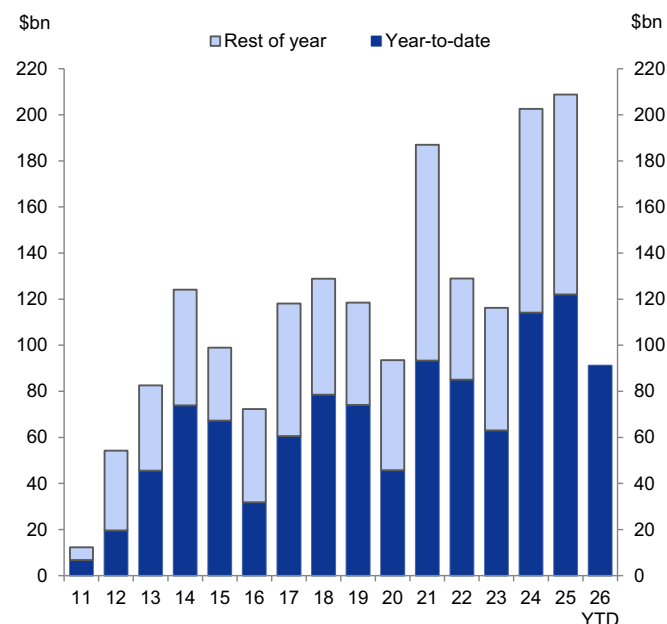
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 96: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

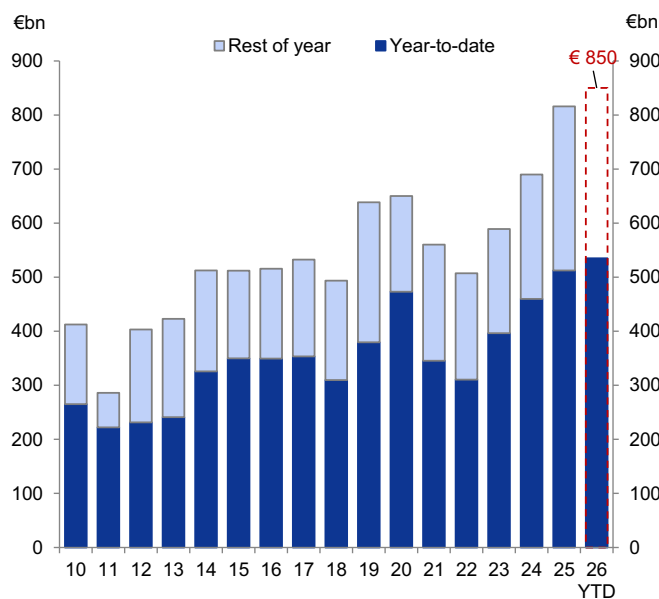


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 97: EUR IG gross issuance

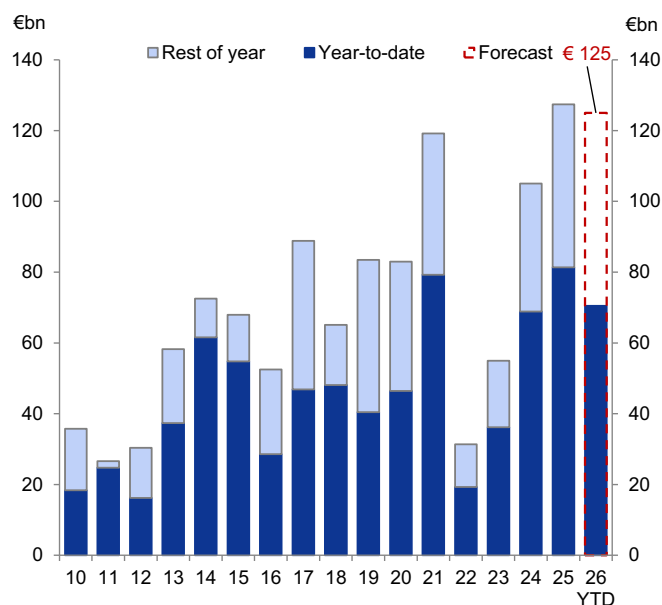
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 98: EUR HY gross issuance

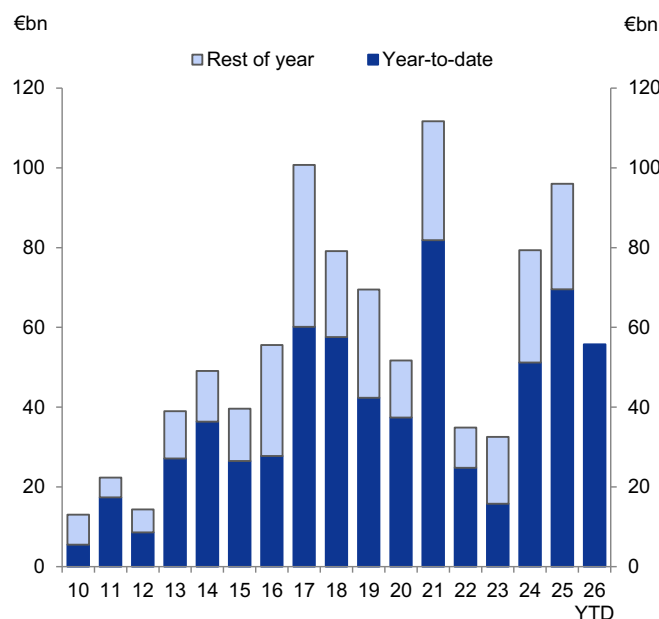
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 99: EUR Leveraged loan gross issuance

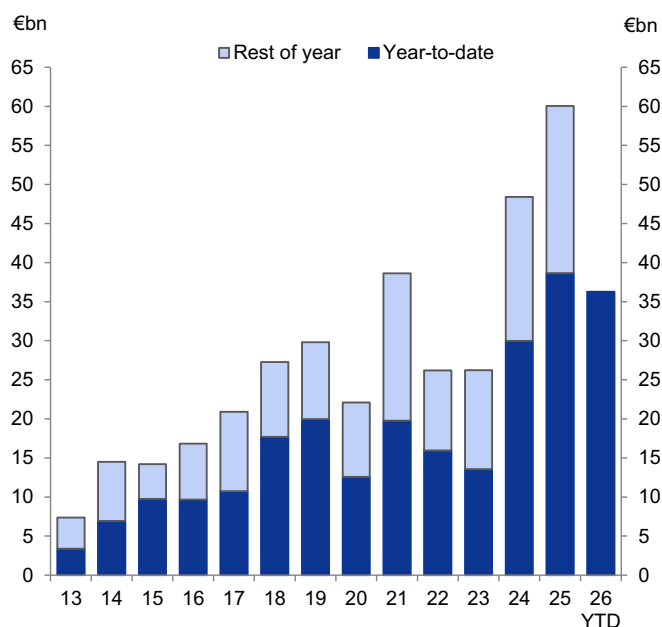
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 100: EUR CLO creation

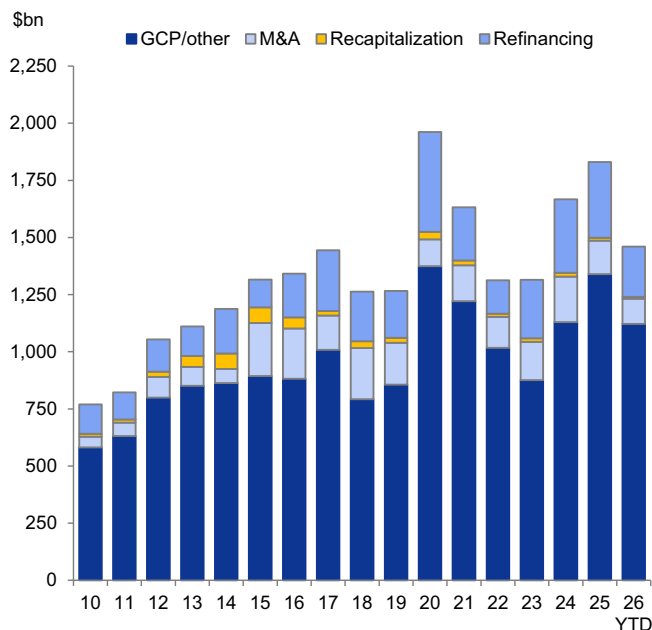
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

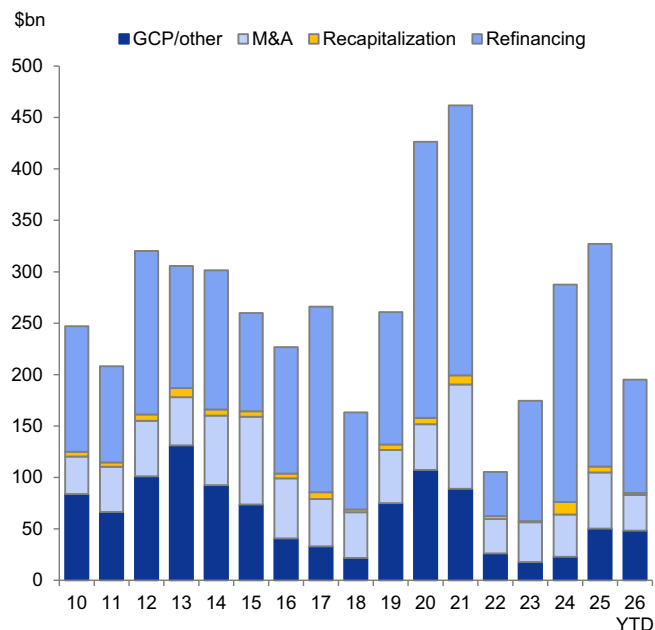
Primary market dashboard: Use of proceeds

Exhibit 101: Annual USD IG issuance volumes by use of proceeds



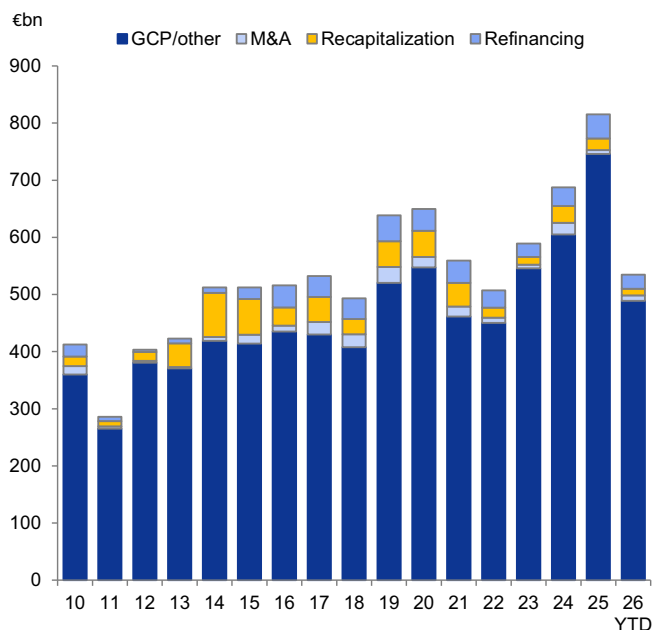
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 102: USD HY issuance volumes by use of proceeds



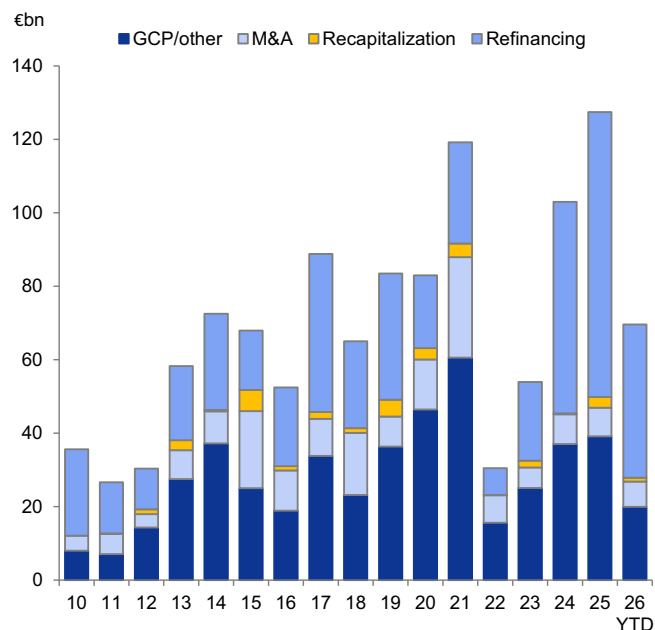
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 103: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 104: EUR HY issuance volumes by use of proceeds

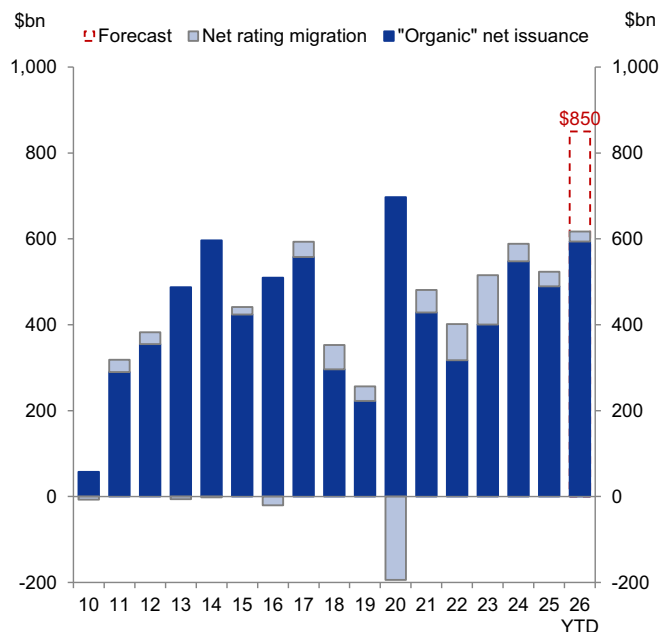


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

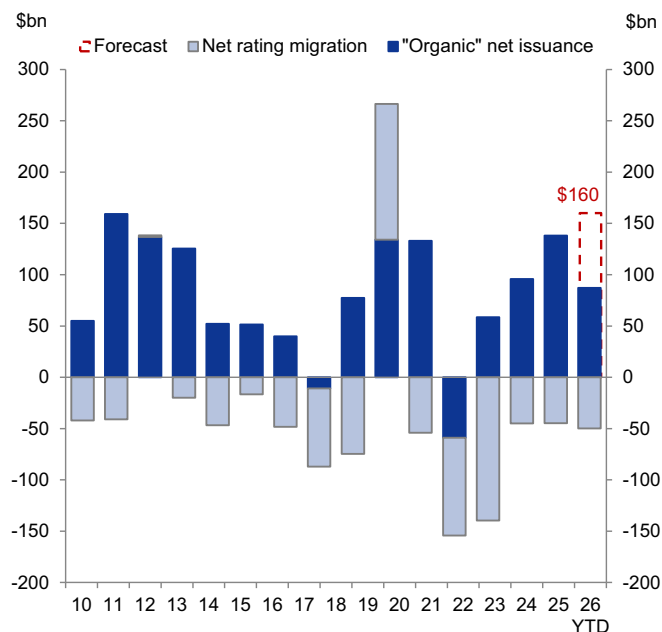
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 105: Annual USD IG net issuance



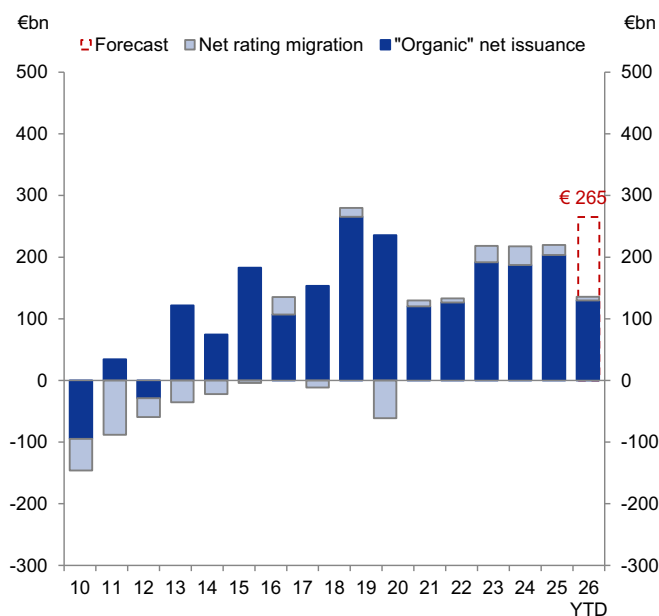
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 106: Annual USD HY net issuance



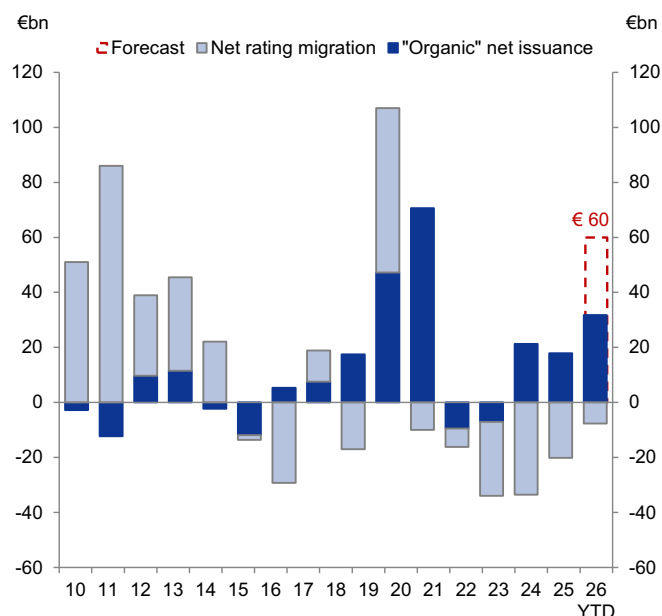
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 107: Annual EUR IG net issuance



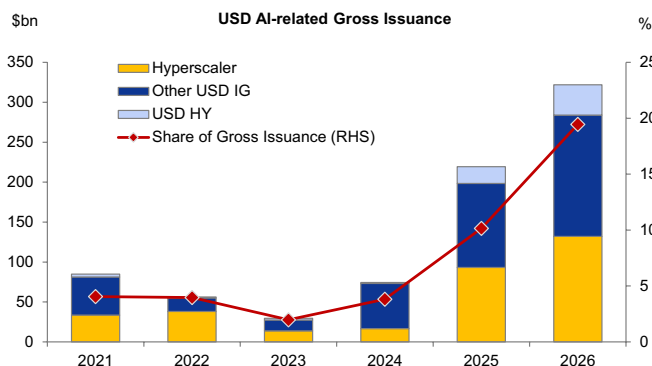
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 108: Annual EUR HY net issuance

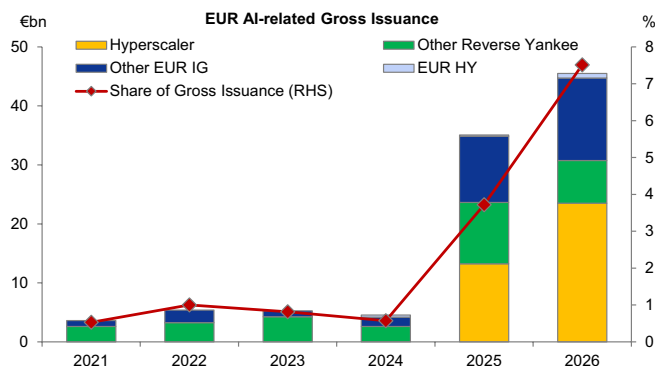


Source: Bloomberg, Goldman Sachs Global Investment Research

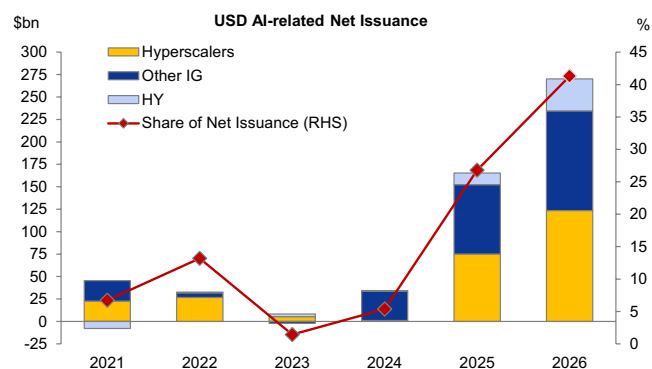
Primary market dashboard: AI-related issuance for the USD and EUR markets

Exhibit 109: USD AI-related Gross Issuance


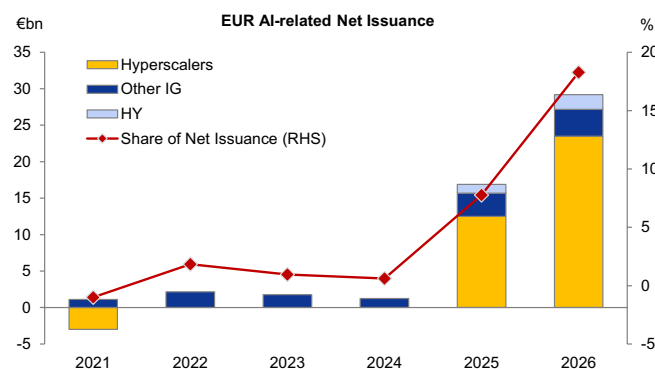
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 110: EUR AI-related Gross Issuance


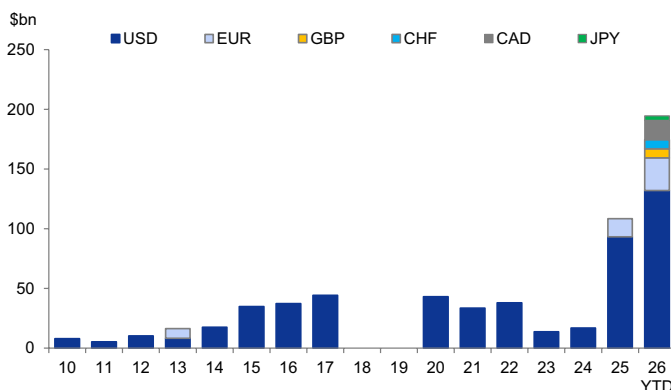
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 111: USD AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 112: EUR AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 113: Hyperscaler debt issuance across currencies (in USD-equivalent)


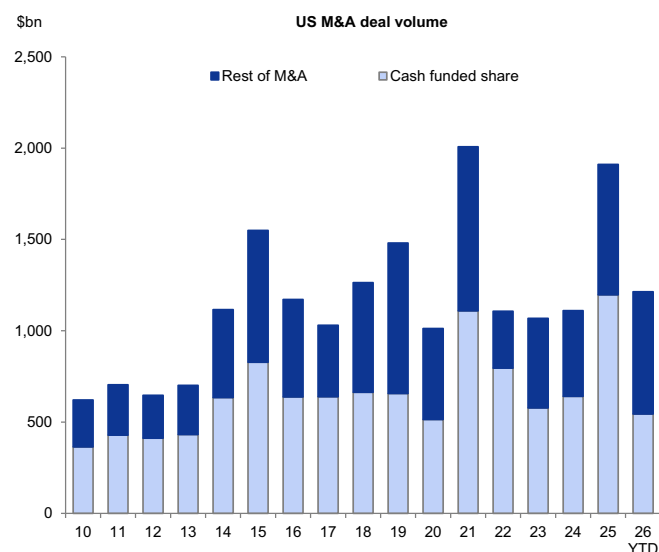
Note: Captures debt issued by the following companies: Alphabet (Google), Amazon, Microsoft, Meta, and Oracle.

Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume

Exhibit 114: Annual US M&A deal volume

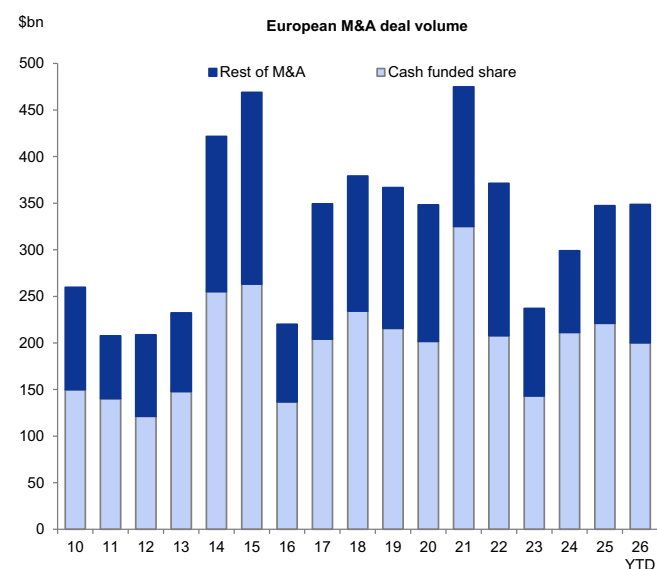
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 115: Annual European M&A deal volume

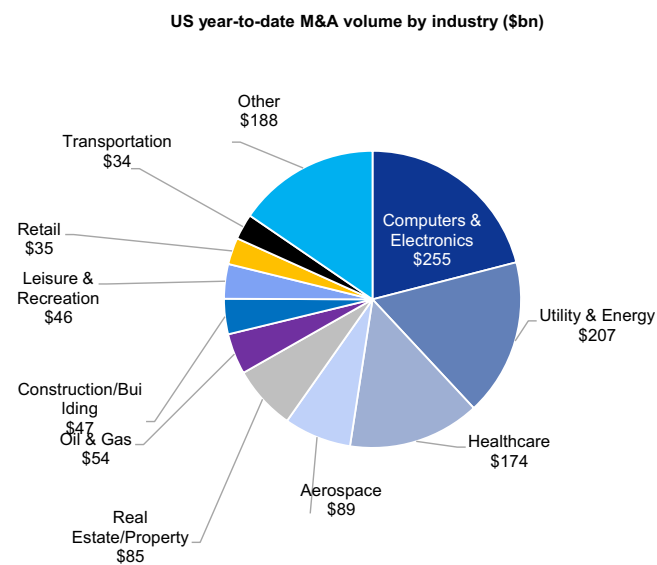
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 116: Year-to-date US M&A volume by sector

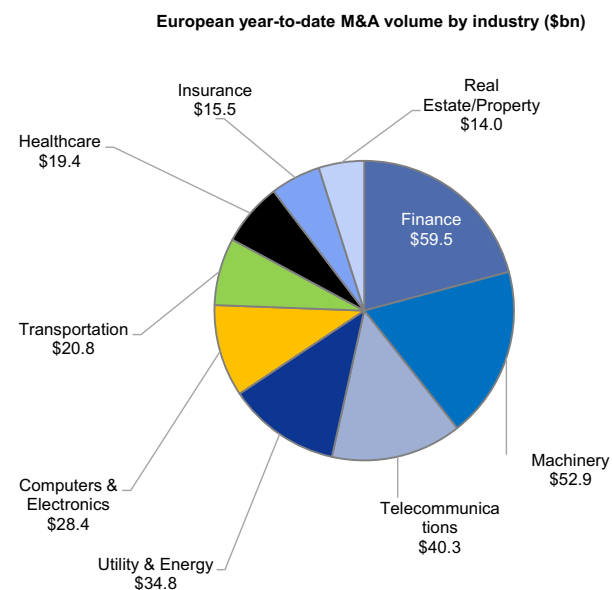
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 117: Year-to-date European M&A volume by sector

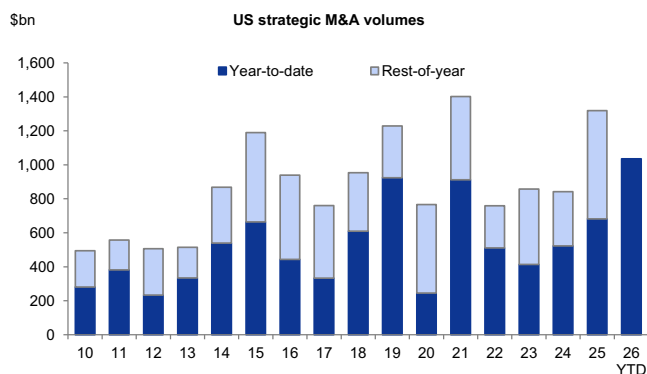
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume (Strategic deals only)

Exhibit 118: Annual US strategic M&A volumes



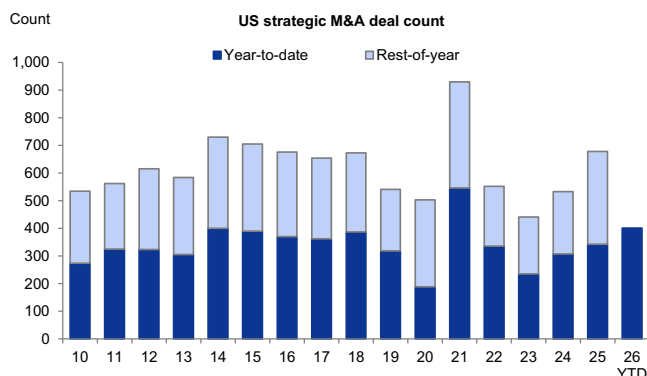
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 119: Annual European strategic M&A volumes



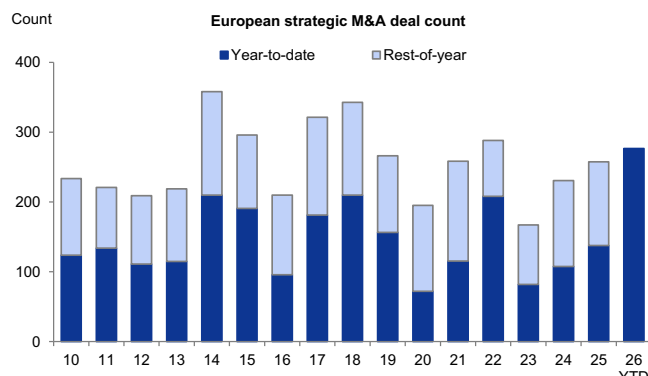
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 120: Annual US strategic M&A deal count



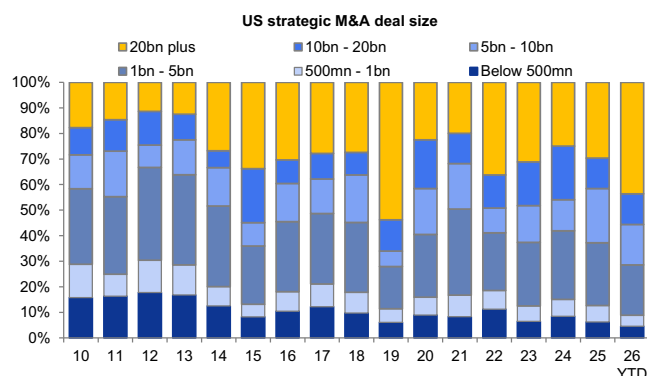
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 121: Annual European strategic M&A deal count



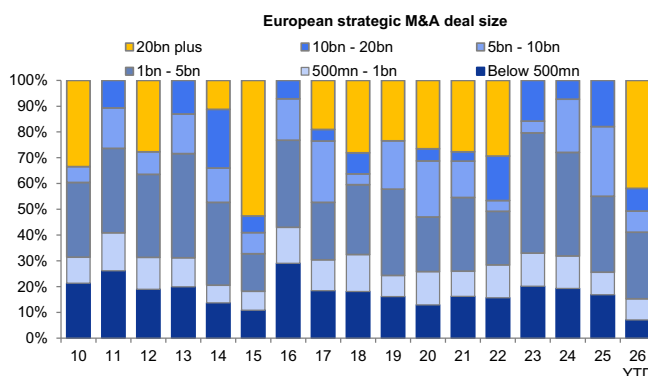
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 122: Annual US strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

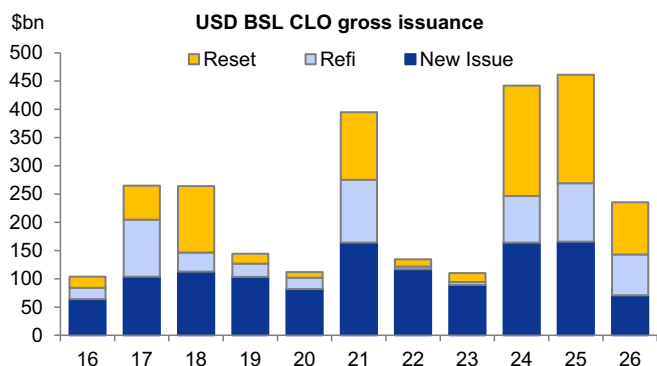
Exhibit 123: Annual European strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

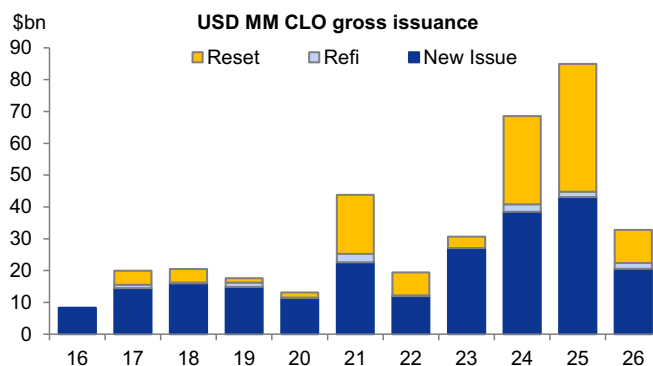
Primary market dashboard: CLO creation and refinancing

Exhibit 124: Annual USD Broadly syndicated loan CLO creation



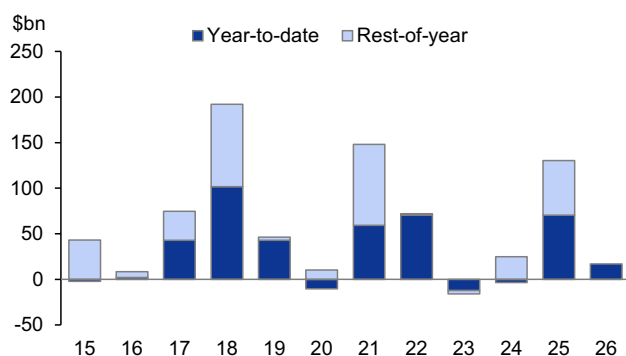
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 125: Annual USD Middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

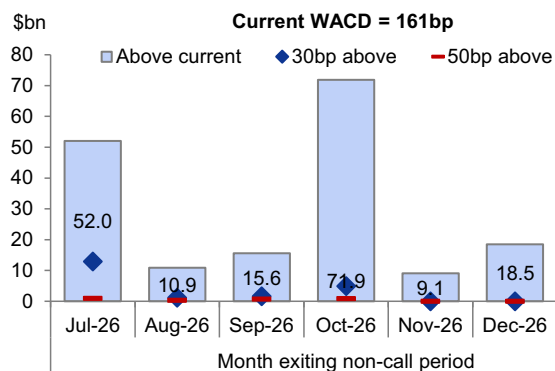
Exhibit 126: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 127: USD CLO refinance monitor

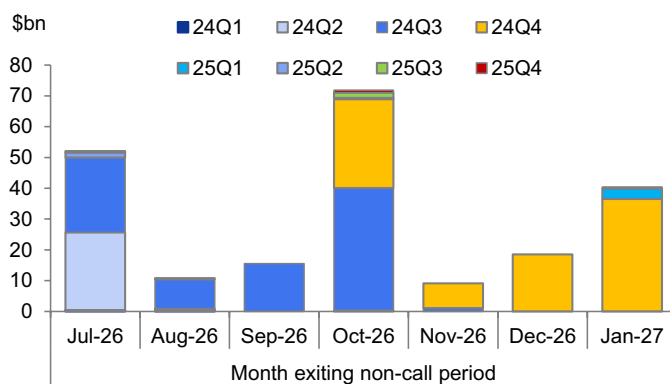
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 128: USD CLO refinance monitor

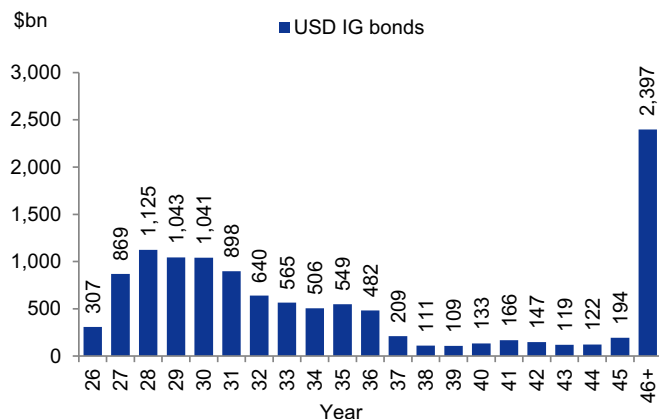
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

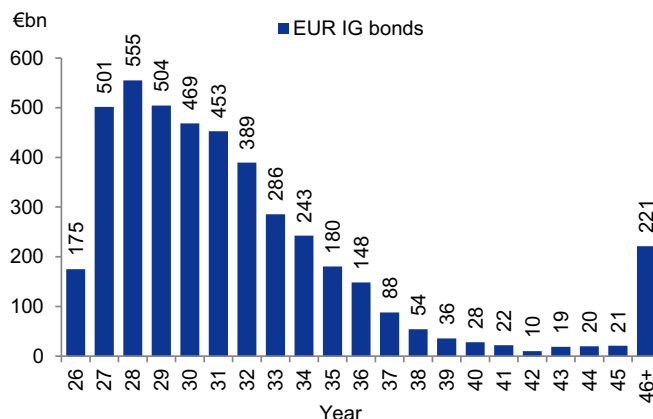
Primary market dashboard: Maturity walls

Exhibit 129: USD IG bond market maturity wall



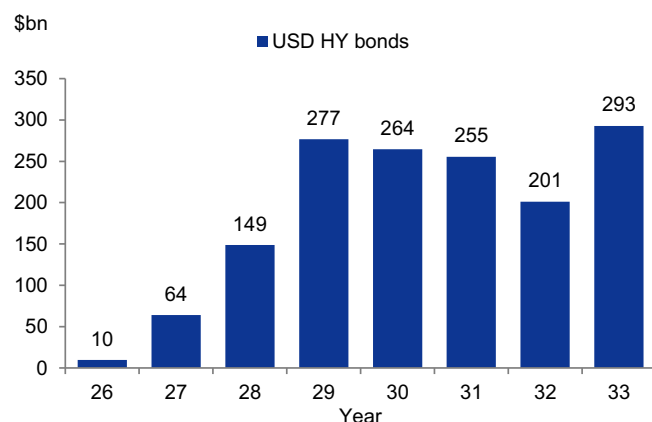
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 130: EUR IG bond market maturity wall



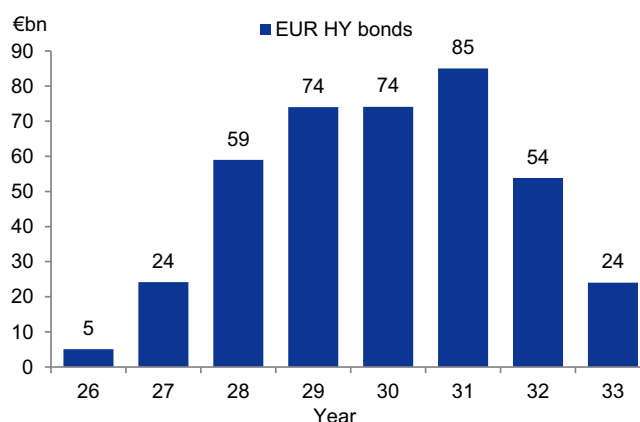
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 131: USD HY bond market maturity wall



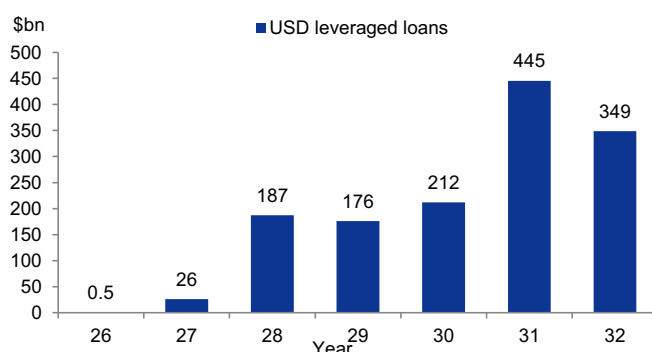
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 132: EUR HY bond market maturity wall



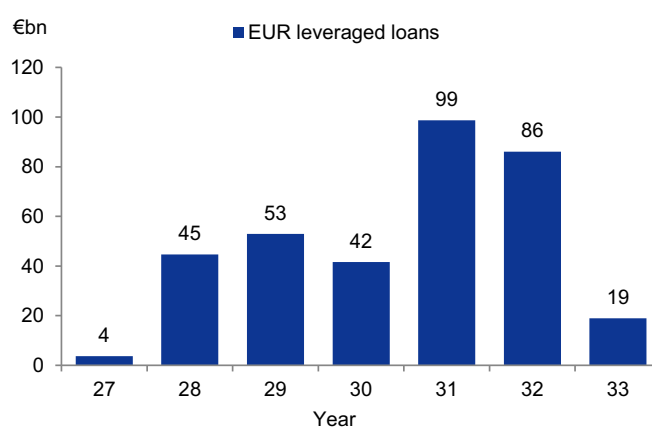
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 133: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

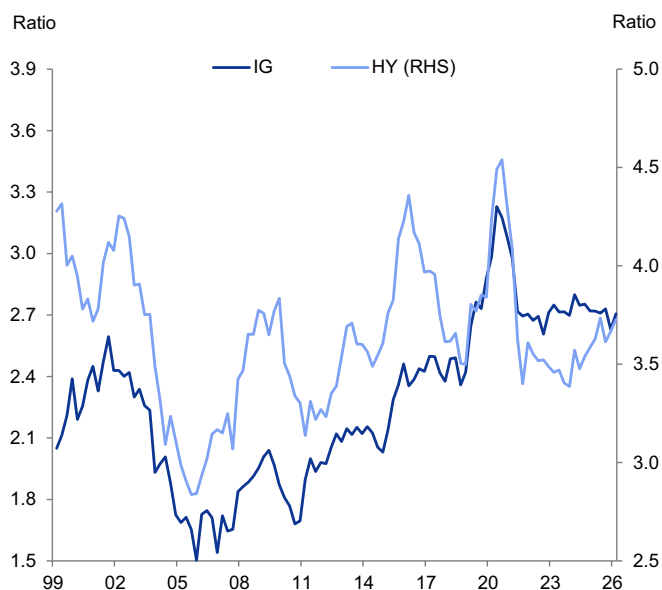
Exhibit 134: EUR leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

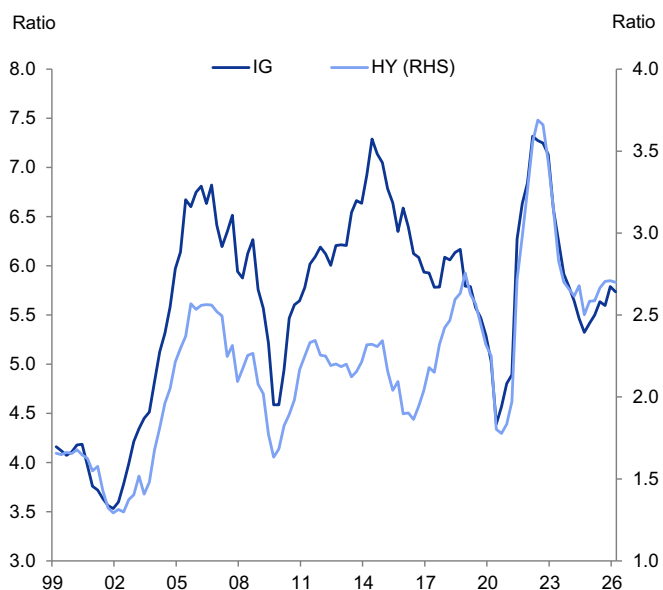
Fundamental dashboard: Credit metrics for US non-financial corporations

Exhibit 135: Net debt to EBITDA ratios for the median US IG and HY-rated non-financial corporations



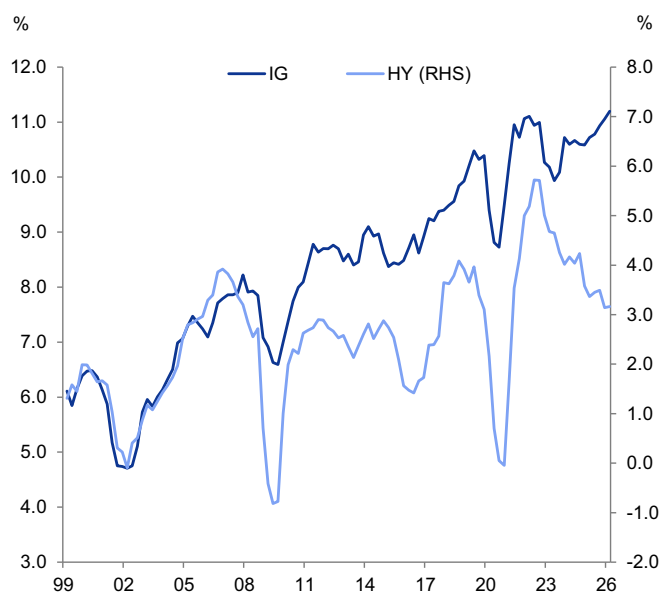
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 136: Interest coverage ratios for the median US IG and HY-rated non-financial corporations



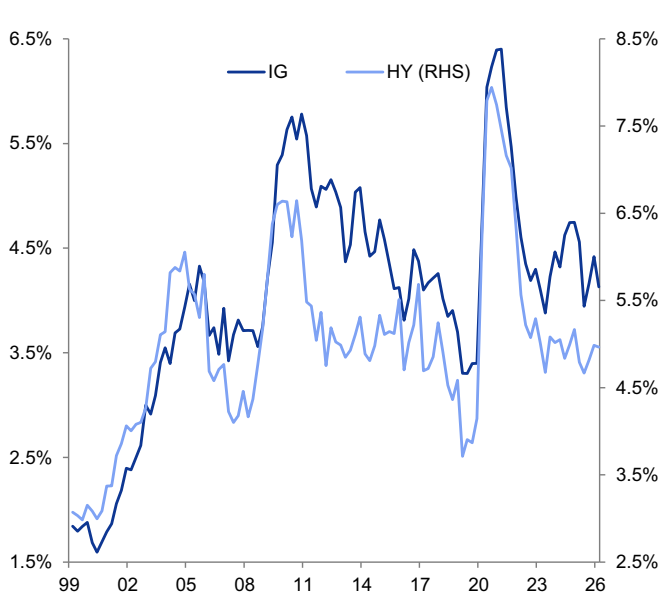
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 137: Net margins for the median US IG and HY-rated non-financial corporations



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 138: Cash to total assets ratios for the median US IG and HY-rated corporations

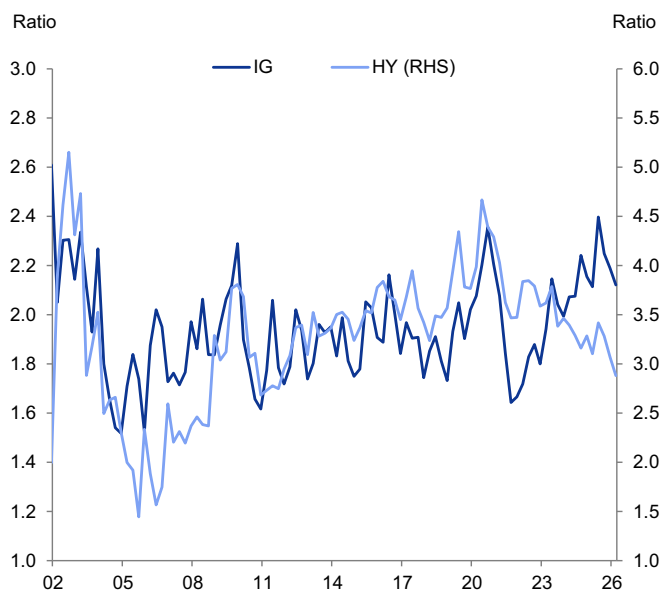


Source: FactSet, Goldman Sachs Global Investment Research

Fundamental dashboard: Credit metrics for European non-financial corporations

Exhibit 139: Net debt to EBITDA ratios for the median European IG and HY-rated non-financial corporations

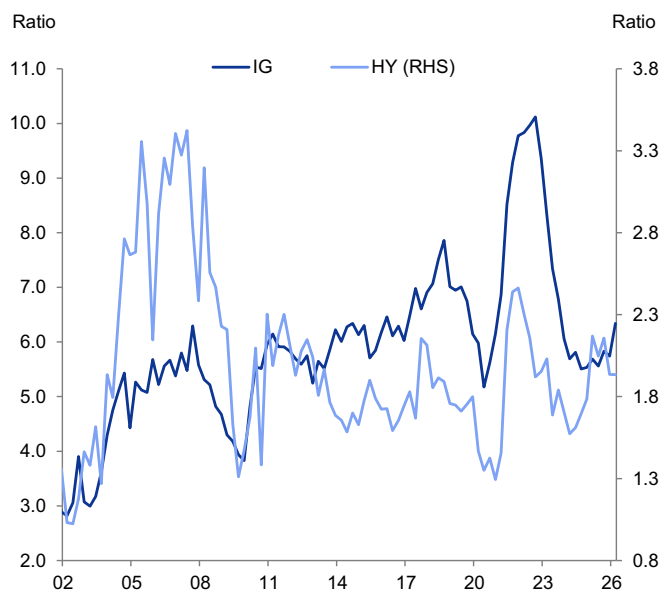
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 140: Interest coverage ratios for the median European IG and HY-rated non-financial corporations

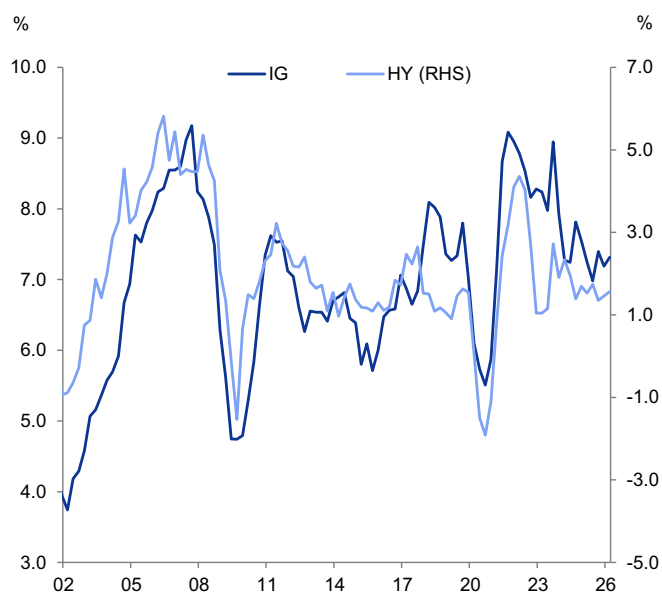
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 141: Net margins for the median European IG and HY-rated non-financial corporations

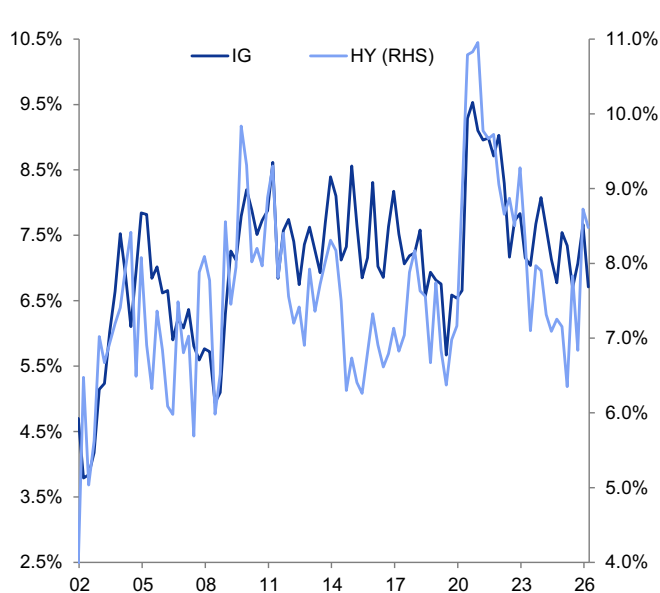
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 142: Cash to total assets ratios for the median European IG and HY-rated non-financial corporations

We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway

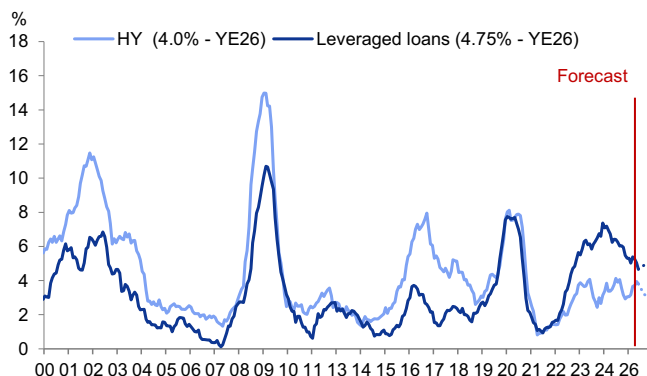


Source: FactSet, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 143: Default rates for US HY bond and leveraged loan issuers

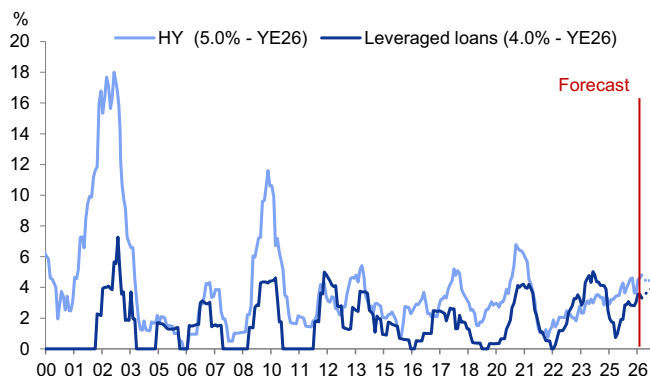
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 144: Default rates for European bond and leveraged loan issuers

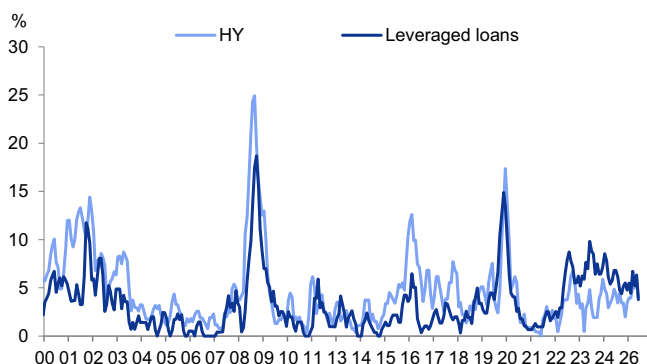
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 145: High frequency default rates for US HY bond and leveraged loan issuers

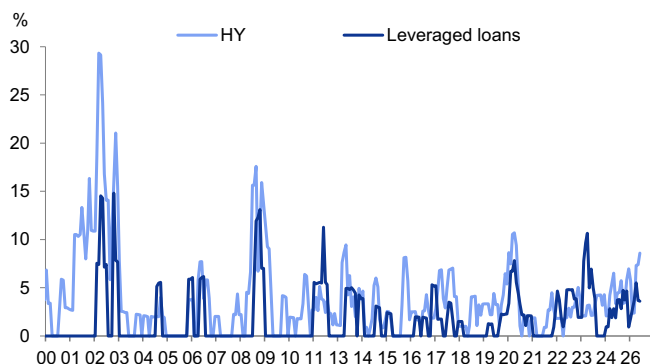
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

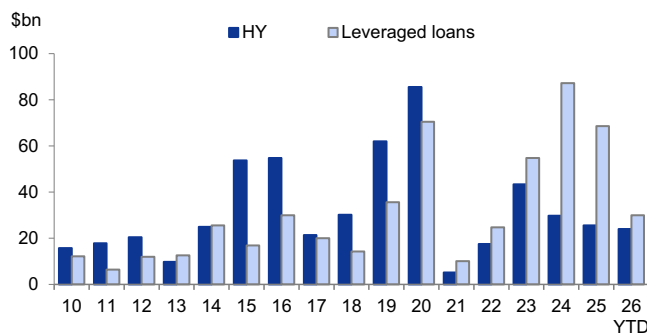
Exhibit 146: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



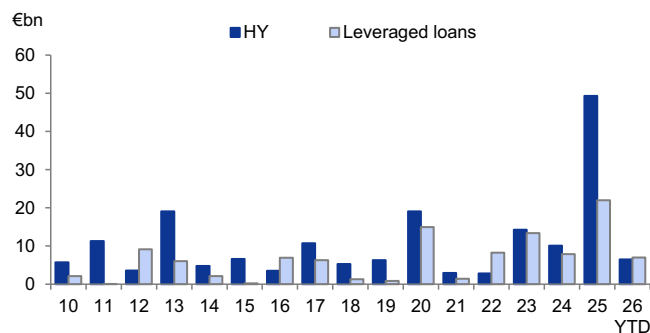
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 147: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 148: Par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 149: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

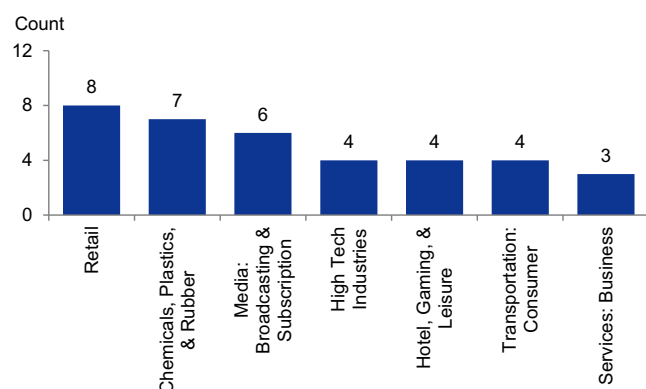
	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	6/30/2026	DISH DBS Corporation	Media: Broadcasting & Subscription	Prepackaged Chapter 11	9.8
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	8/8/2025	Saks Global Enterprises LLC	Retail	Distressed exchange	2.1
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.9
	2/17/2026	Vibrant Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6
	12/8/2025	Newfold Digital Holdings Group, Inc	High Tech Industries	Distressed exchange	2.5
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.3

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

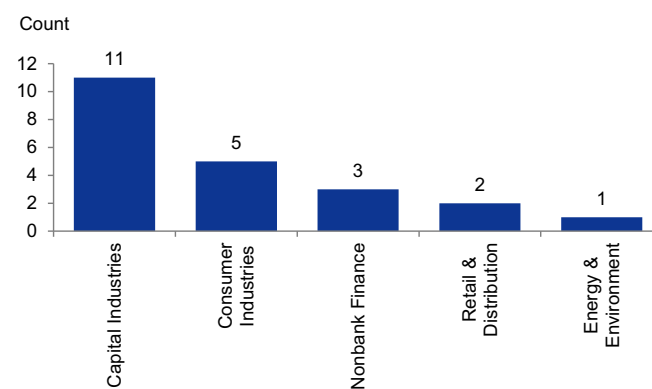
Exhibit 150: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
	4/29/2026	La Financiere ATALIAN S.A.S.	Services: Business	Missed interest payment	1.0
	11/12/2025	ARD Finance S.A.	Containers, Packaging, & Glass	Distressed exchange	0.9
	11/4/2025	Klockner Pentaplast of America, Inc	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8
Loans	4/30/2026	Poseidon BidCo S.A.S	High Tech Industries	Missed interest payment	1.6
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	4/7/2026	Biscuit Holding S.A.S.	Beverage, Food, & Tobacco	Missed interest payment	0.8
	11/4/2025	Klockner Pentaplast of America, Inc	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8
	2/18/2026	Accell Group Holding B.V.	Consumer goods: Durable	Distressed exchange	0.7

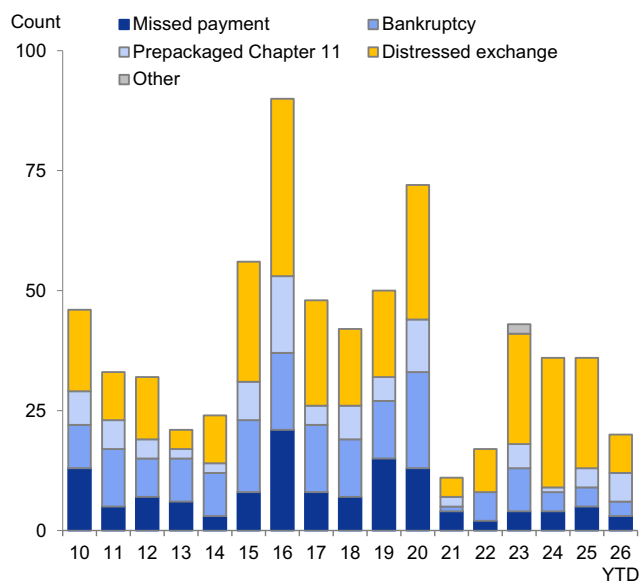
Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 151: Industry default count in the US

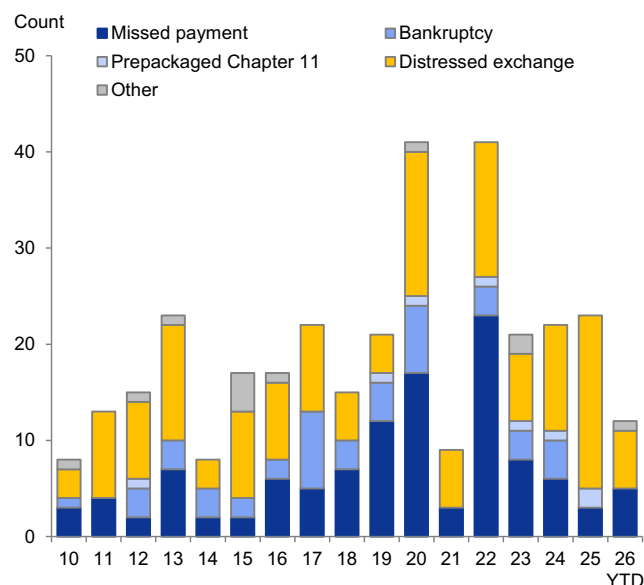
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 152: Industry default count in Europe

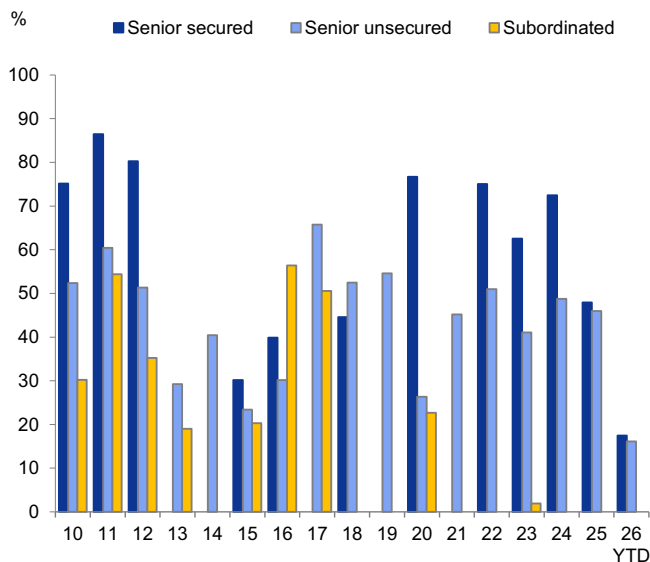
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 153: Composition of defaults in the USD HY bond market by type

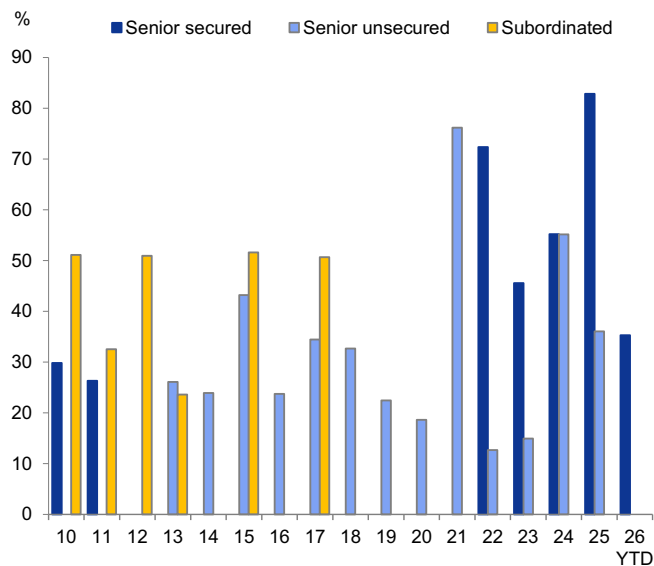
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 154: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

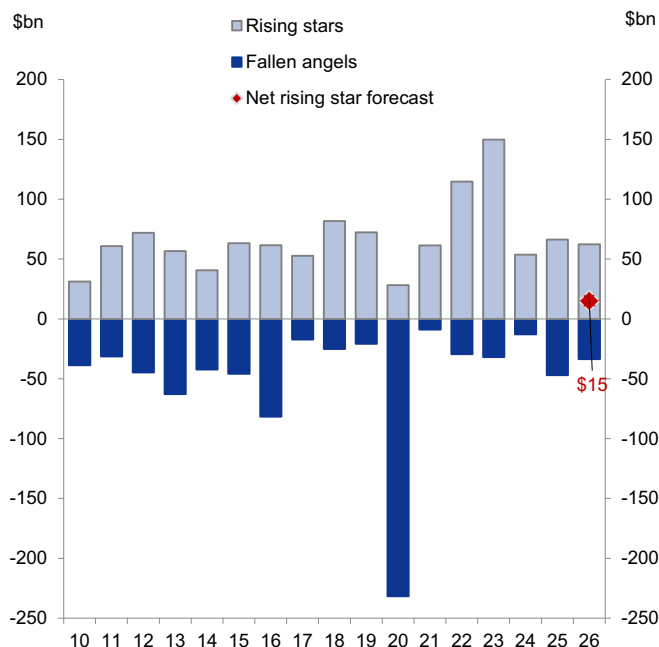
Exhibit 155: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

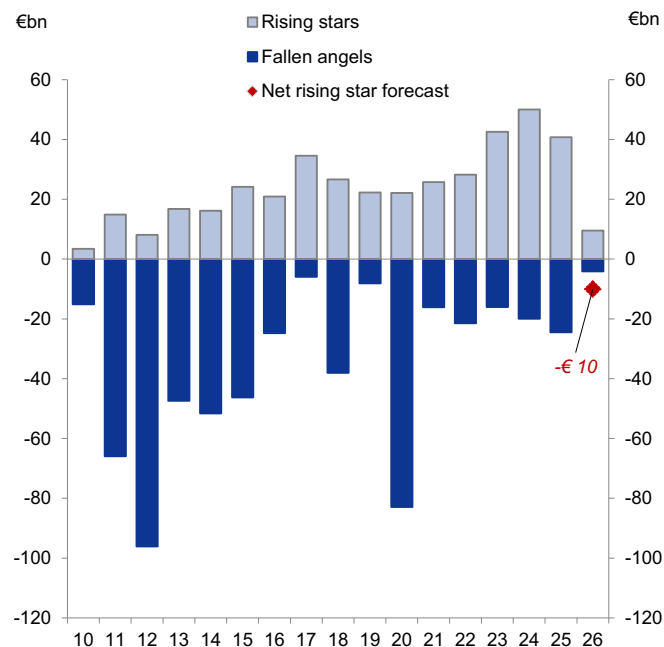
Exhibit 156: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

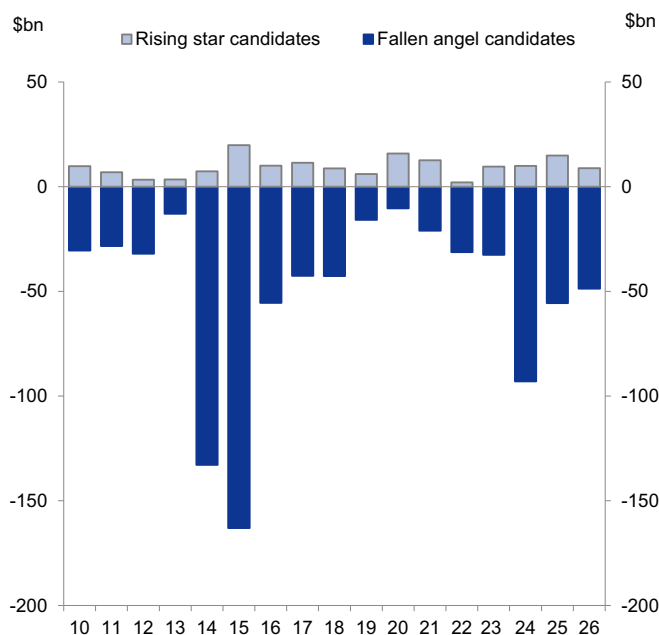
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 157: Fallen angels vs. rising stars in the USD market


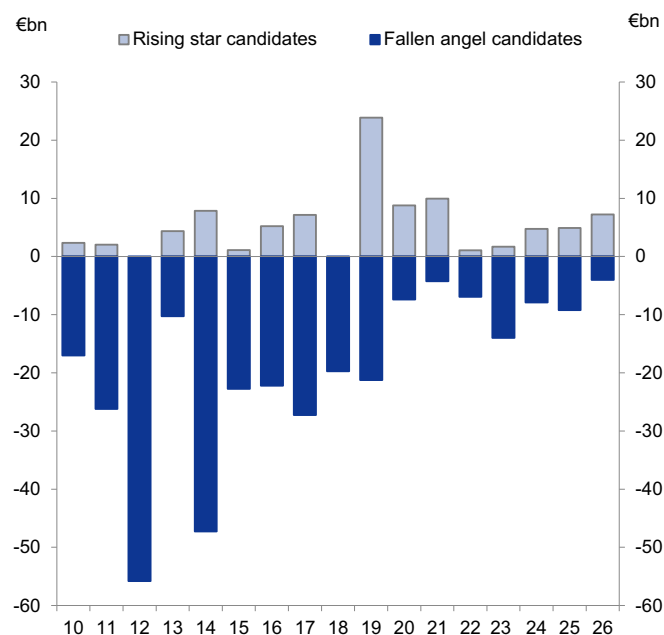
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 158: Fallen angels vs. rising stars in the EUR market


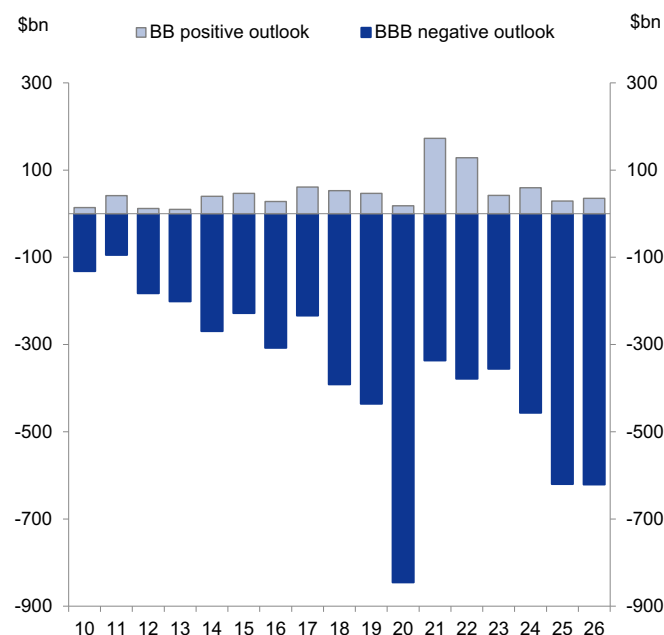
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 159: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


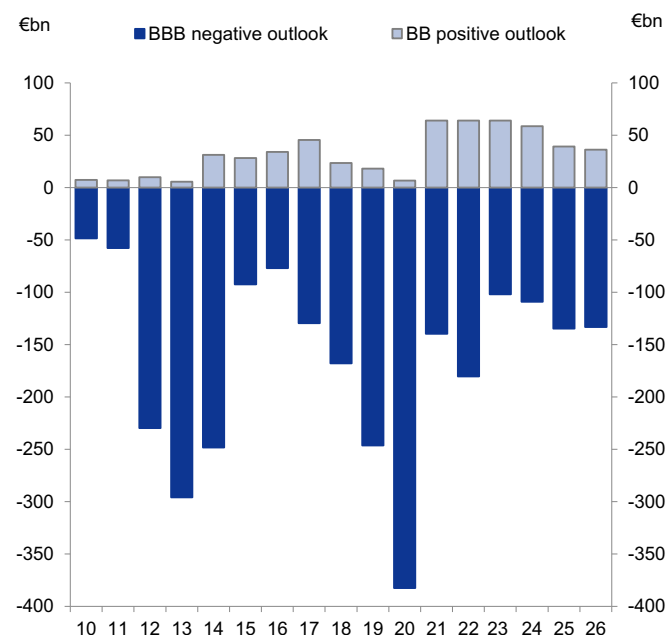
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 160: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 161: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 162: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 163: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Coty/Hfc Prestige/Int Us	1.7
Banco Santander Mexico	1.3
Horizon Mutual Holdings	0.6
Coty/Hfc Prestige/Int Us	0.5
Ses Americom Inc	0.5
Ses Sa	0.3

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 164: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Carnival Corp Ltd	8.5
Vistra Operations Co Llc	6.3
Medline Borrower Lp	4.5
Office Cherifien Des Pho	4.1
1011778 Bc / New Red Fin	4.0
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Sba Communications Corp	3.0
Frontier Communications	2.8
Frontier Communications	2.8

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 165: Top EUR Fallen angel issues

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Adecco Int Financial Svs	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 166: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Carnival Uk Ltd	1.6
Unipol Assicurazioni Spa	1.3
Metro Ag	1.1
Bper Banca	1.0
National Bank Greece Sa	1.0
Piraeus Bank Sa	1.0
Czechoslovak Group	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Amanda Lynam, CPA, Spencer Rogers, CFA, Sara Grut and Shamshad Ali, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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